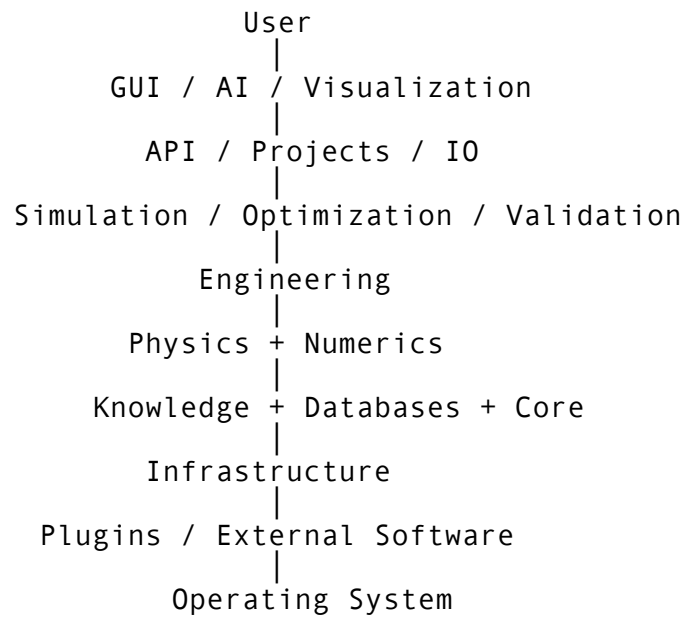


Dependency Structure



Part 1 → Root Repository Architecture

Part 2 → core/

Part 3 → knowledge/

Part 4 → physics/

Part 5 → numerics/

Part 6 → engineering/

Part 7 → simulation/

Part 8 → optimization/

Part 9 → ai/

Part 10 → gui/

Part 11 → visualization/

Part 12 → databases/

Part 13 → plugins/

Part 14 → api/

Part 15 → io/

Part 16 → validation/

Part 17 → projects/

Part 18 → examples/

Part 19 → docs/

Part 20 → infrastructure/

Part 21 → governance/

Part 22 → scripts/

Part 23 → tests/

Part 24 → external/

Architectural layering

Organized into six layers

Layer-1 : Foundation

Layer-2: Scientific Computing

Layer-3: Engineering Workflows

Layer-4 : integration

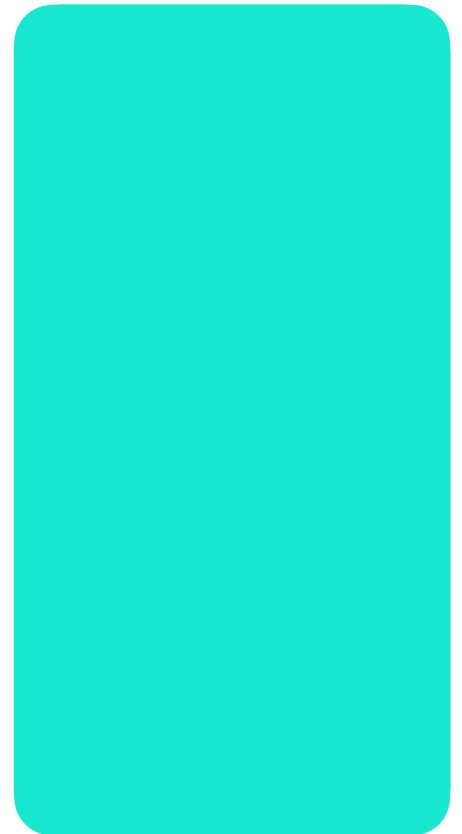
Layer-5 : Presentation

Layer-6 : Governance

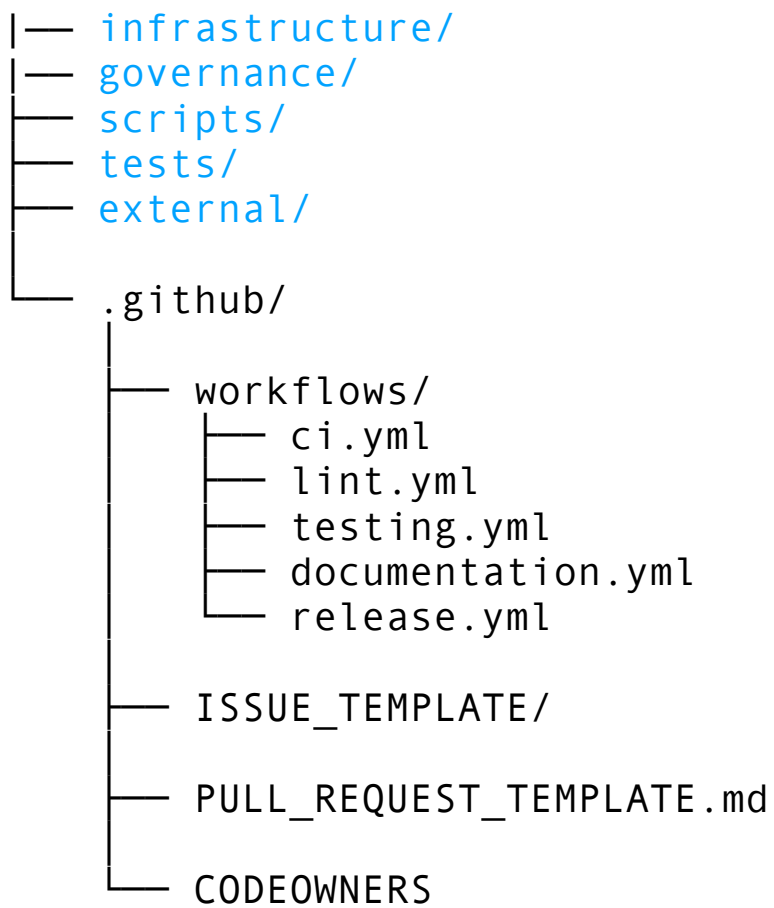
COSMOS_0.1/

Root Repository Architecture

- README.md
- LICENSE
- CHANGELOG.md
- CONTRIBUTING.md
- CODE_OF_CONDUCT.md
-
- pyproject.toml
- requirements.txt
- requirements-dev.txt
- setup.py
- setup.cfg
- MANIFEST.in
-
- .gitignore
- .gitattributes
- .editorconfig
- .pre-commit-config.yaml
-
- main.py
- launcher.py
-
- core/
- knowledge/
- physics/
- numerics/
- engineering/
- simulation/
- optimization/
- ai/
- gui/
- visualization/
- databases/
- plugins/
- api/
- io/
- validation/
- projects/
- examples/
- docs/



COSMOS_0.1 CODE STRUCTURE



COSMOS_0.1 CODE STRUCTURE

core/

- `__init__.py`
- `constants.py`
- `units.py`
- `validation.py`
- `logger.py`
- `config.py`
- `exceptions.py`
- `settings.py`
- `version.py`
- `metadata.py`
- `registry.py`
- `profiler.py`
- `environment.py`
- `paths.py`
- `platform.py`
- `resources.py`
- `dependency_manager.py`
- `plugin_registry.py`
- `serialization.py`
- `hashing.py`
- `timers.py`
- `decorators.py`
- `utilities.py`
- `math_utils.py`
- `string_utils.py`
- `file_utils.py`
- `collection_utils.py`
- `typing.py`
- `enums.py`
- `state_machine.py`
- `event_bus.py`
- `progress.py`
- `diagnostics.py`
- `startup.py`
- `shutdown.py`
- `licensing.py`
- `telemetry.py`

knowledge/

— `__init__.py`

— `models/`

— `document.py`

— `chapter.py`

— `section.py`

— `paragraph.py`

— `sentence.py`

— `figure.py`

— `table.py`

— `appendix.py`

— `glossary.py`

— `reference.py`

— `citation.py`

— `equation.py`

— `variable.py`

— `constant.py`

— `unit.py`

— `dimension.py`

— `quantity.py`

— `physical_law.py`

— `correlation.py`

— `empirical_relation.py`

— `assumption.py`

— `boundary_condition.py`

— `material.py`

— `property.py`

— `component.py`

— `subsystem.py`

— `engineering_domain.py`

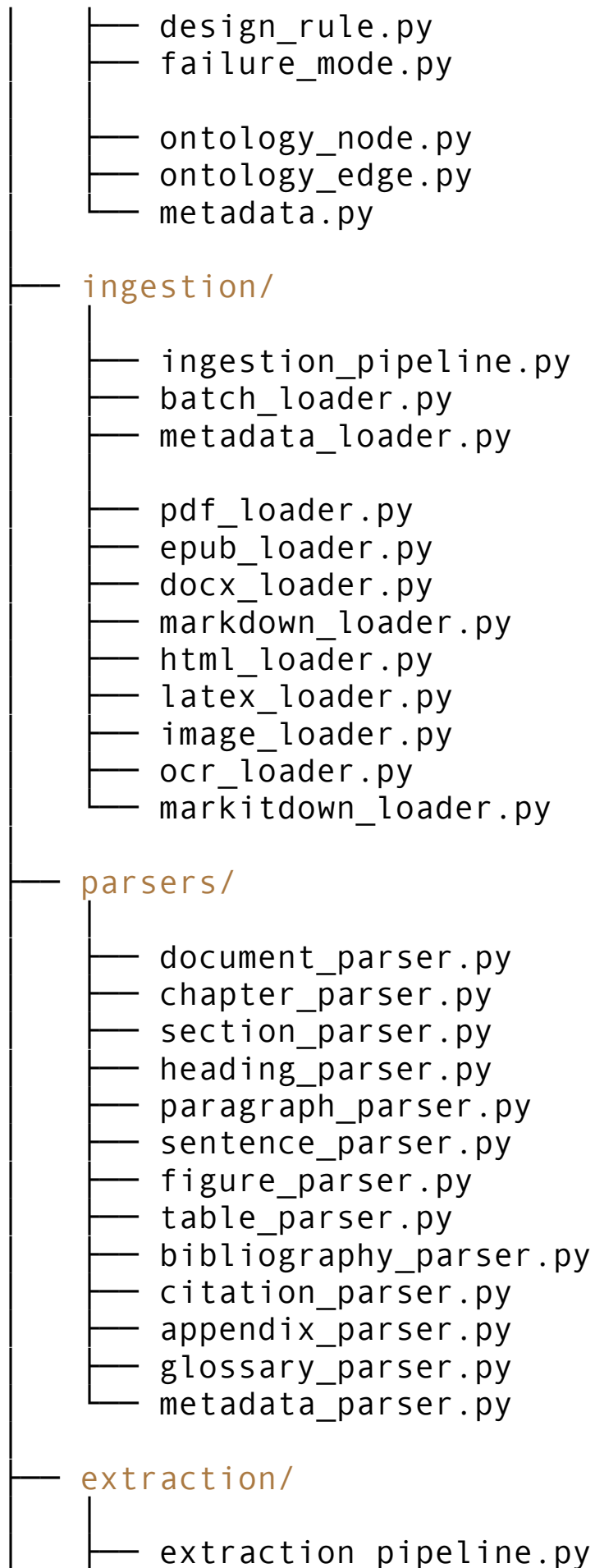
— `process.py`

— `manufacturing_process.py`

— `experiment.py`

— `simulation.py`

COSMOS_0.1 CODE STRUCTURE



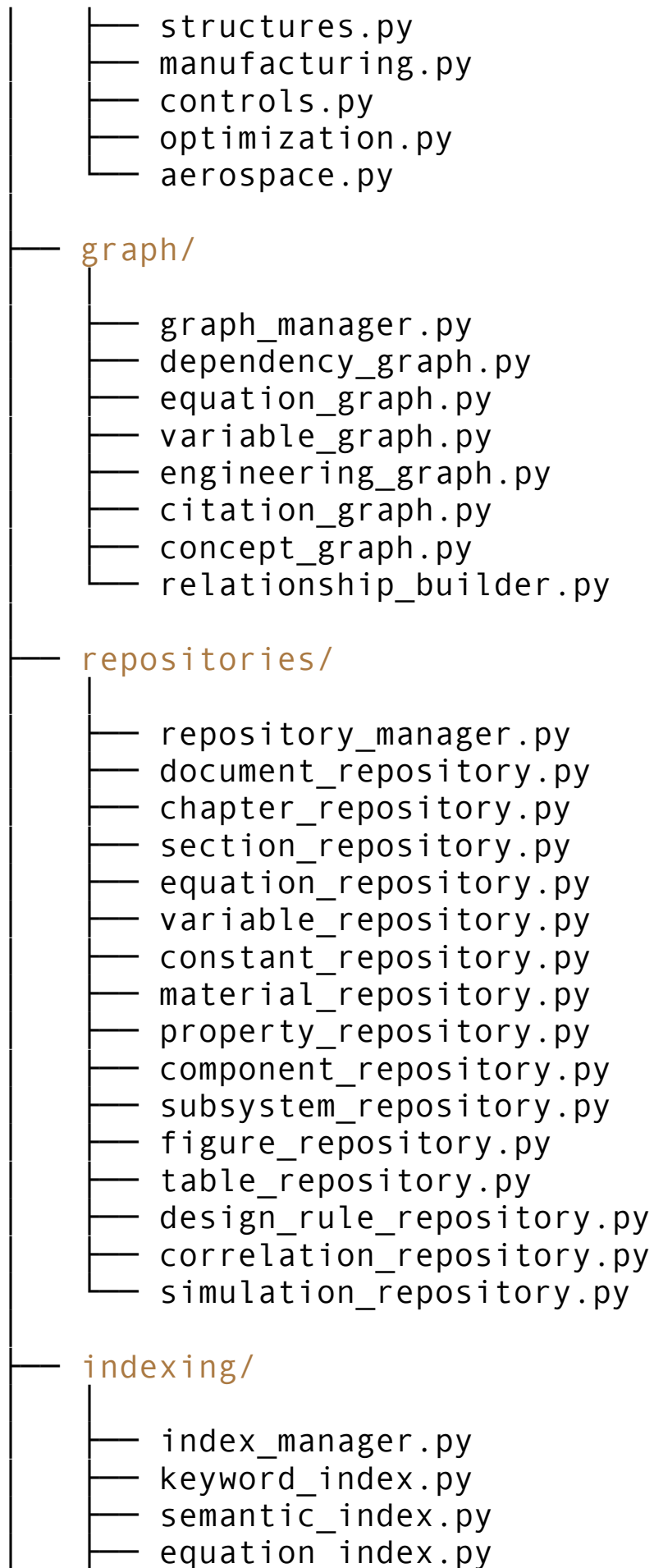
COSMOS_0.1 CODE STRUCTURE

- equation_extractor.py
- variable_extractor.py
- constant_extractor.py
- unit_extractor.py
- dimension_extractor.py
- quantity_extractor.py
- material_extractor.py
- property_extractor.py
- component_extractor.py
- subsystem_extractor.py
- engineering_domain_extractor.py
- process_extractor.py
- manufacturing_extractor.py
- experiment_extractor.py
- simulation_extractor.py
- failure_mode_extractor.py
- design_rule_extractor.py
- physical_law_extractor.py
- correlation_extractor.py
- assumption_extractor.py
- boundary_condition_extractor.py
- glossary_extractor.py
- abbreviation_extractor.py

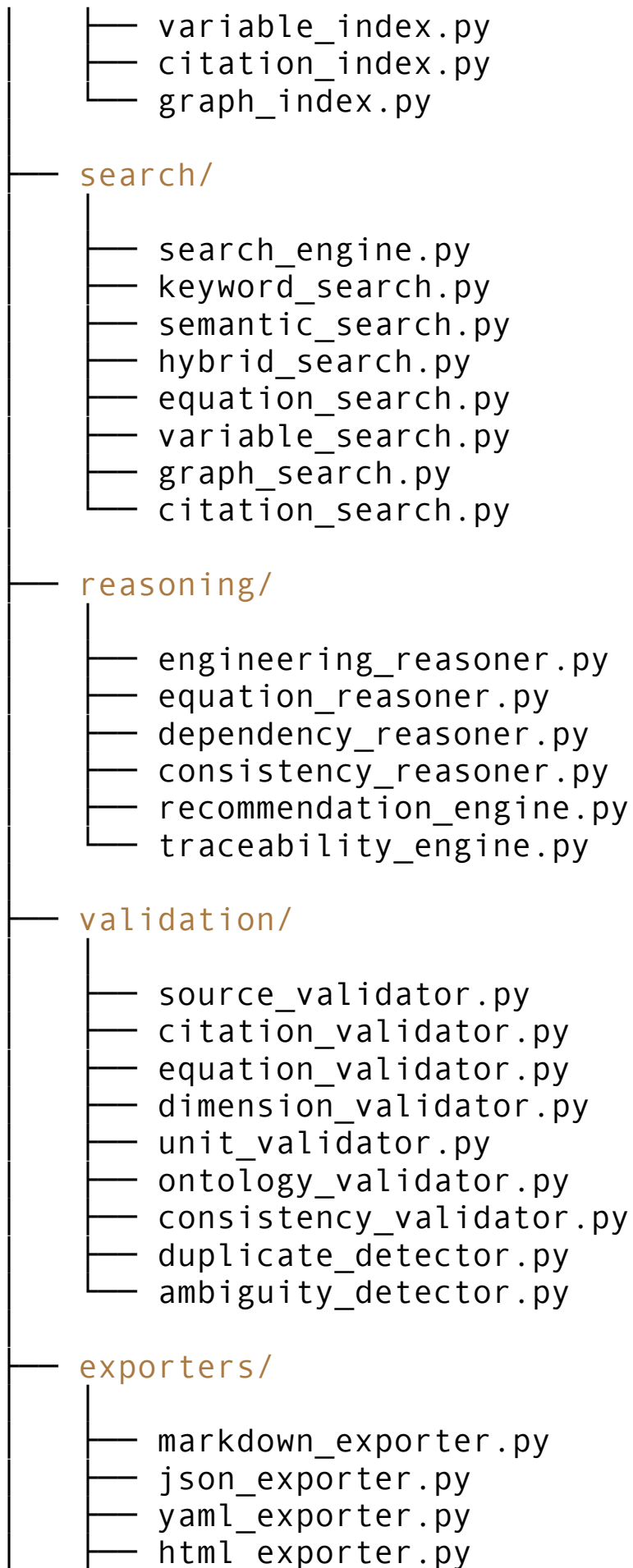
ontology/

- ontology_manager.py
- engineering_domains.py
- propulsion.py
- thermodynamics.py
- thermochemistry.py
- combustion.py
- fluid_mechanics.py
- compressible_flow.py
- heat_transfer.py
- cryogenics.py
- materials.py

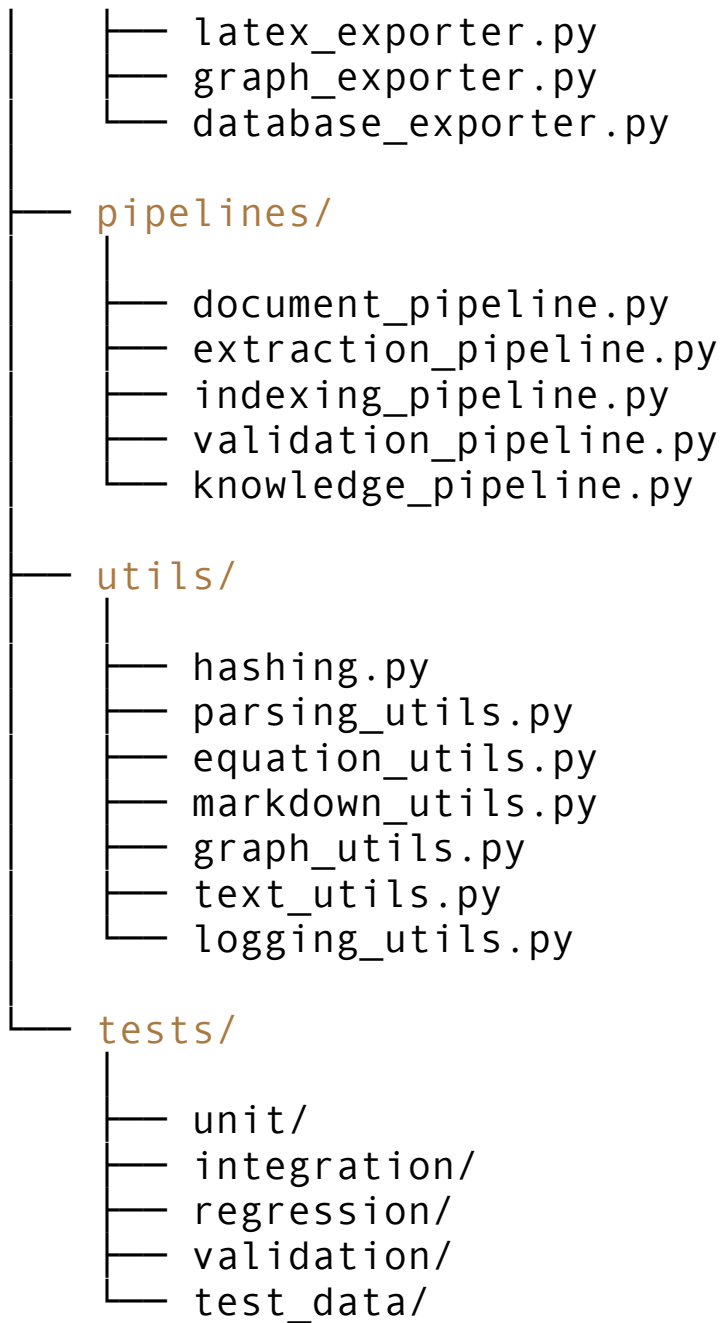
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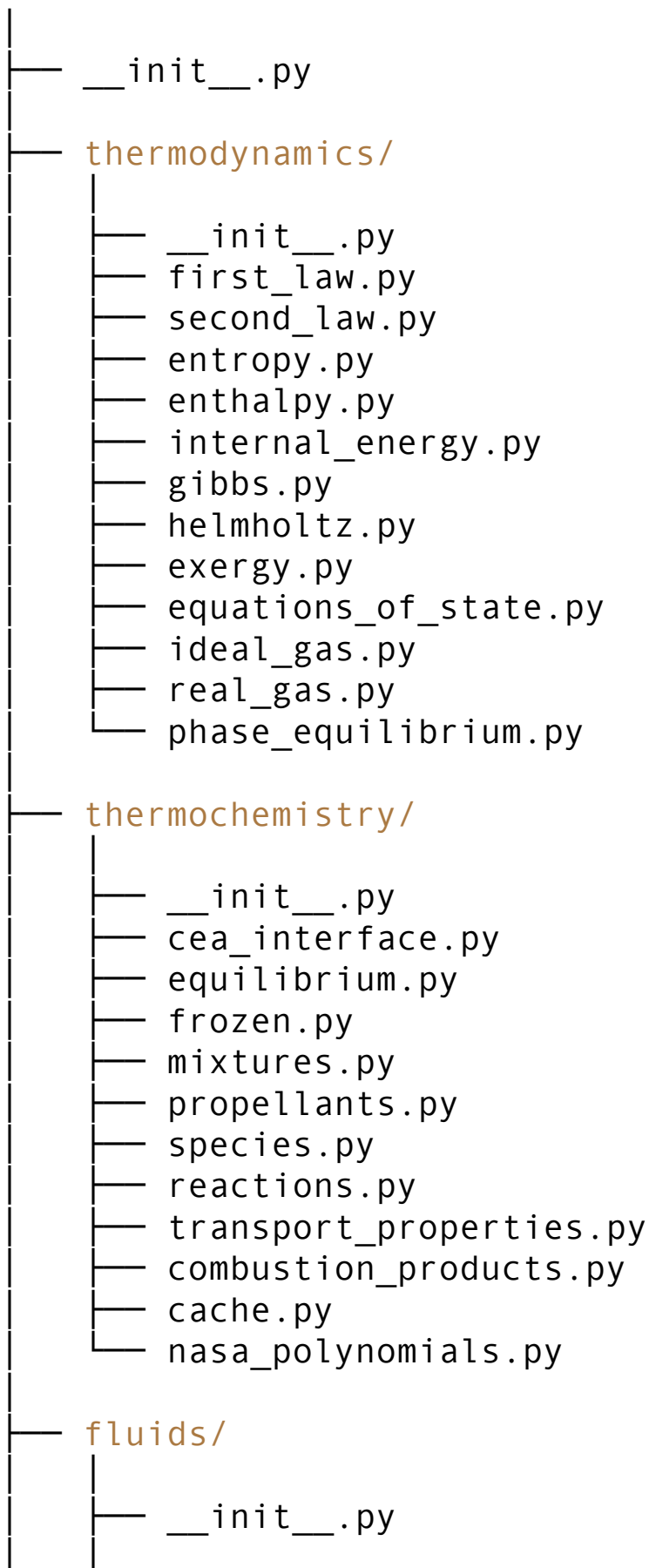
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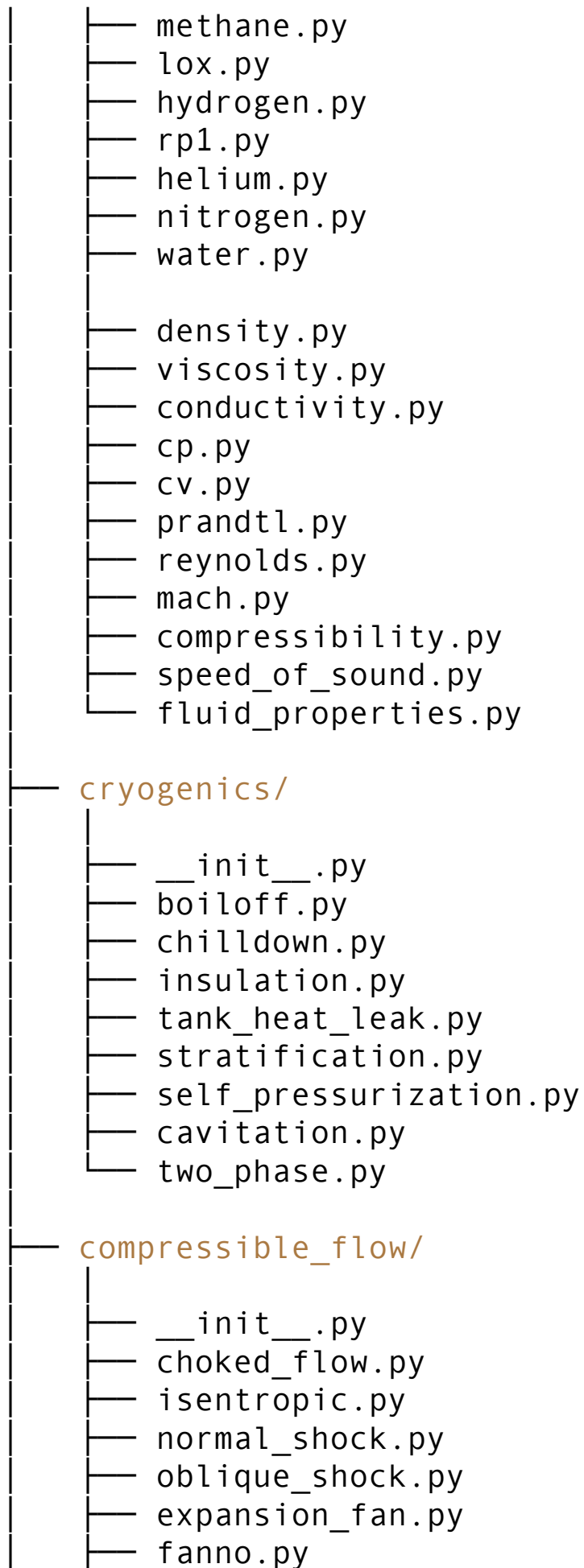
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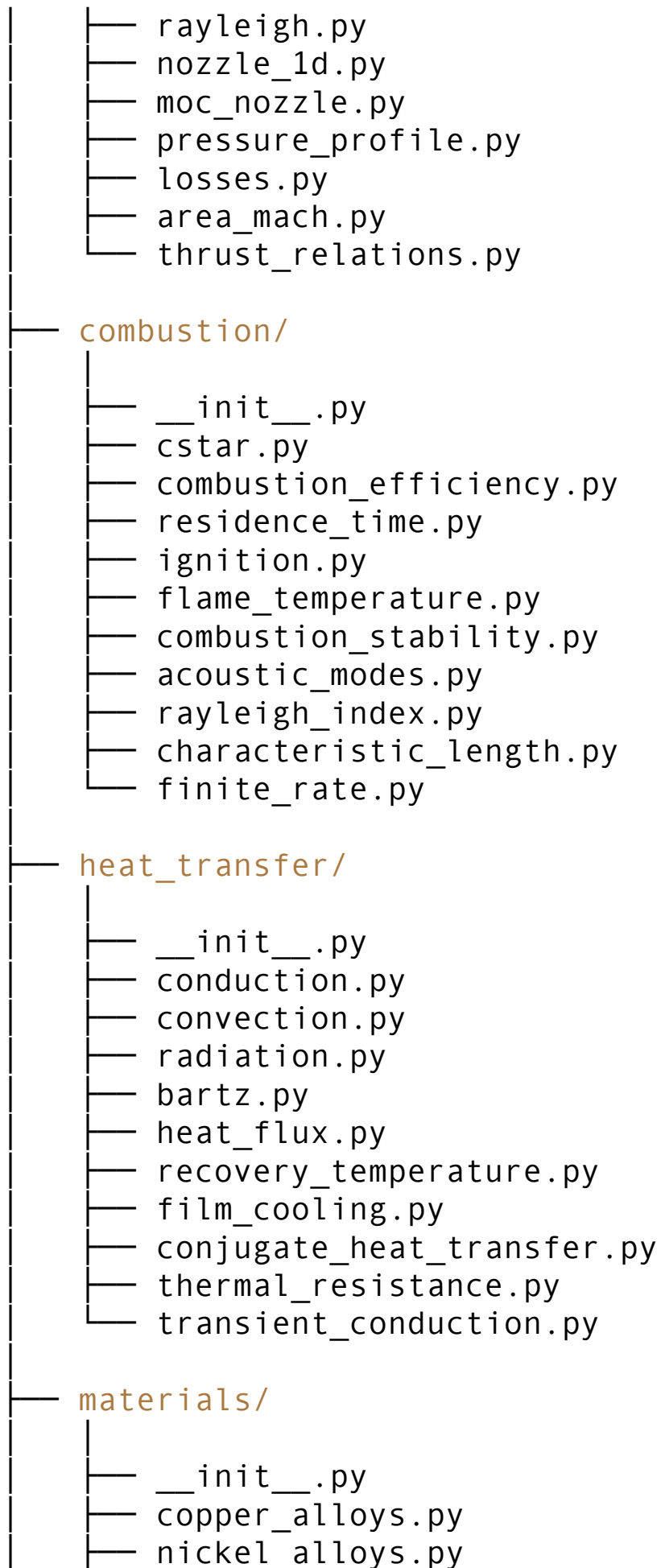
physics/



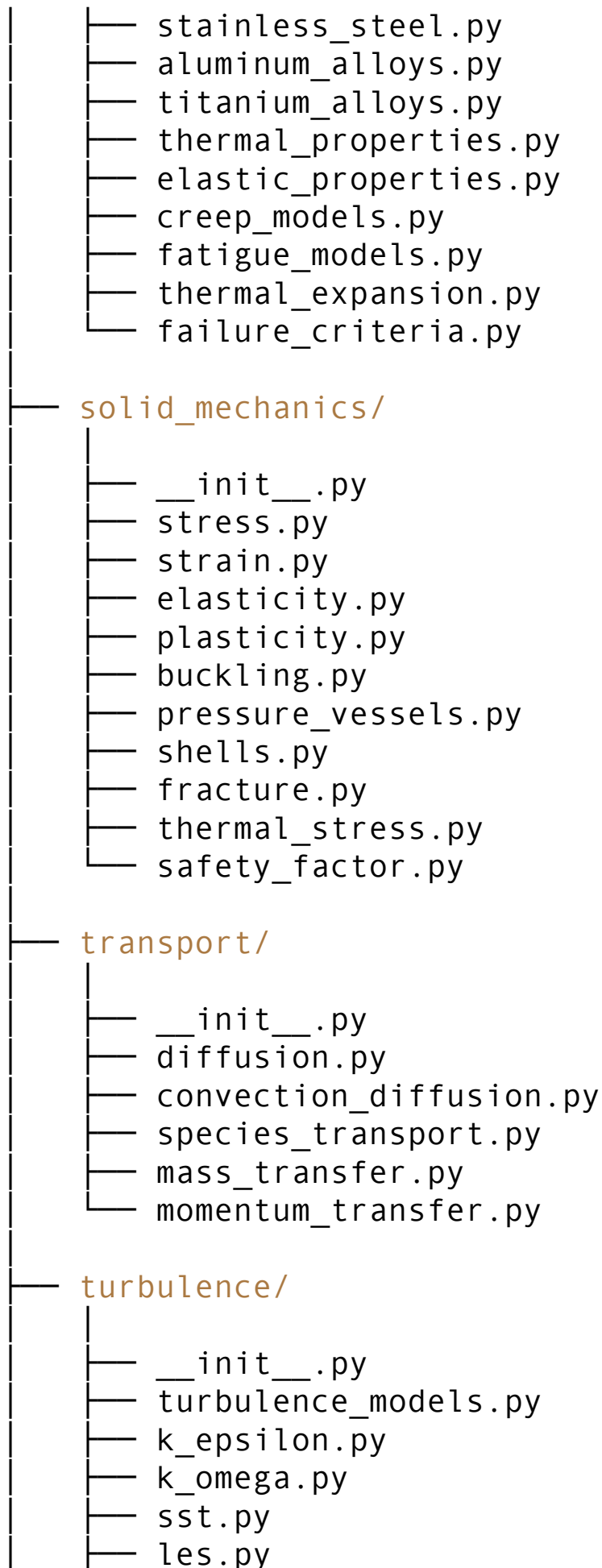
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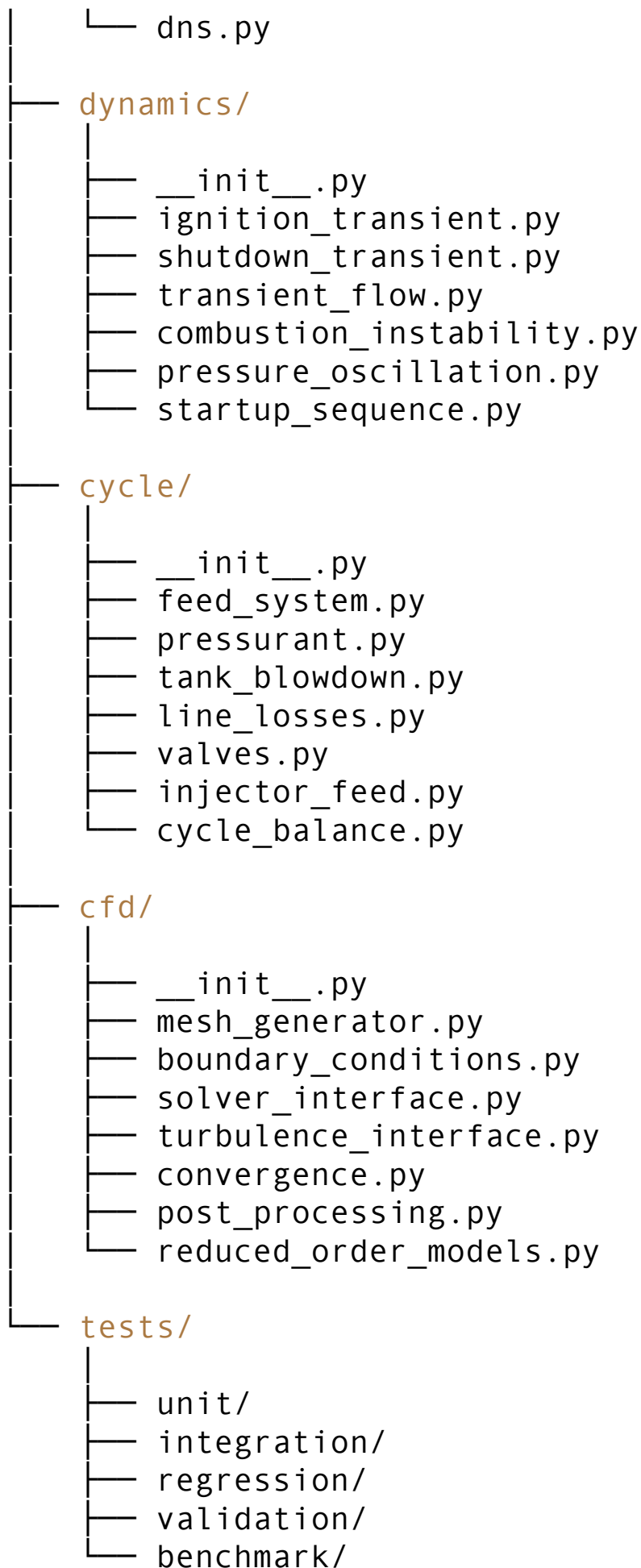
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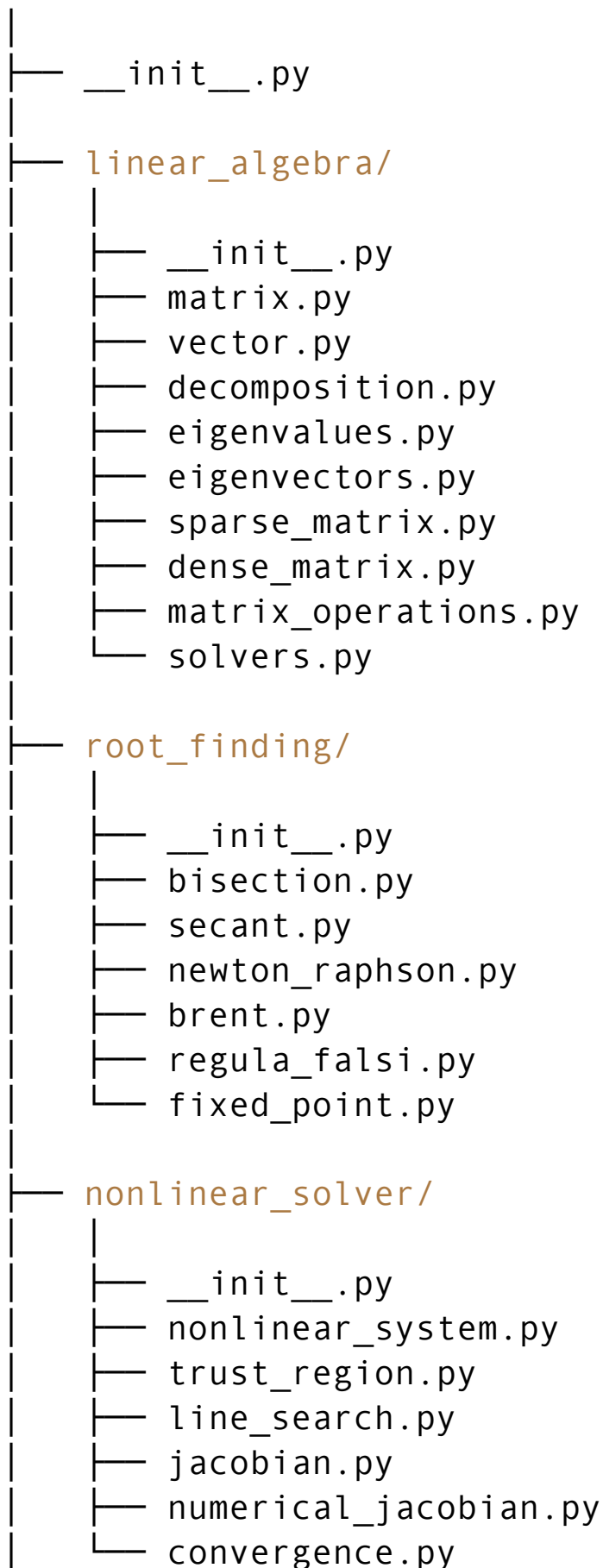
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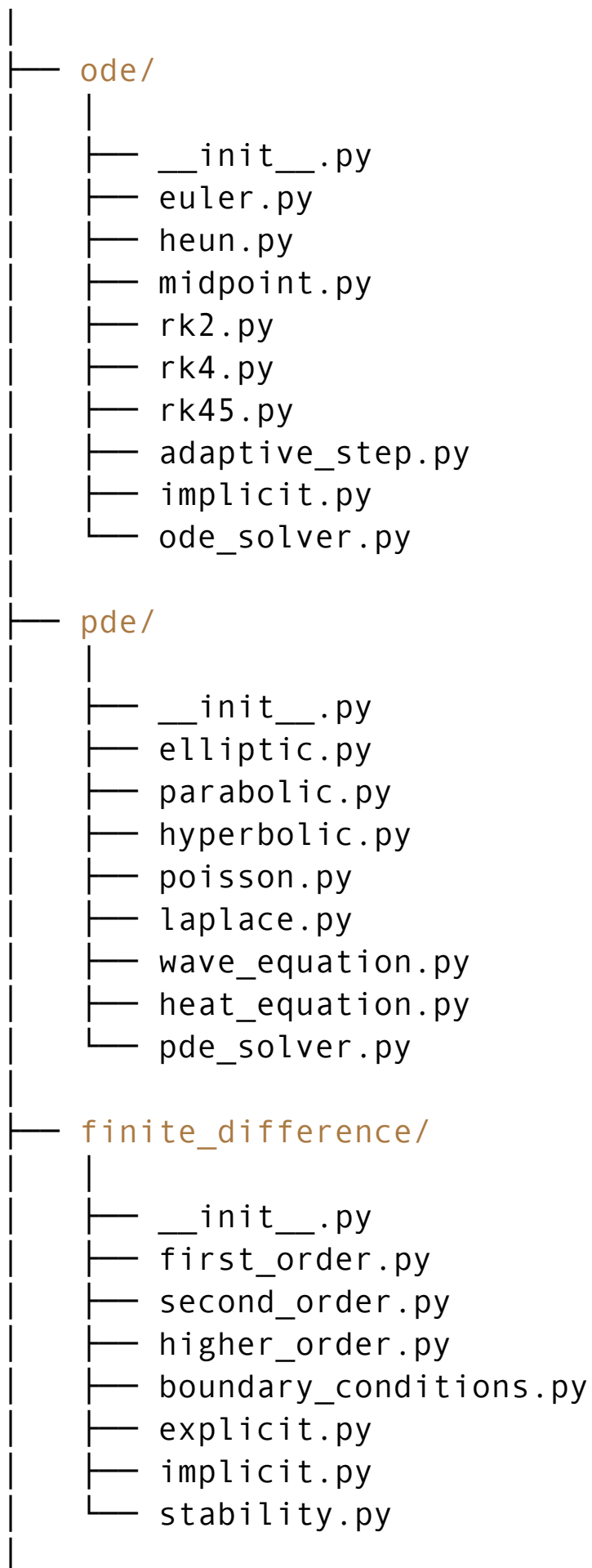
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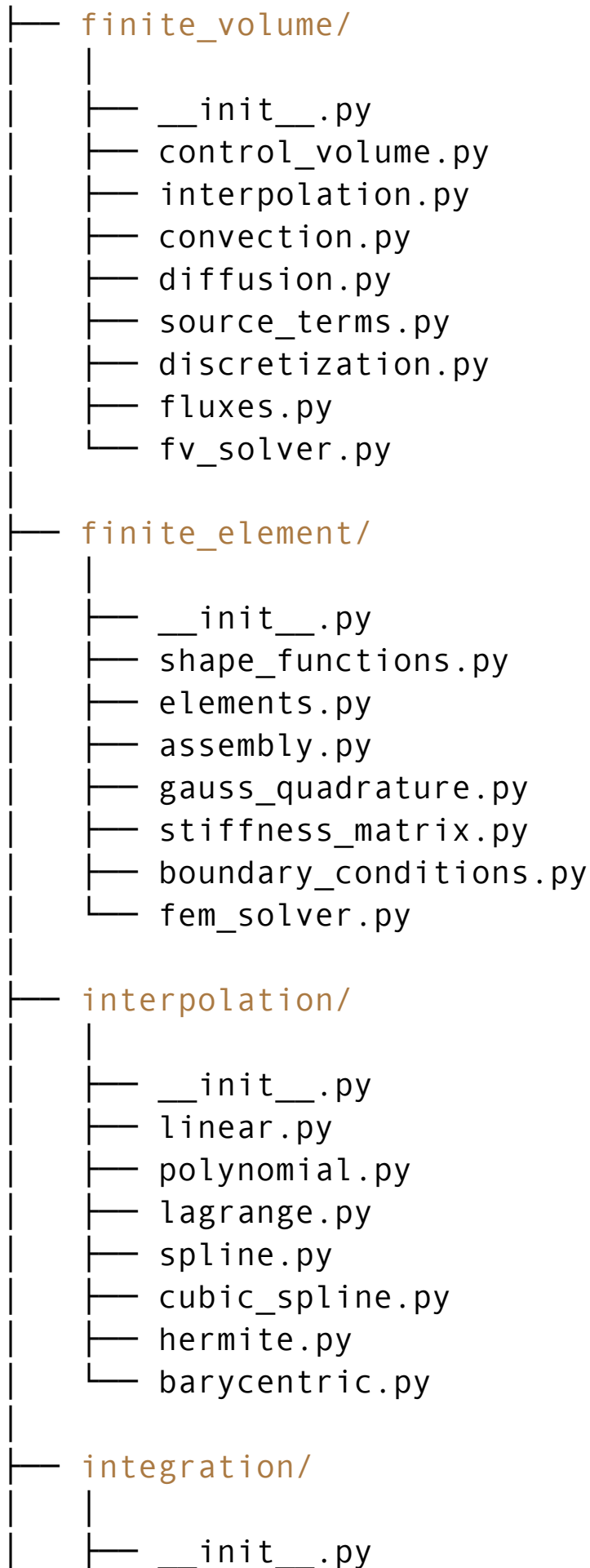
numerics/



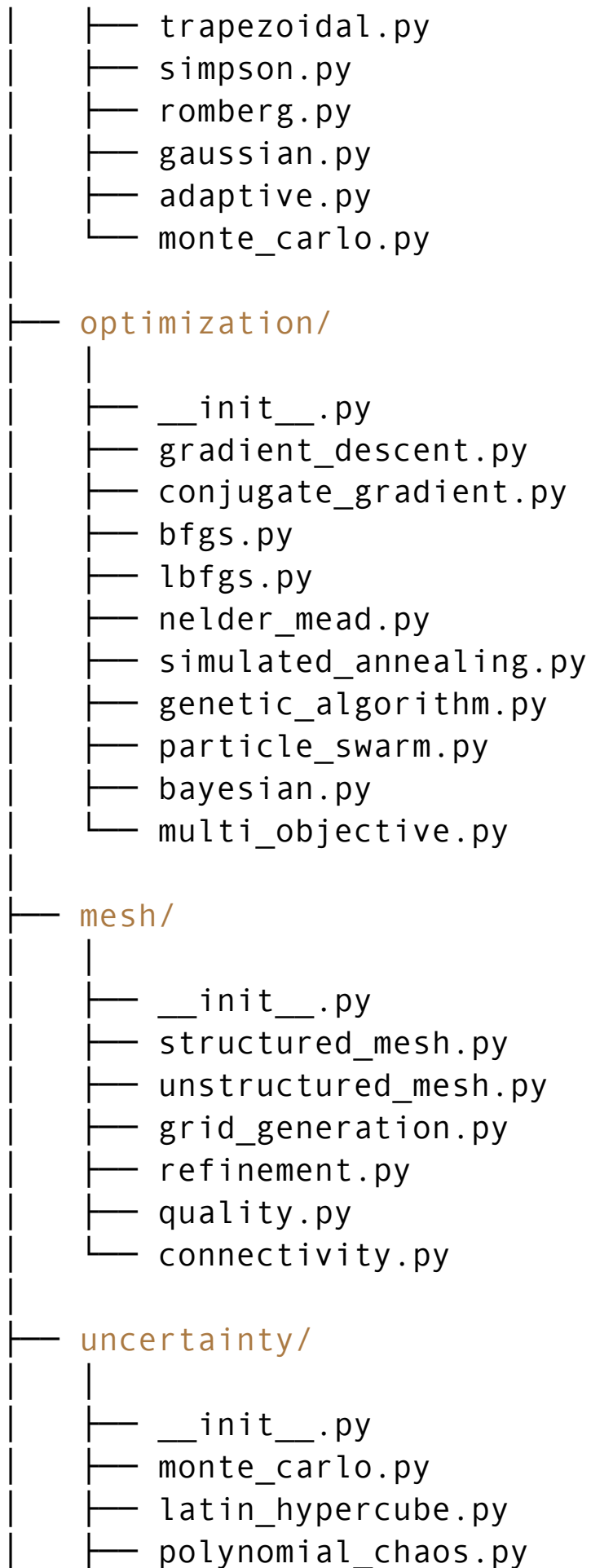
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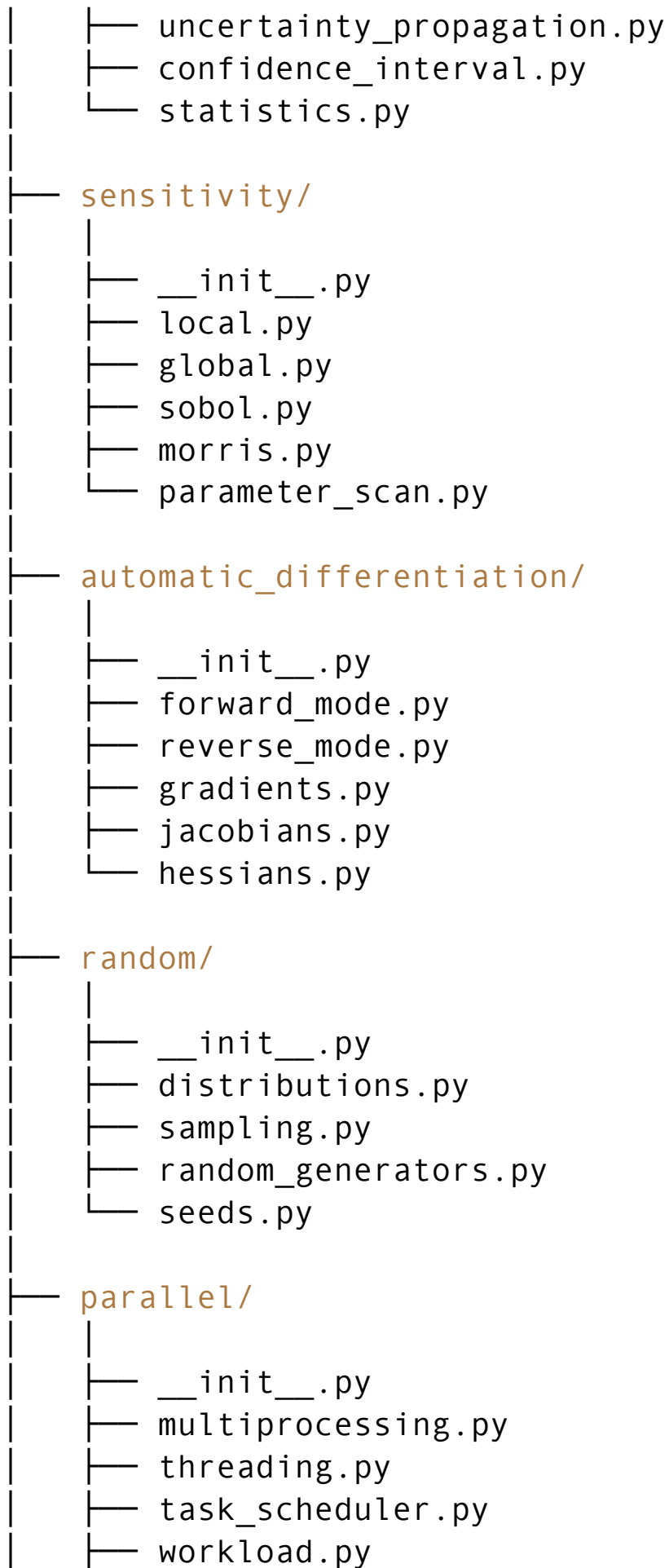
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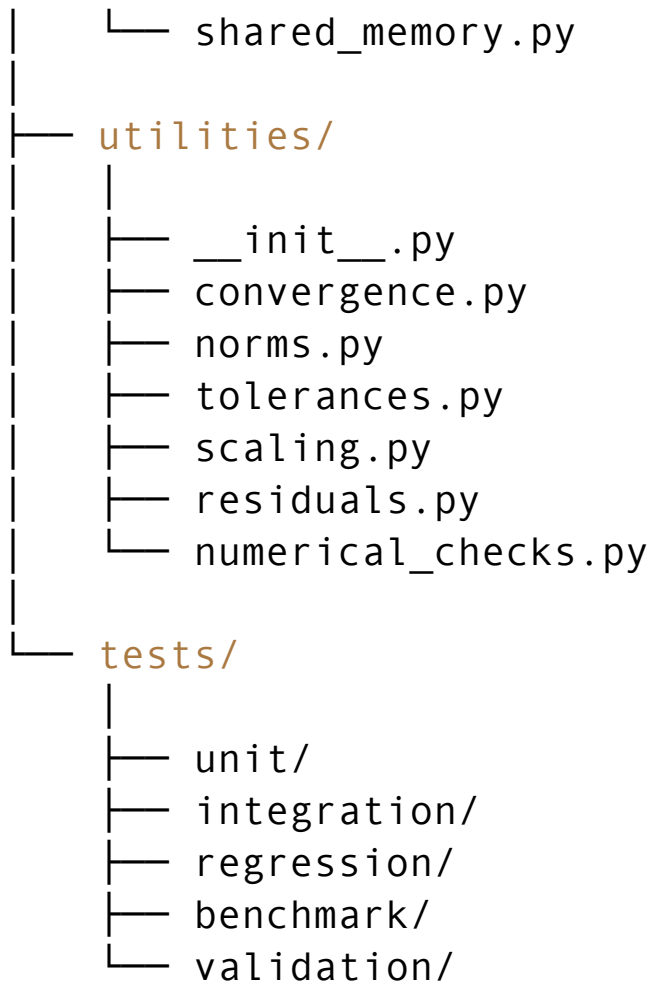
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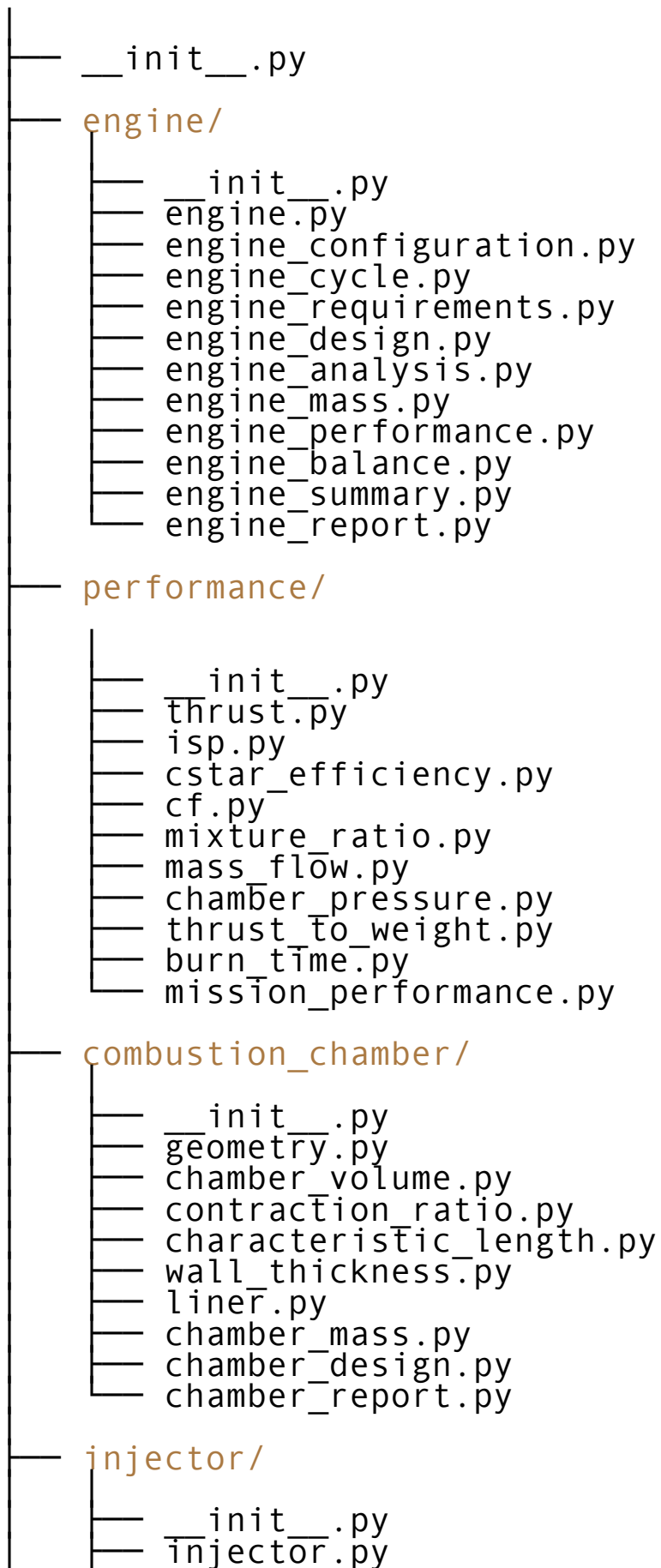
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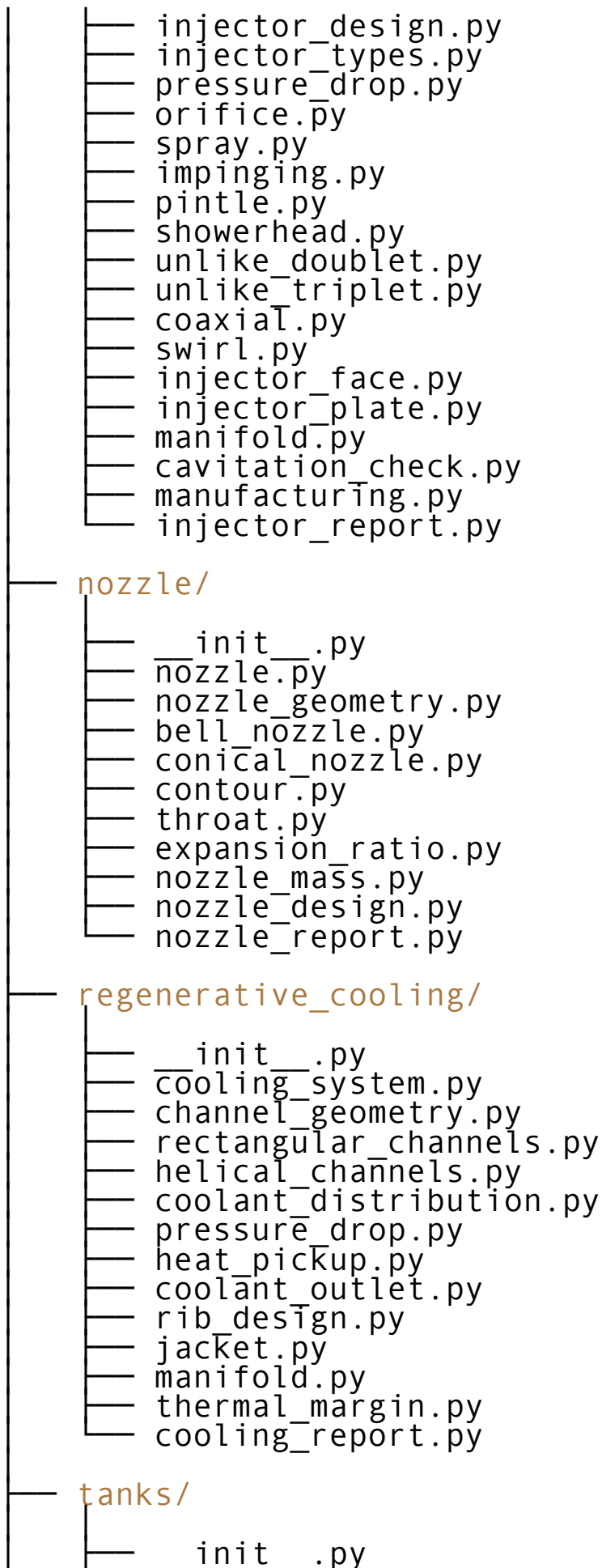
COSMOS_0.1 CODE STRUCTURE



engineering/



COSMOS_0.1 CODE STRUCTURE



COSMOS_0.1 CODE STRUCTURE

- tank.py
- lox_tank.py
- fuel_tank.py
- geometry.py
- sizing.py
- wall_thickness.py
- dome.py
- supports.py
- insulation.py
- tank_mass.py
- tank_report.py

pressurization/

- __init__.py
- pressurant.py
- helium_system.py
- regulator.py
- pressure_schedule.py
- bottle.py
- bottle_sizing.py
- blowdown.py
- line_losses.py
- pressurization_report.py

feed_system/

- __init__.py
- feed_system.py
- piping.py
- bends.py
- fittings.py
- valves.py
- filters.py
- flexible_hoses.py
- line_sizing.py
- pressure_losses.py
- feed_report.py

structures/

- __init__.py
- structural_design.py
- chamber_structure.py
- nozzle_structure.py
- tank_structure.py
- support_structure.py
- bolted_joints.py
- welds.py
- fasteners.py
- safety_factor.py
- structure_report.py

COSMOS_0.1 CODE STRUCTURE

instrumentation/

- `__init__.py`
- `pressure_sensors.py`
- `temperature_sensors.py`
- `flowmeters.py`
- `load_cells.py`
- `data_acquisition.py`
- `wiring.py`
- `calibration.py`
- `instrumentation_report.py`

manufacturing/

- `__init__.py`
- `machining.py`
- `additive_manufacturing.py`
- `casting.py`
- `forging.py`
- `brazing.py`
- `welding.py`
- `heat_treatment.py`
- `tolerances.py`
- `inspection.py`
- `manufacturing_report.py`

reliability/

- `__init__.py`
- `failure_modes.py`
- `fault_tree.py`
- `fmea.py`
- `margins.py`
- `design_limits.py`
- `reliability_analysis.py`
- `reliability_report.py`

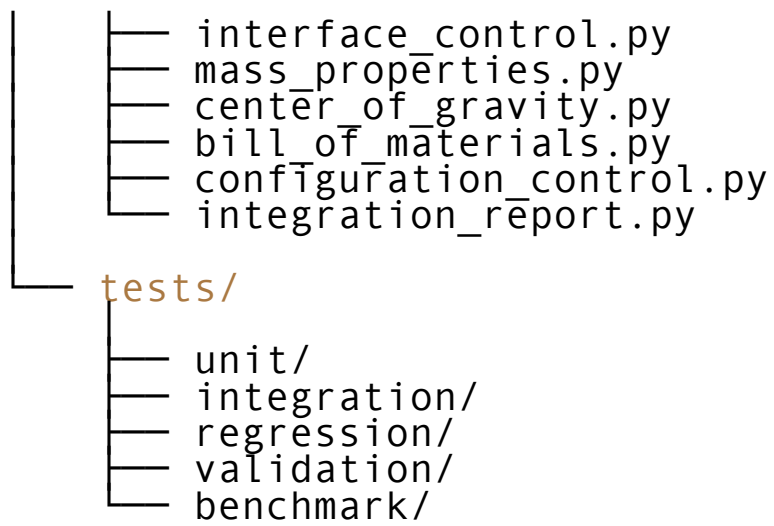
testing/

- `__init__.py`
- `test_plan.py`
- `cold_flow.py`
- `hot_fire.py`
- `acceptance.py`
- `instrumentation_setup.py`
- `test_procedure.py`
- `test_analysis.py`
- `test_report.py`

integration/

- `__init__.py`
- `engine_assembly.py`

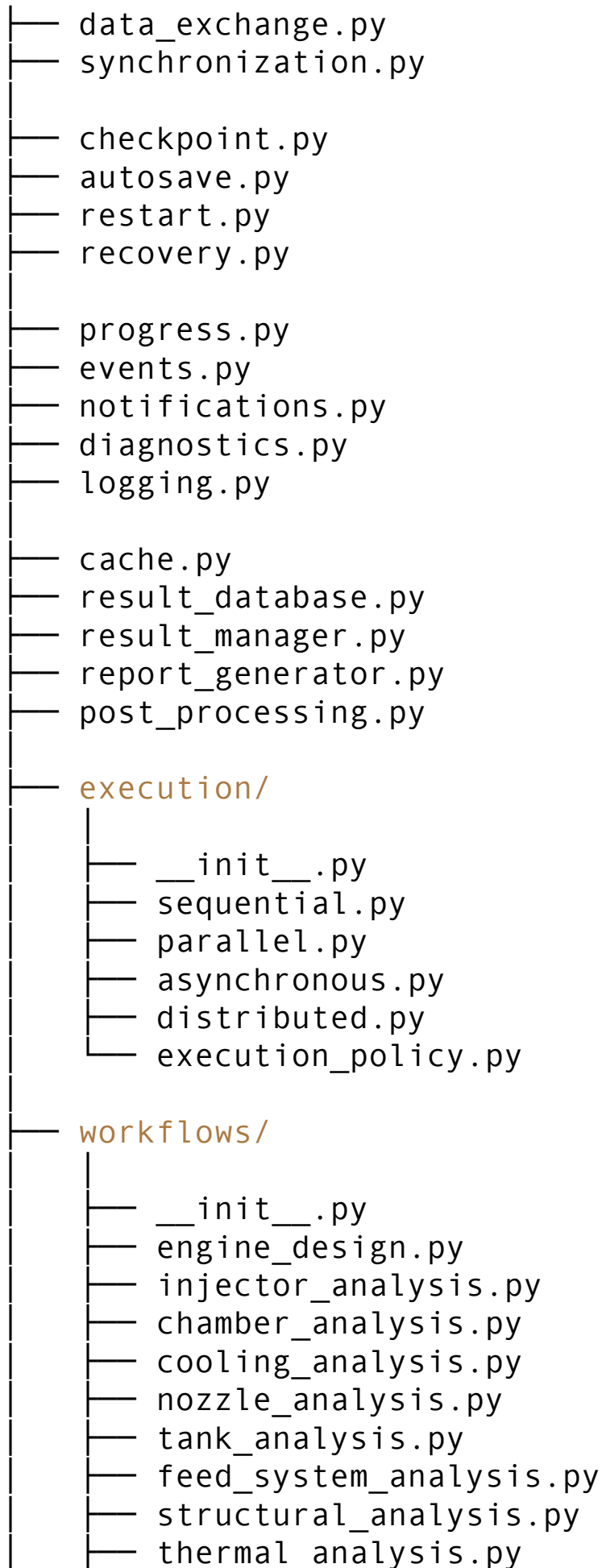
COSMOS_0.1 CODE STRUCTURE



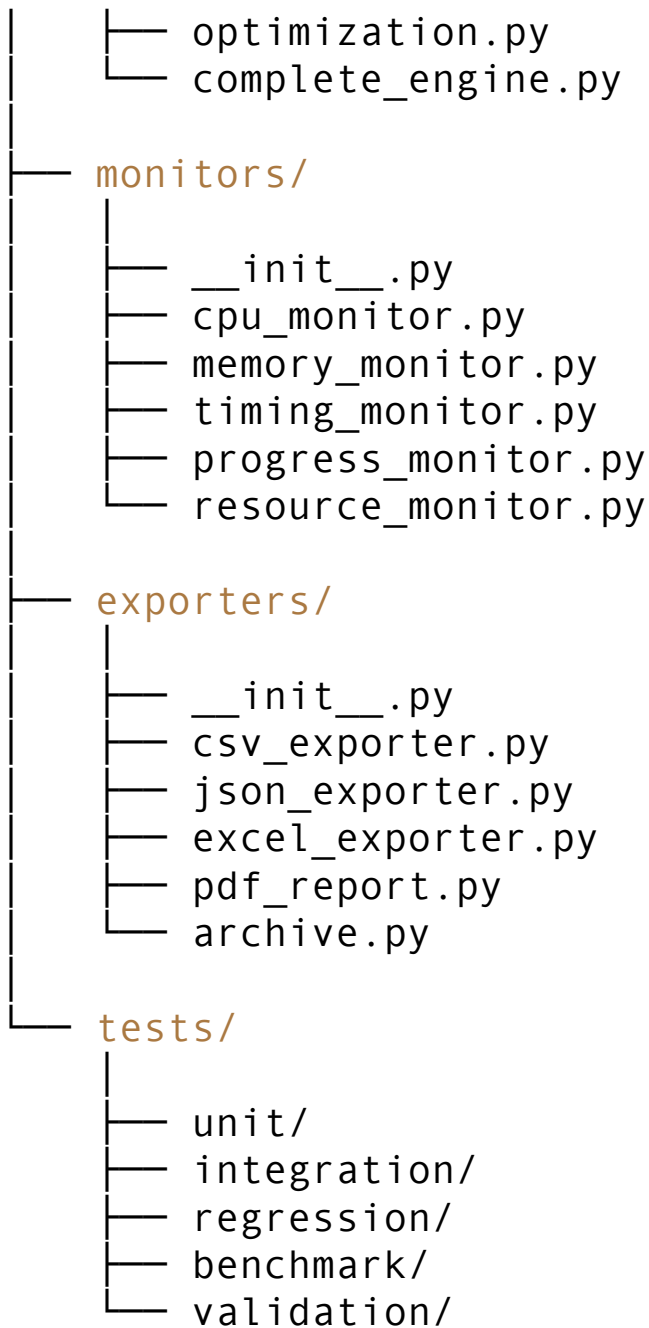
simulation/

- |
- |— __init__.py
- |
- |— simulation.py
- |— simulation_manager.py
- |— simulation_session.py
- |— simulation_state.py
- |— simulation_context.py
- |— simulation_result.py
- |— simulation_history.py
- |
- |— orchestrator.py
- |— scheduler.py
- |— dependency_manager.py
- |— execution_graph.py
- |— workflow.py
- |— task.py
- |— task_queue.py
- |— pipeline.py
- |
- |— solver_controller.py
- |— solver_factory.py
- |— solver_registry.py
- |— solver_interface.py
- |— solver_result.py
- |— solver_monitor.py
- |— solver_statistics.py
- |
- |— convergence.py
- |— convergence_history.py
- |— convergence_monitor.py
- |— convergence_criteria.py
- |— residuals.py
- |— iteration.py
- |
- |— coupling.py
- |— coupling_manager.py
- |— coupling_interface.py

COSMOS_0.1 CODE STRUCTURE



COSMOS_0.1 CODE STRUCTURE



optimization/

- `__init__.py`

- `optimization_manager.py`

- `optimization_problem.py`

- `optimization_case.py`

- `optimization_session.py`

- `optimization_history.py`

- `optimization_report.py`

objectives/

- `__init__.py`

- `objective.py`

- `maximize_isp.py`

- `maximize_thrust.py`

- `maximize_cstar.py`

- `minimize_mass.py`

- `minimize_pressure_drop.py`

- `minimize_wall_temperature.py`

- `minimize_cost.py`

- `maximize_reliability.py`

- `multi_objective.py`

design_variables/

- `__init__.py`

- `design_variable.py`

- `chamber_variables.py`

- `injector_variables.py`

- `nozzle_variables.py`

- `cooling_variables.py`

- `tank_variables.py`

- `feed_system_variables.py`

- `material_variables.py`

- `engine_variables.py`

constraints/

- `__init__.py`

- `constraint.py`

- `geometry_constraints.py`

- `structural_constraints.py`

- `thermal_constraints.py`

- `pressure_constraints.py`

COSMOS_0.1 CODE STRUCTURE

- manufacturing_constraints.py
- material_constraints.py
- safety_constraints.py
- performance_constraints.py
- operational_constraints.py

studies/

- __init__.py
- parameter_study.py
- design_of_experiments.py
- sensitivity_study.py
- trade_study.py
- optimization_study.py
- robustness_study.py
- uncertainty_study.py

evaluators/

- __init__.py
- evaluator.py
- engine_evaluator.py
- injector_evaluator.py
- chamber_evaluator.py
- cooling_evaluator.py
- nozzle_evaluator.py
- structural_evaluator.py
- thermal_evaluator.py

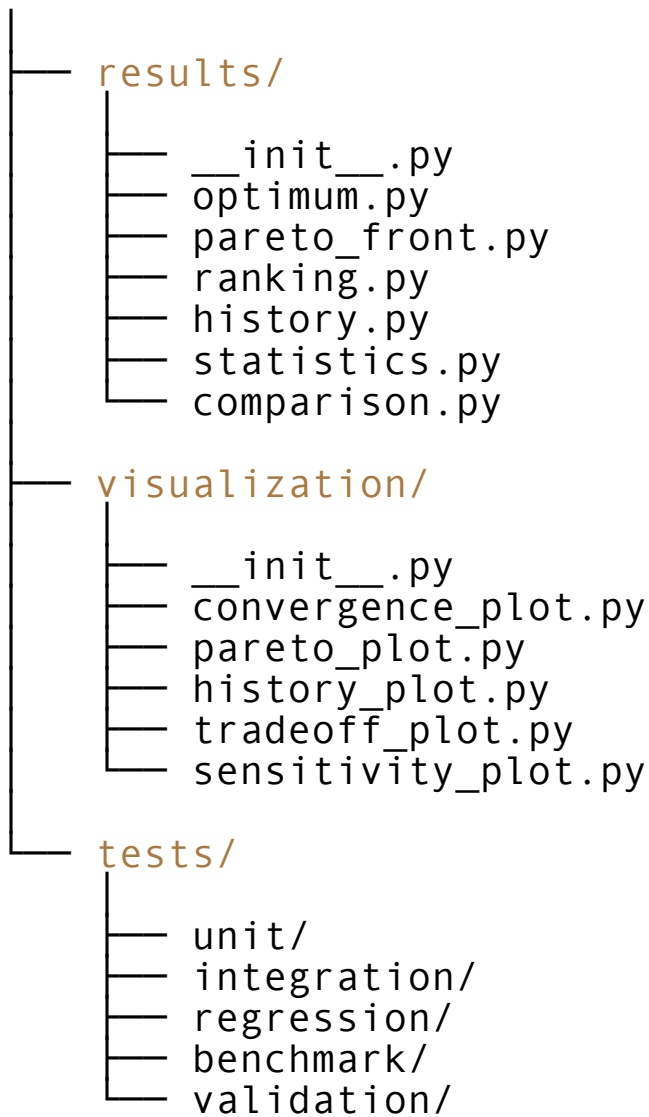
workflows/

- __init__.py
- single_objective.py
- multi_objective.py
- constrained.py
- unconstrained.py
- sequential.py
- coupled.py
- robust_design.py

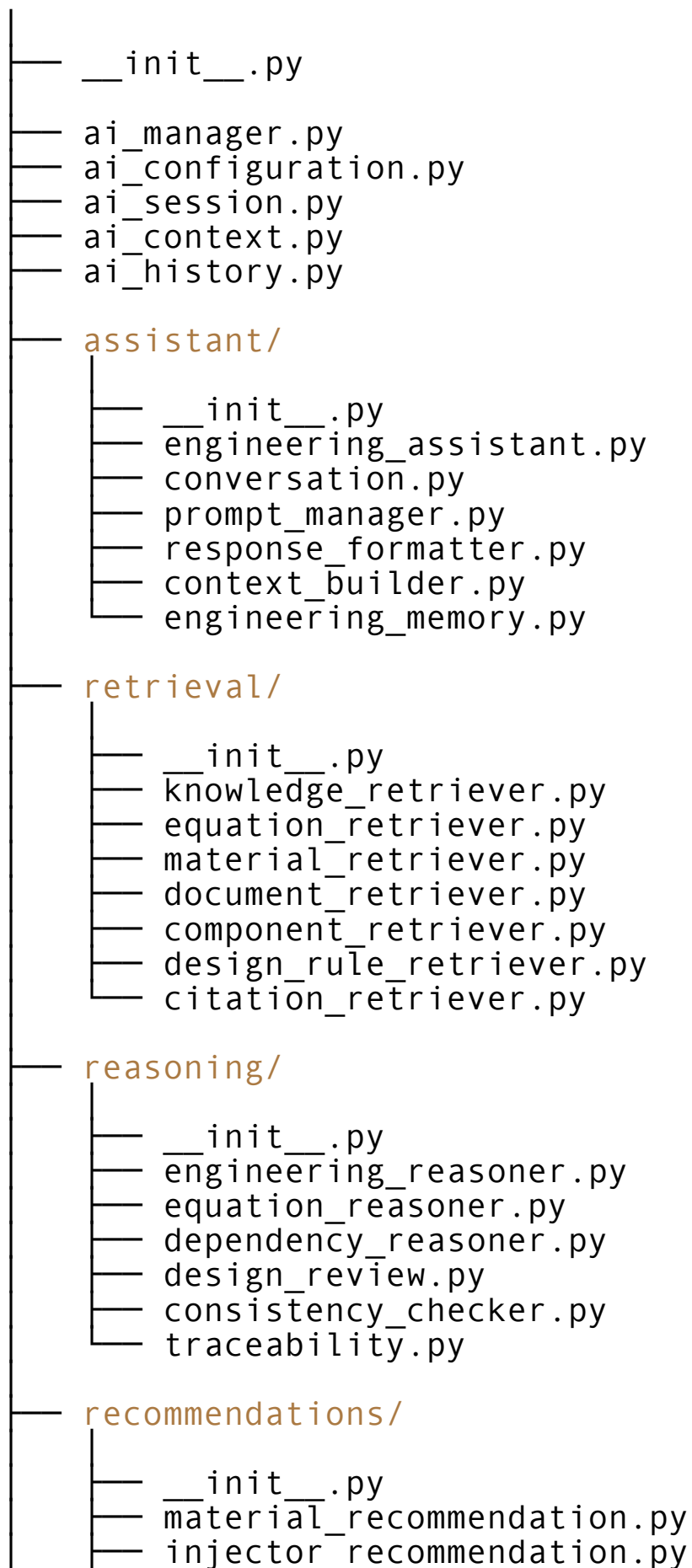
convergence/

- __init__.py
- convergence.py
- stopping_criteria.py
- convergence_history.py
- monitoring.py

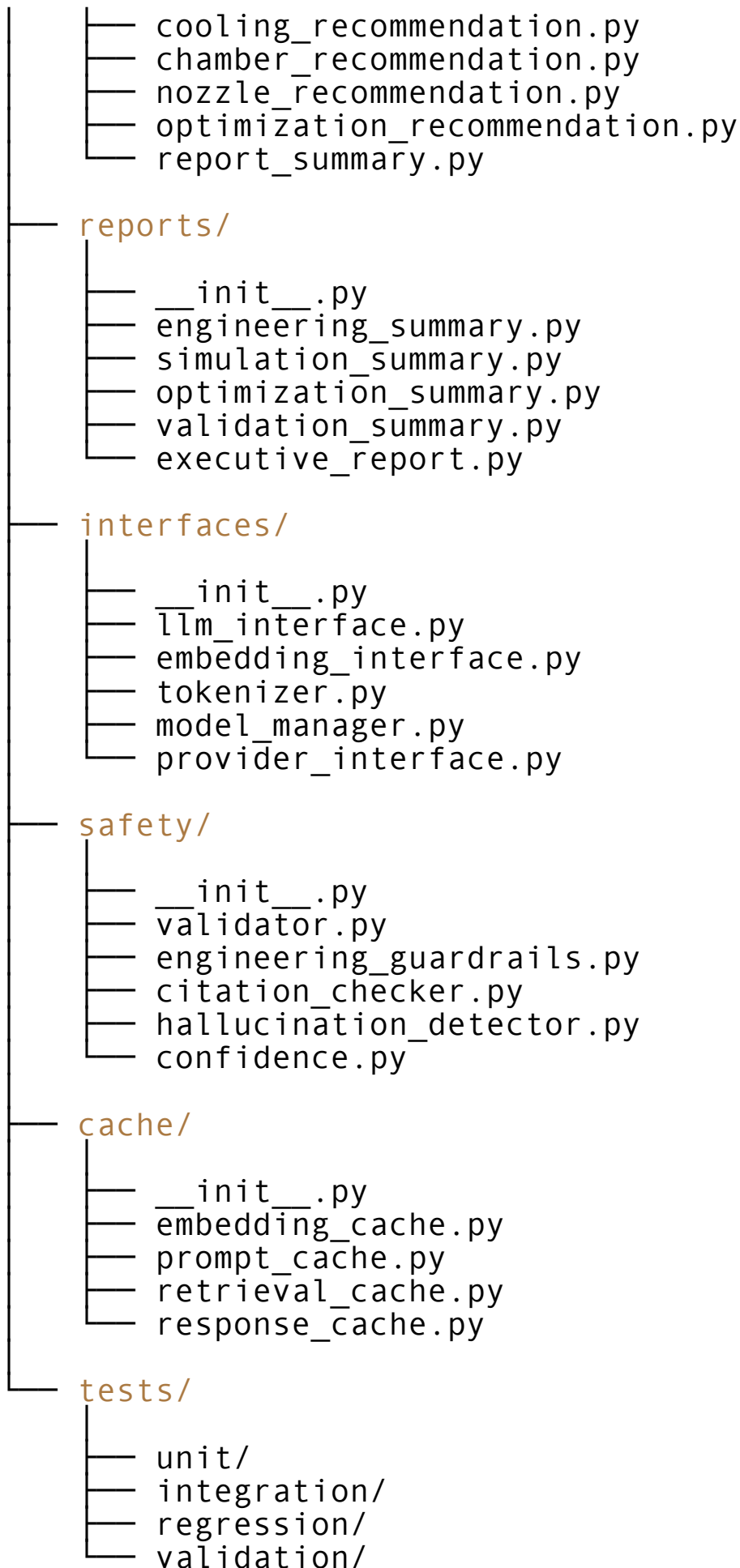
COSMOS_0.1 CODE STRUCTURE



ai/

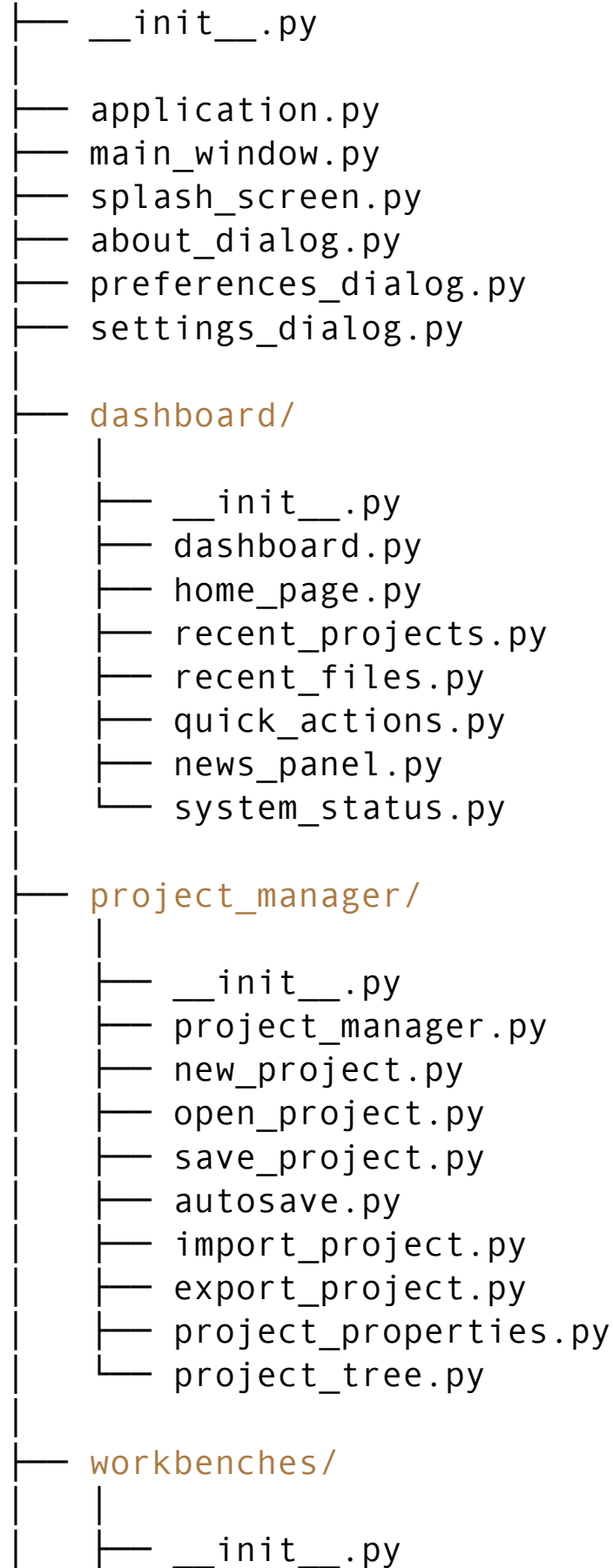


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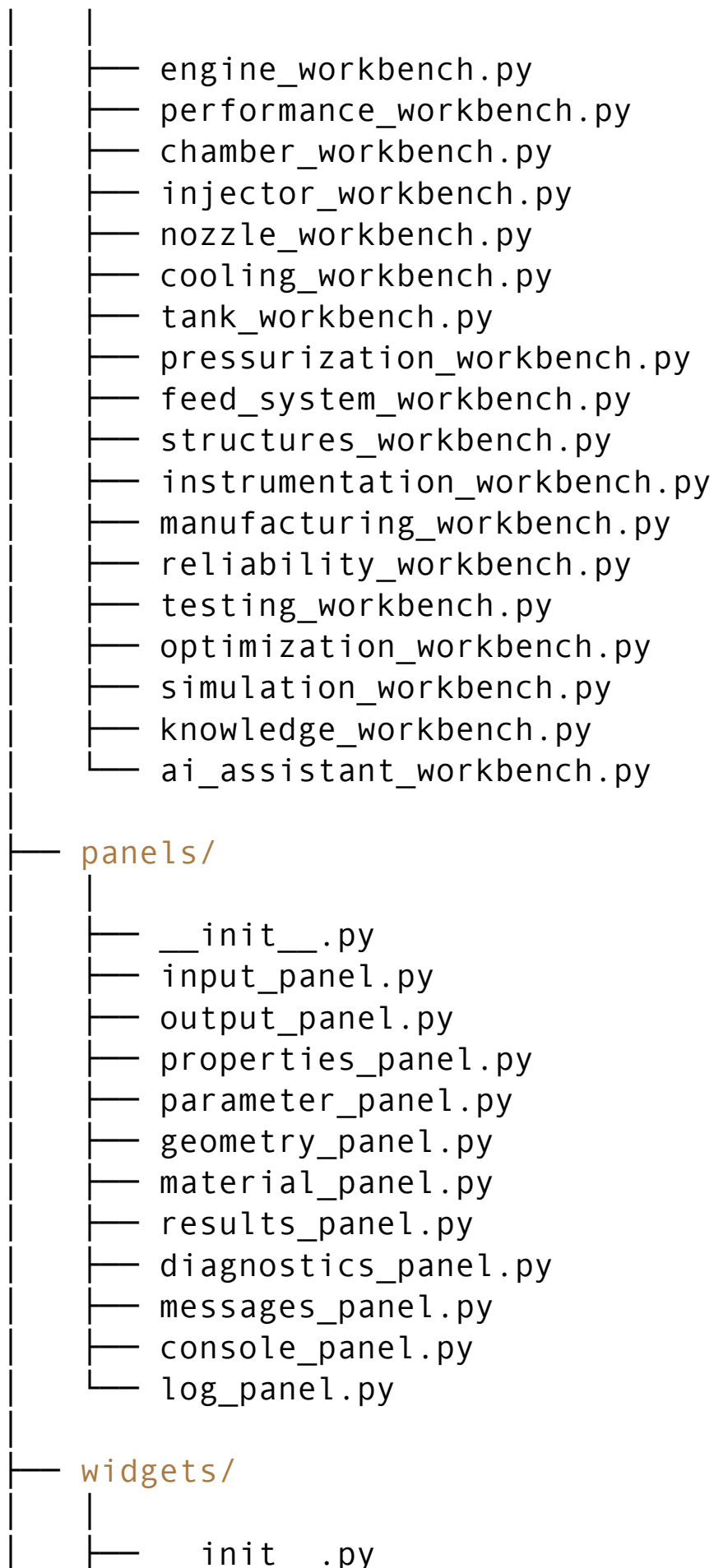


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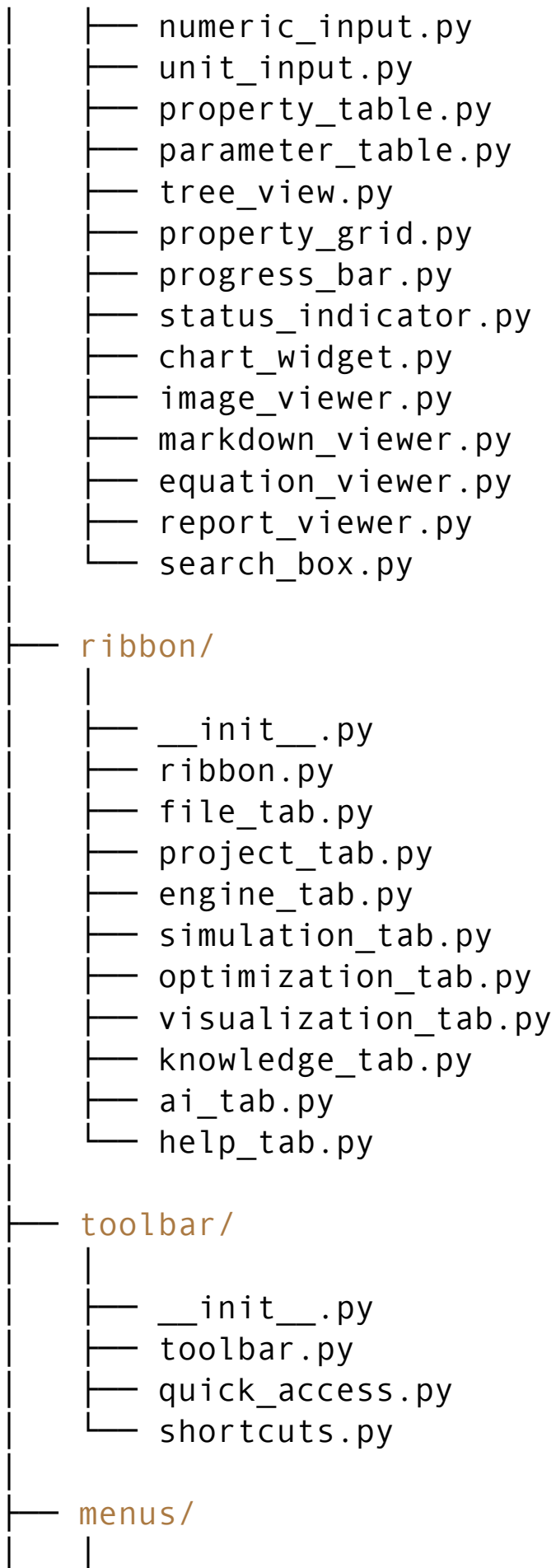
gui/



COSMOS_0.1 CODE STRUCTURE



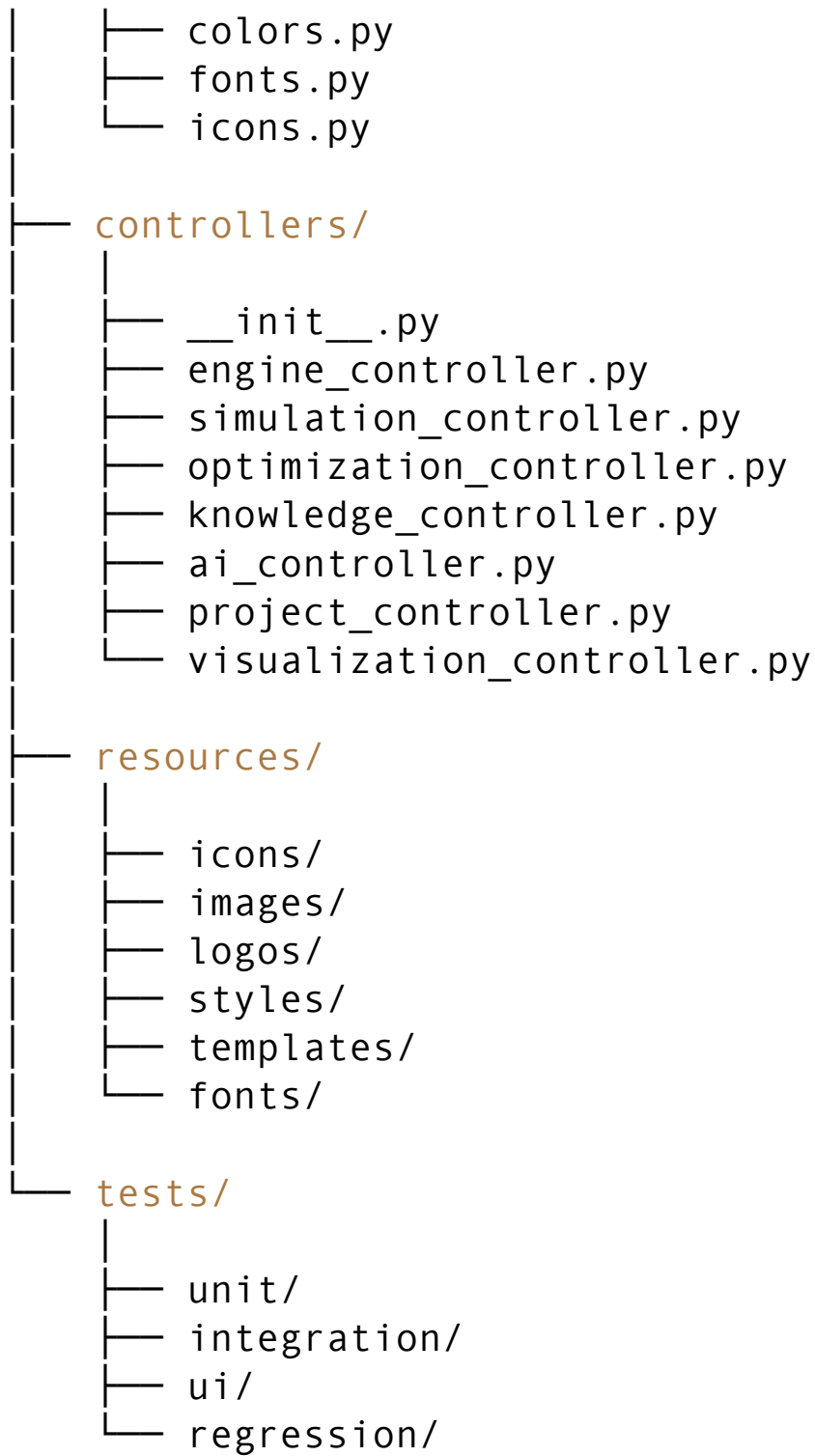
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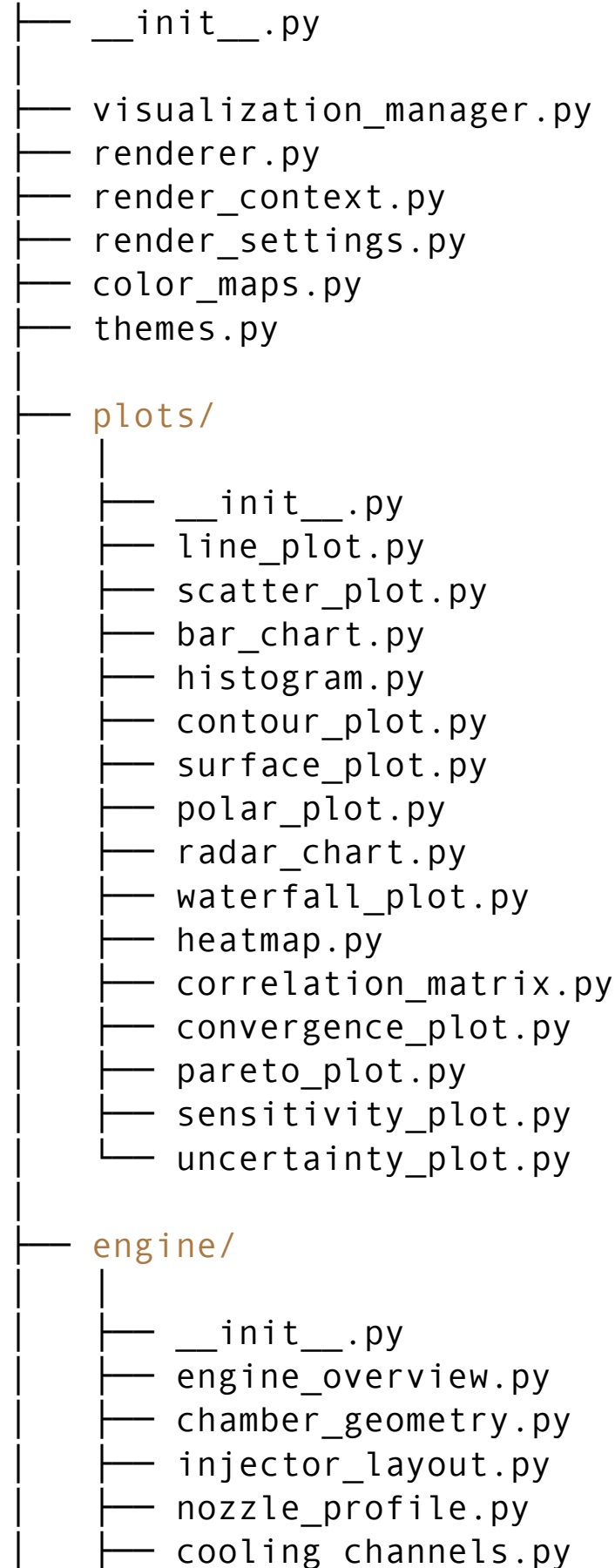
COSMOS_0.1 CODE STRUCTURE

```
├── __init__.py
├── file_menu.py
├── edit_menu.py
├── view_menu.py
├── tools_menu.py
├── simulation_menu.py
├── optimization_menu.py
├── knowledge_menu.py
├── window_menu.py
├── help_menu.py
├── docking/
│   ├── __init__.py
│   ├── dock_manager.py
│   ├── dock_layout.py
│   ├── floating_window.py
│   └── workspace.py
├── dialogs/
│   ├── __init__.py
│   ├── confirmation_dialog.py
│   ├── error_dialog.py
│   ├── warning_dialog.py
│   ├── progress_dialog.py
│   ├── units_dialog.py
│   ├── material_selector.py
│   ├── component_selector.py
│   ├── solver_dialog.py
│   └── export_dialog.py
├── themes/
│   ├── __init__.py
│   ├── theme_manager.py
│   ├── dark_theme.py
│   └── light_theme.py
```

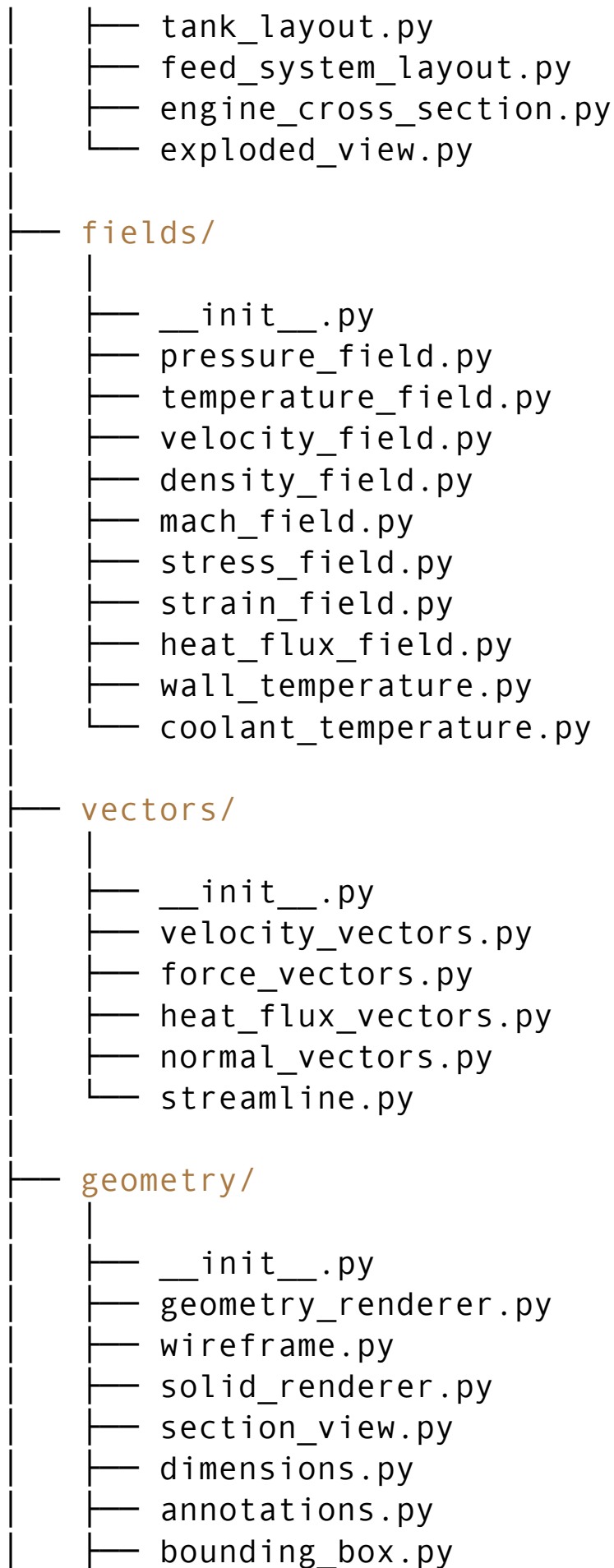
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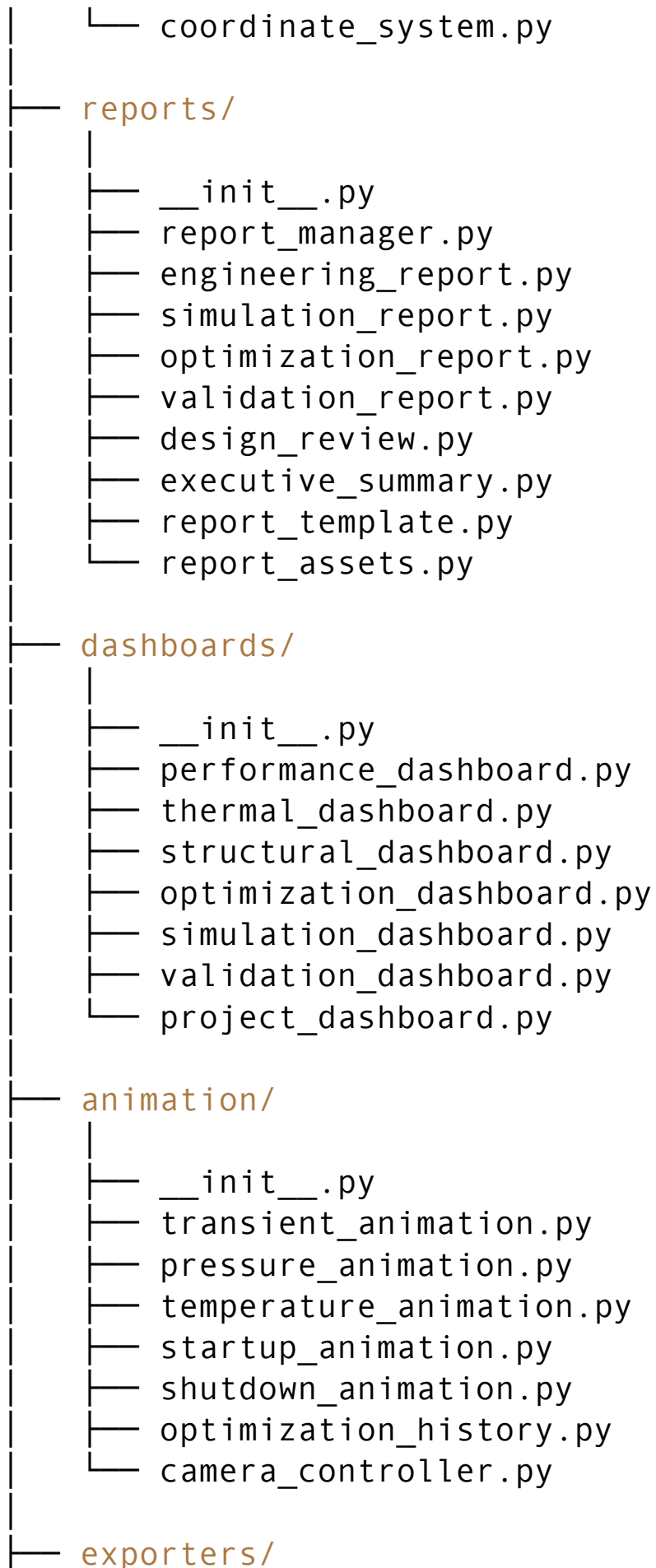
visualization/



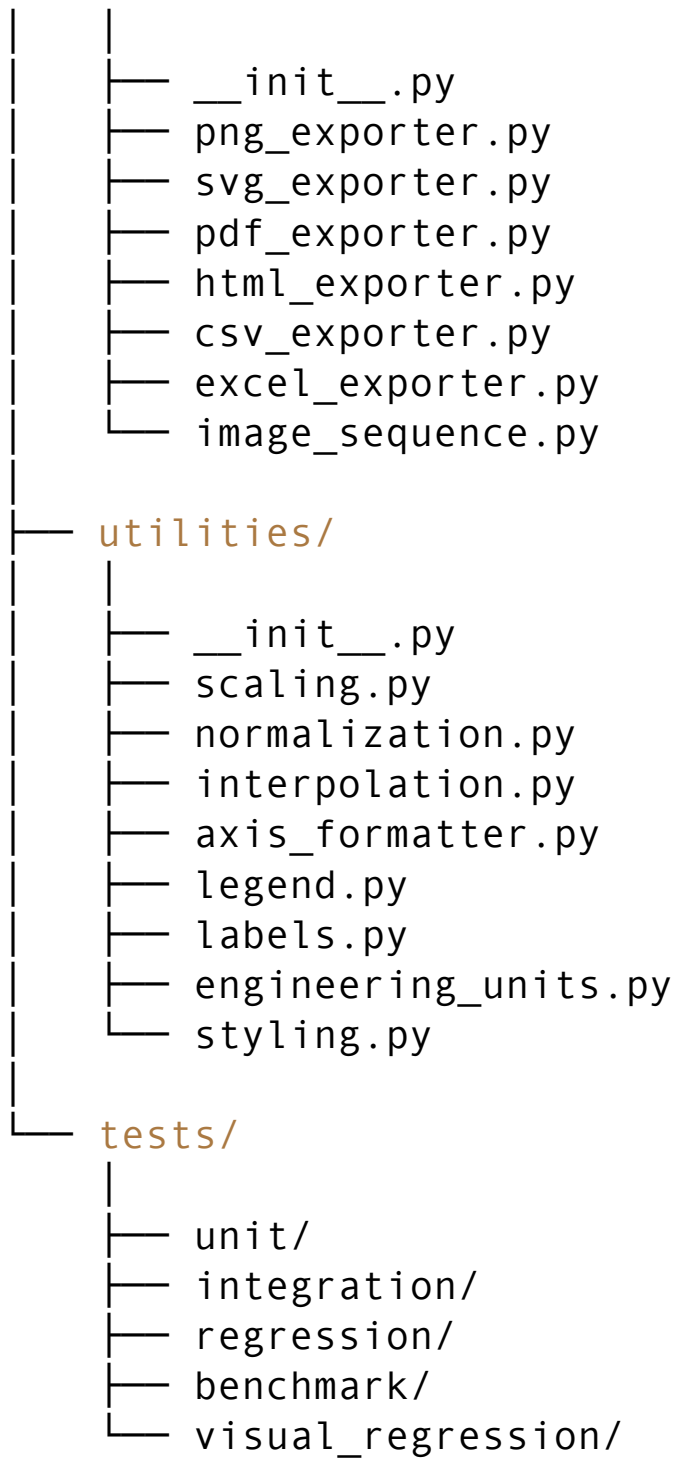
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COSMOS_0.1 CODE STRUCTURE

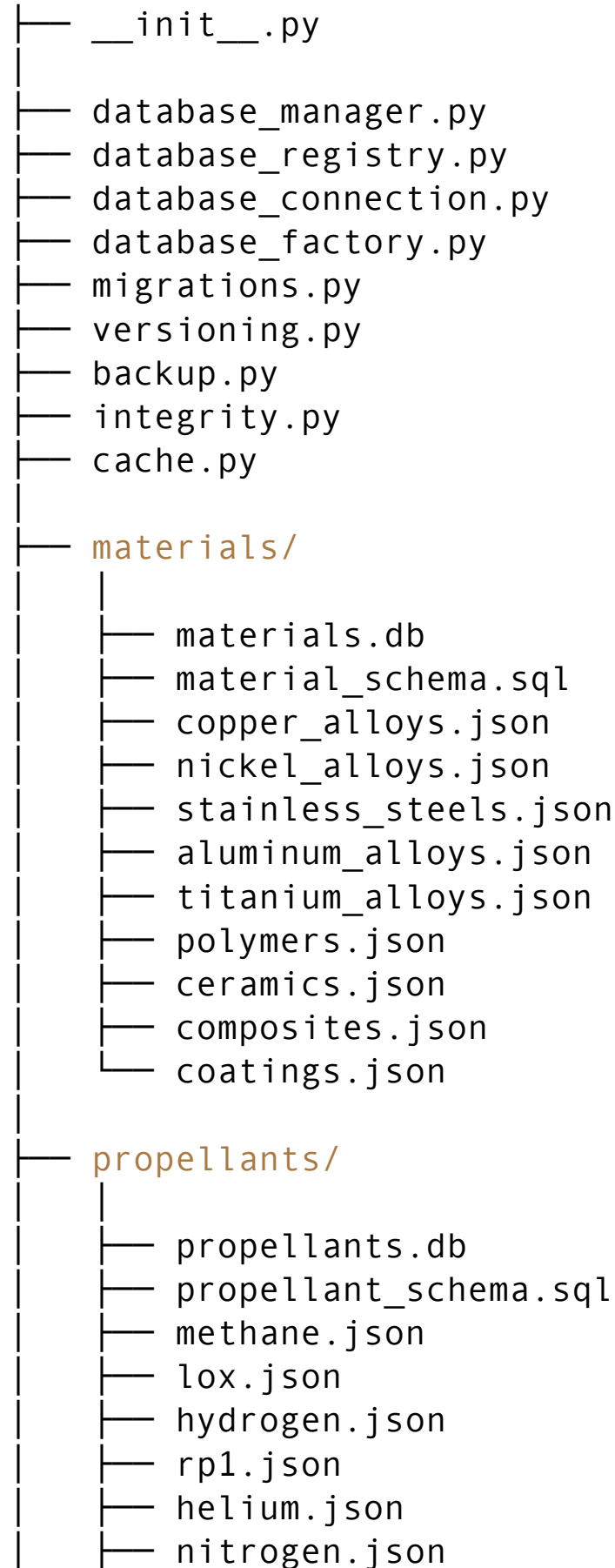


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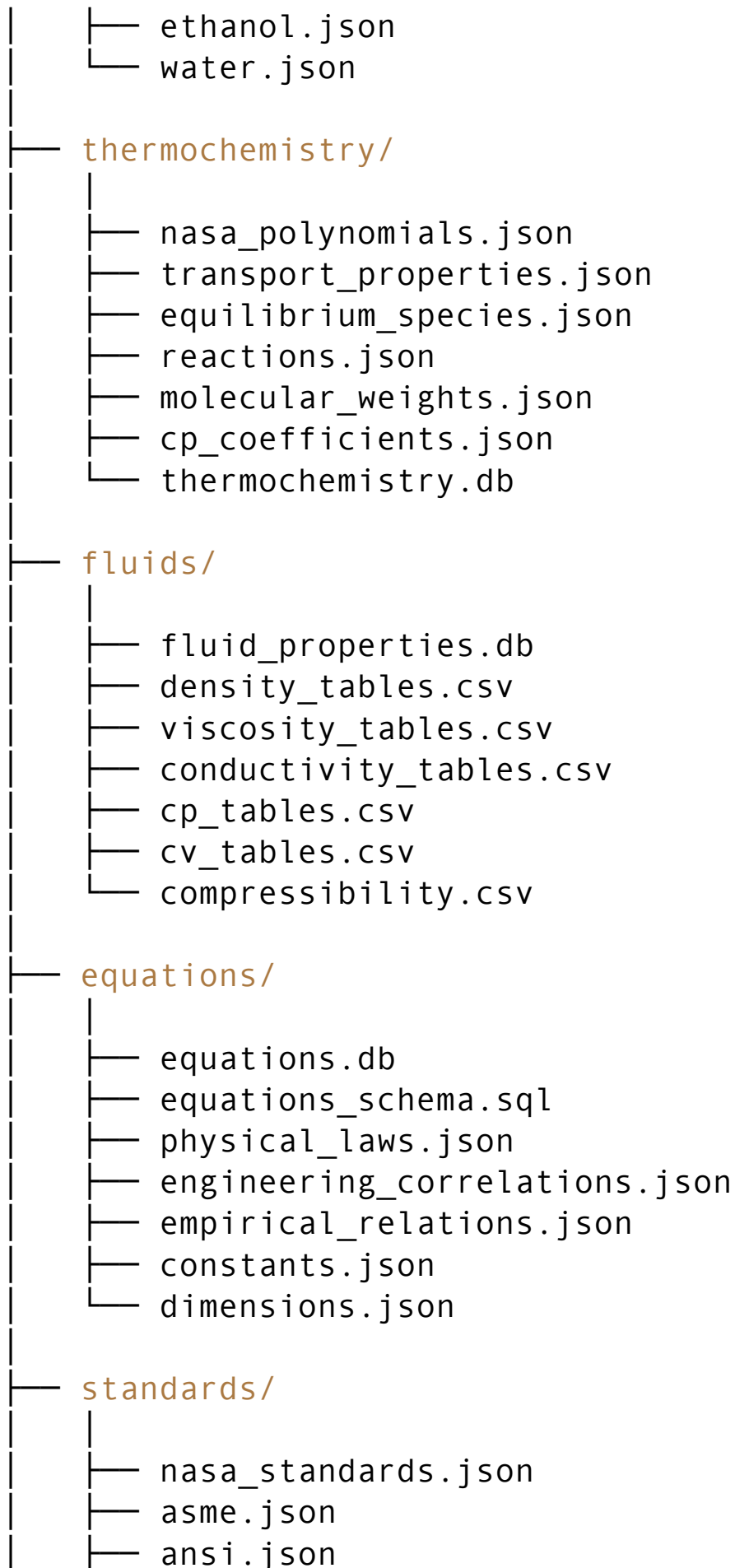


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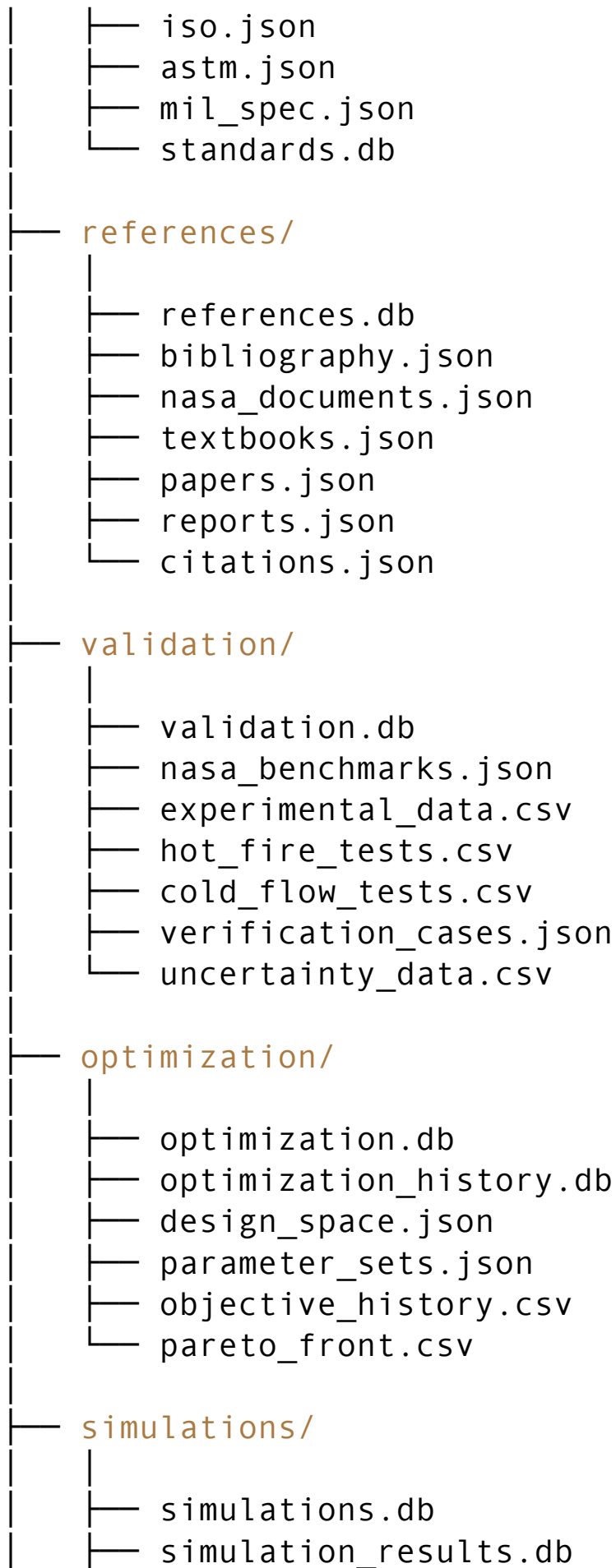
databases/



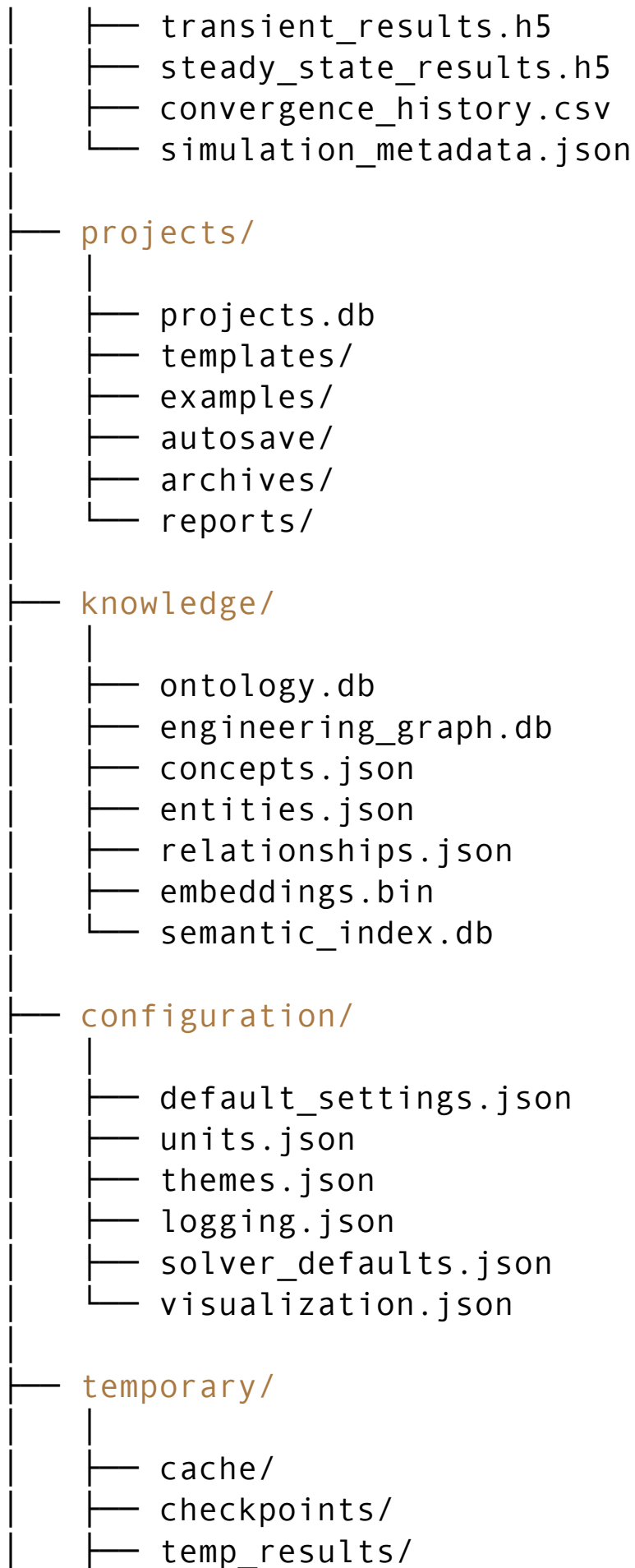
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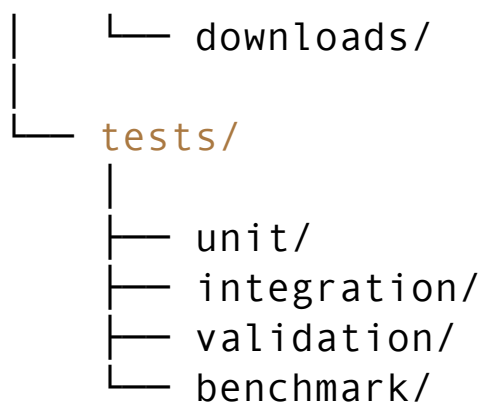
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COSMOS_0.1 CODE STRUCTURE

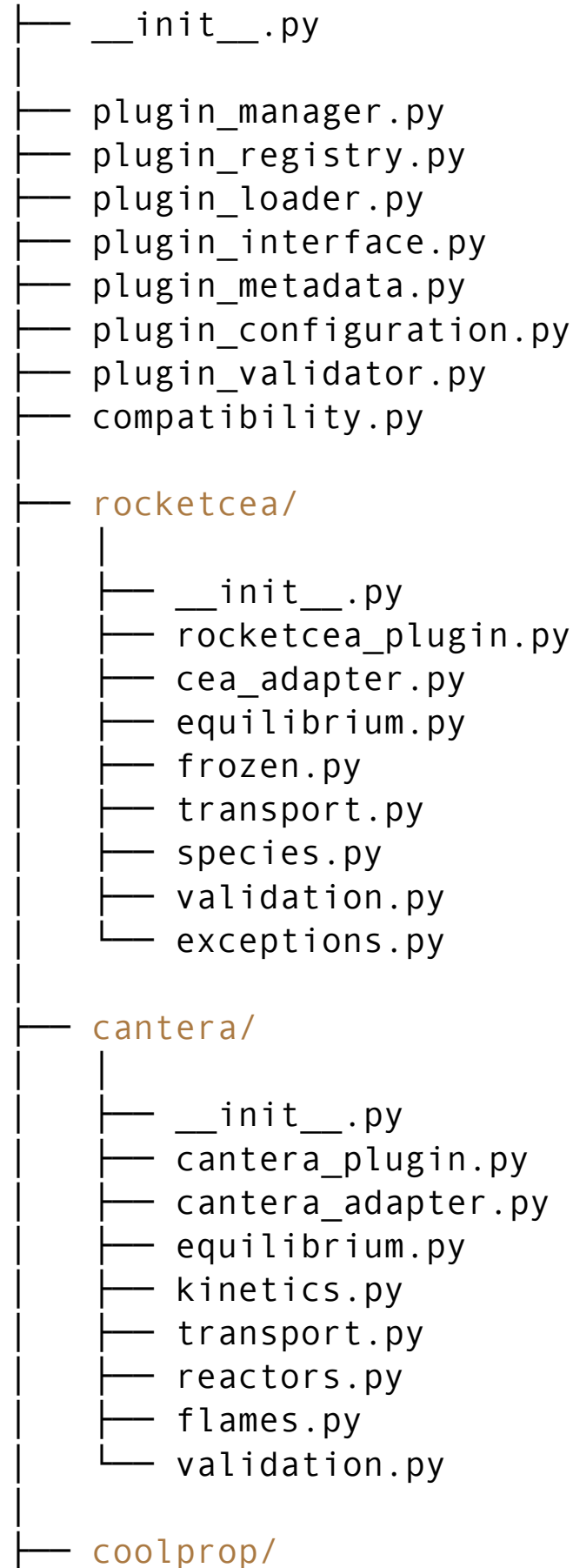


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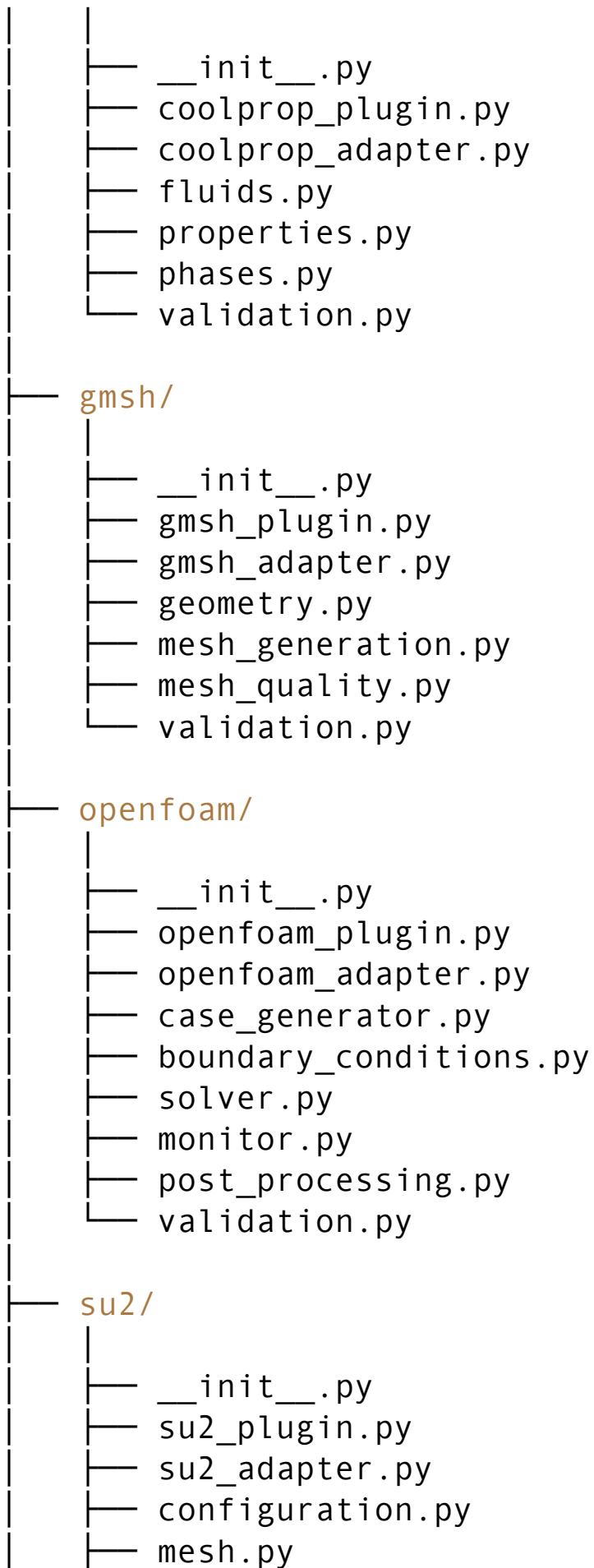


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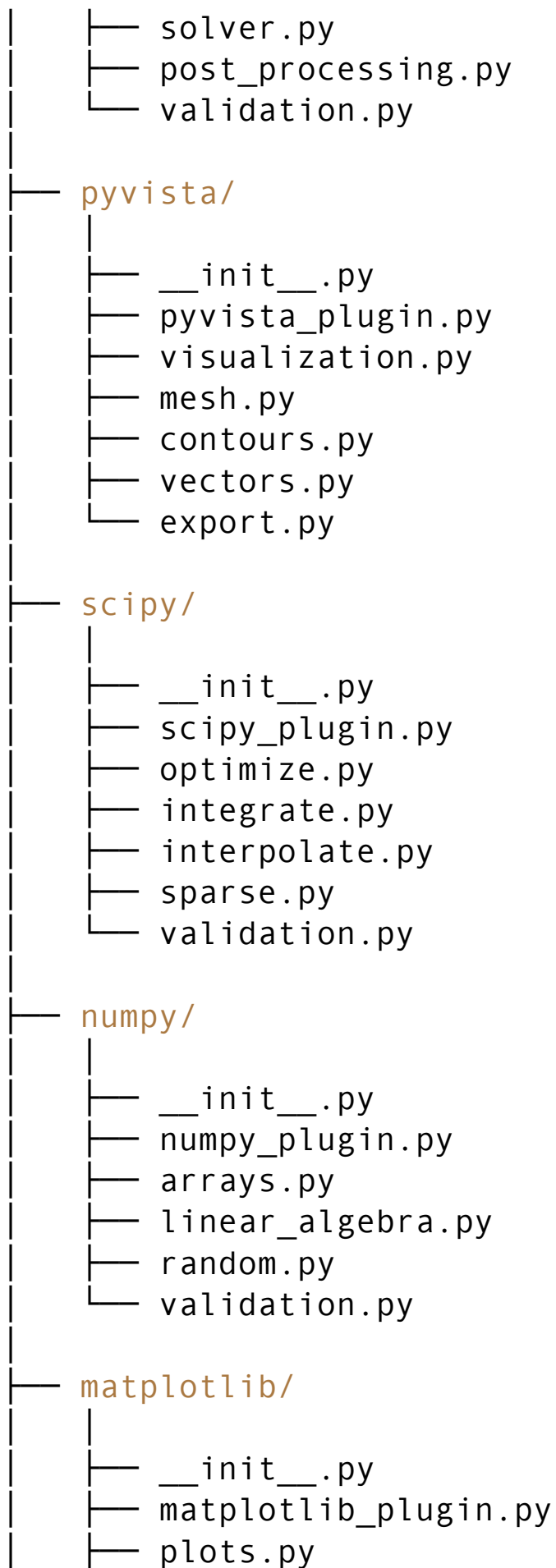
plugins/



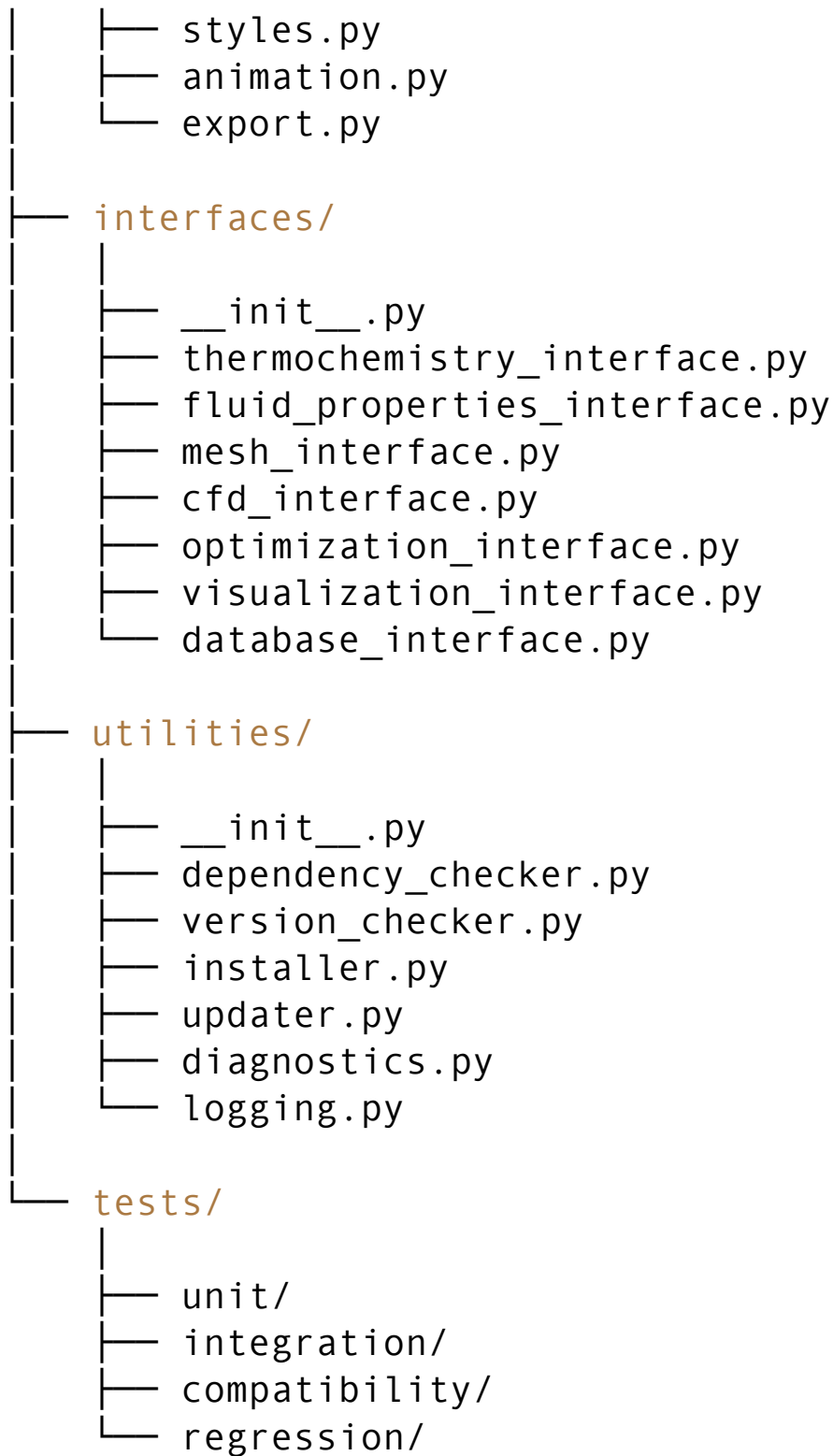
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COSMOS_0.1 CODE STRUCTURE

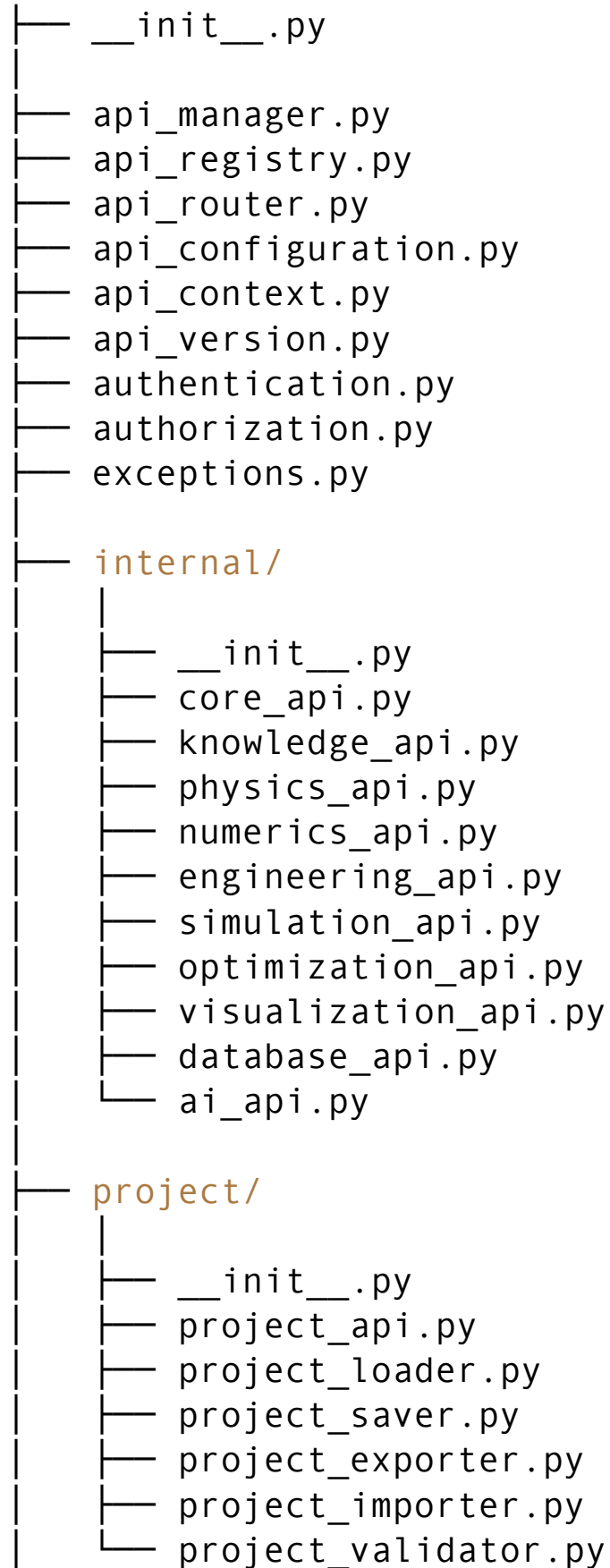


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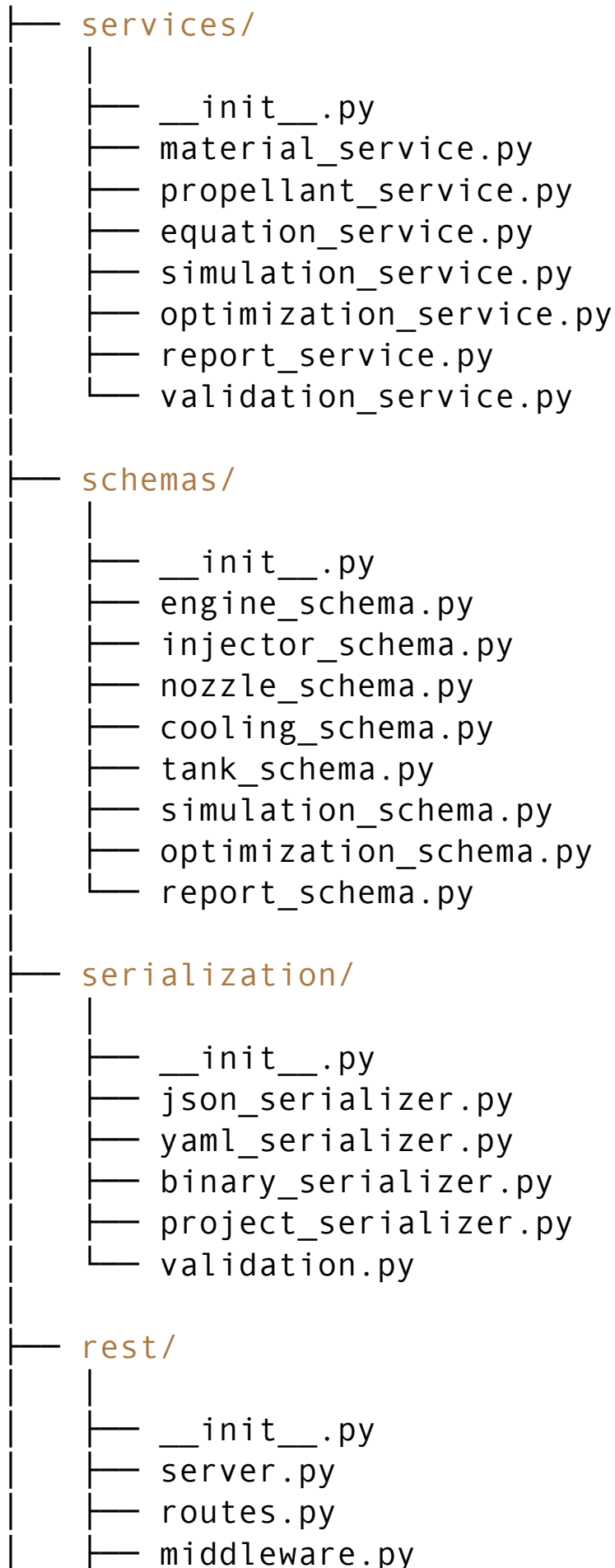


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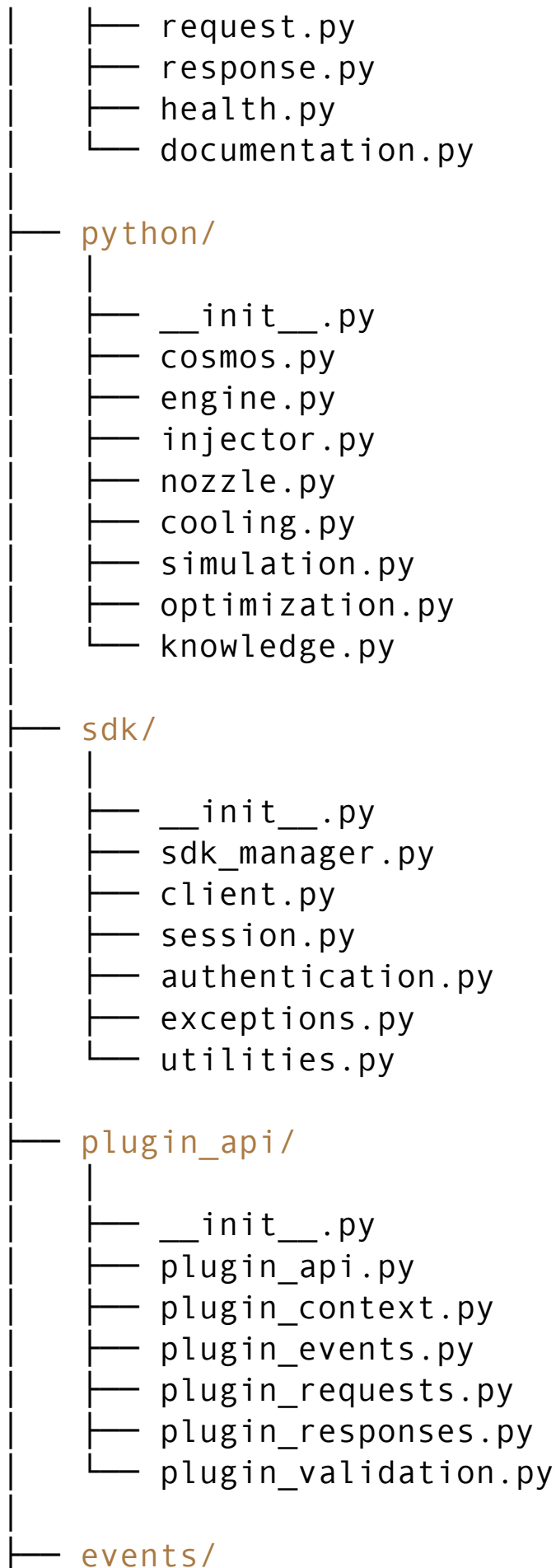
api/



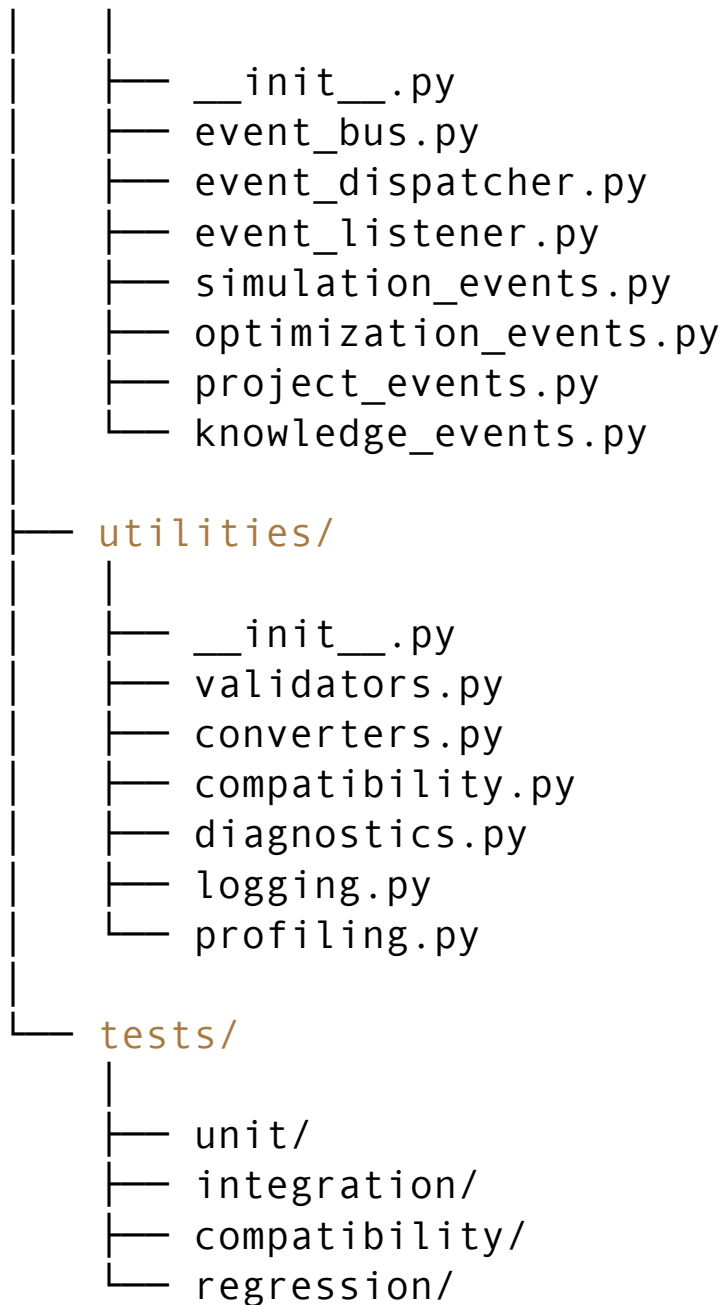
COSMOS_0.1 CODE STRUCTURE



COSMOS_0.1 CODE STRUCTURE

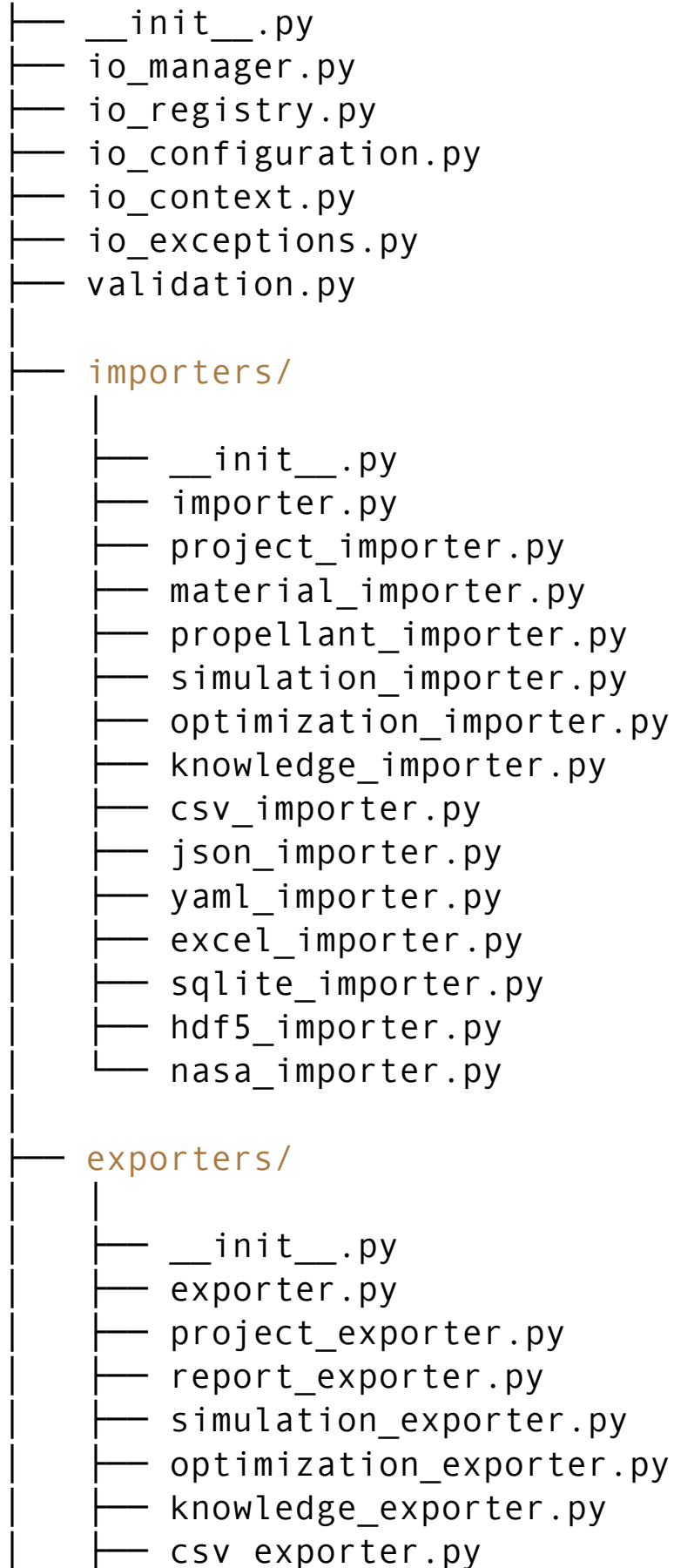


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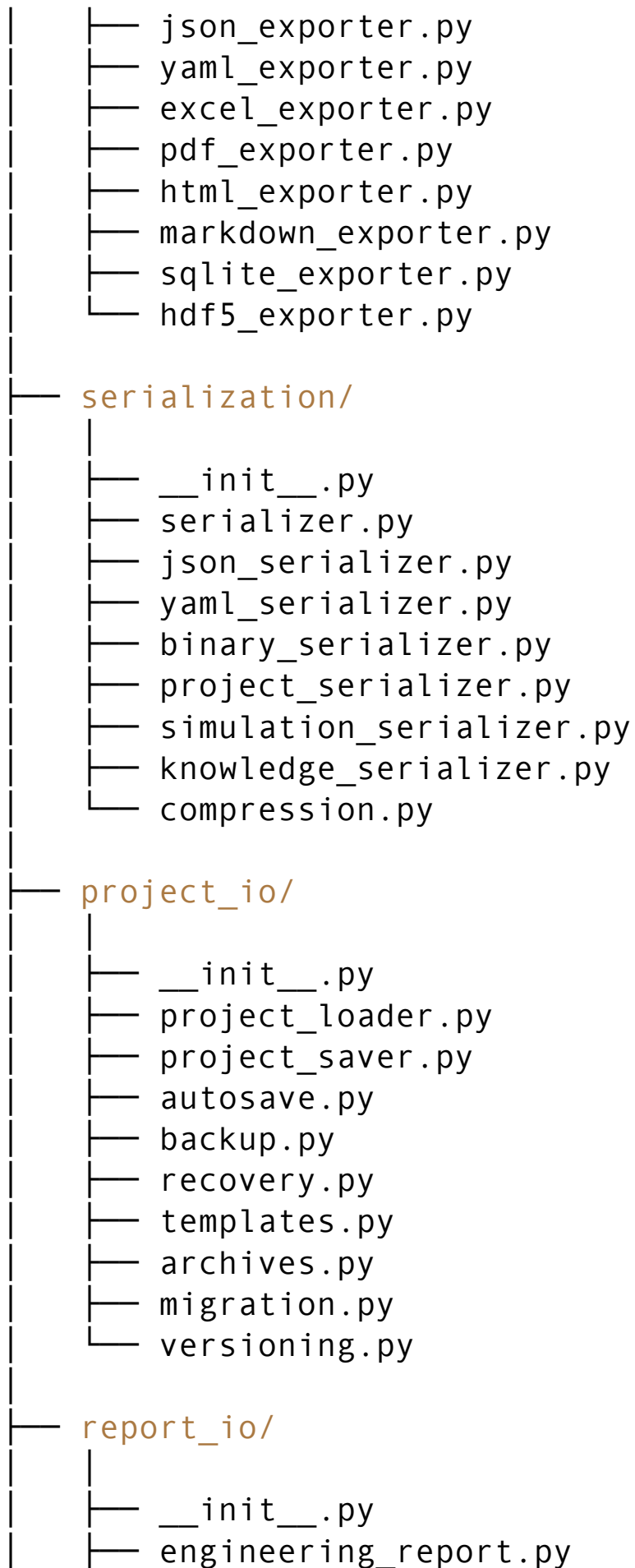


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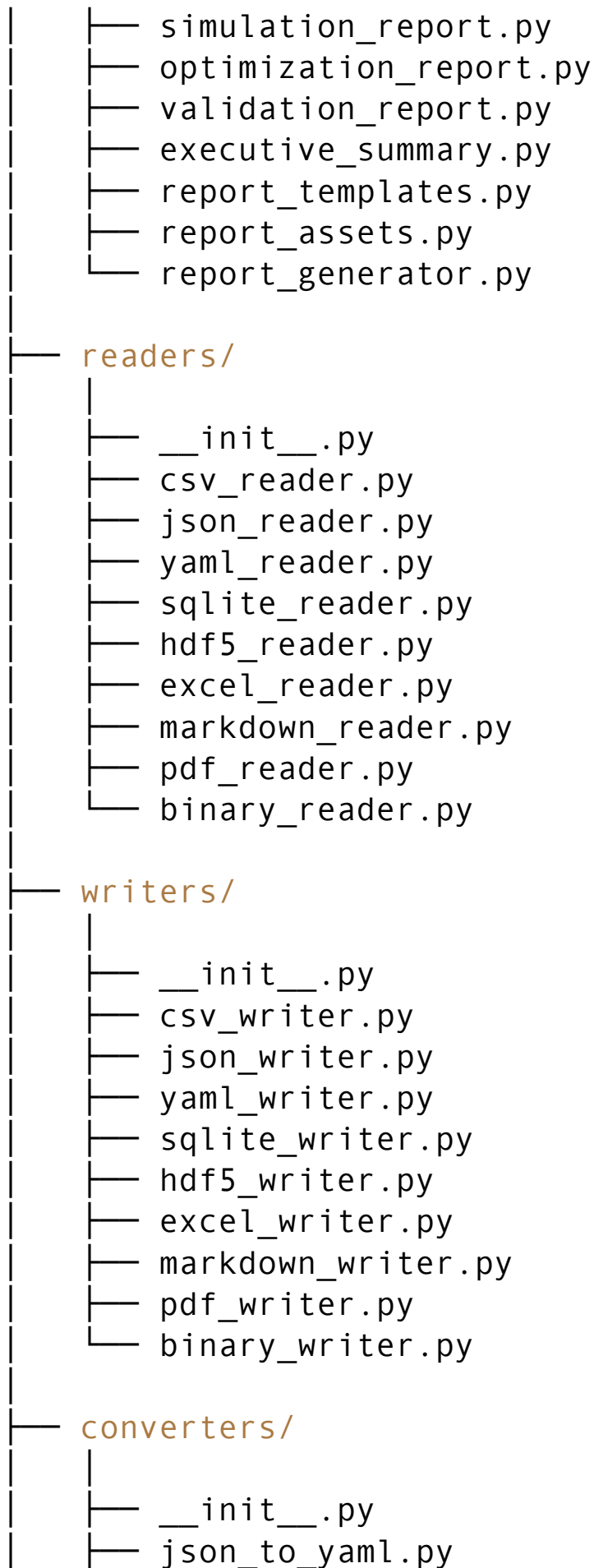
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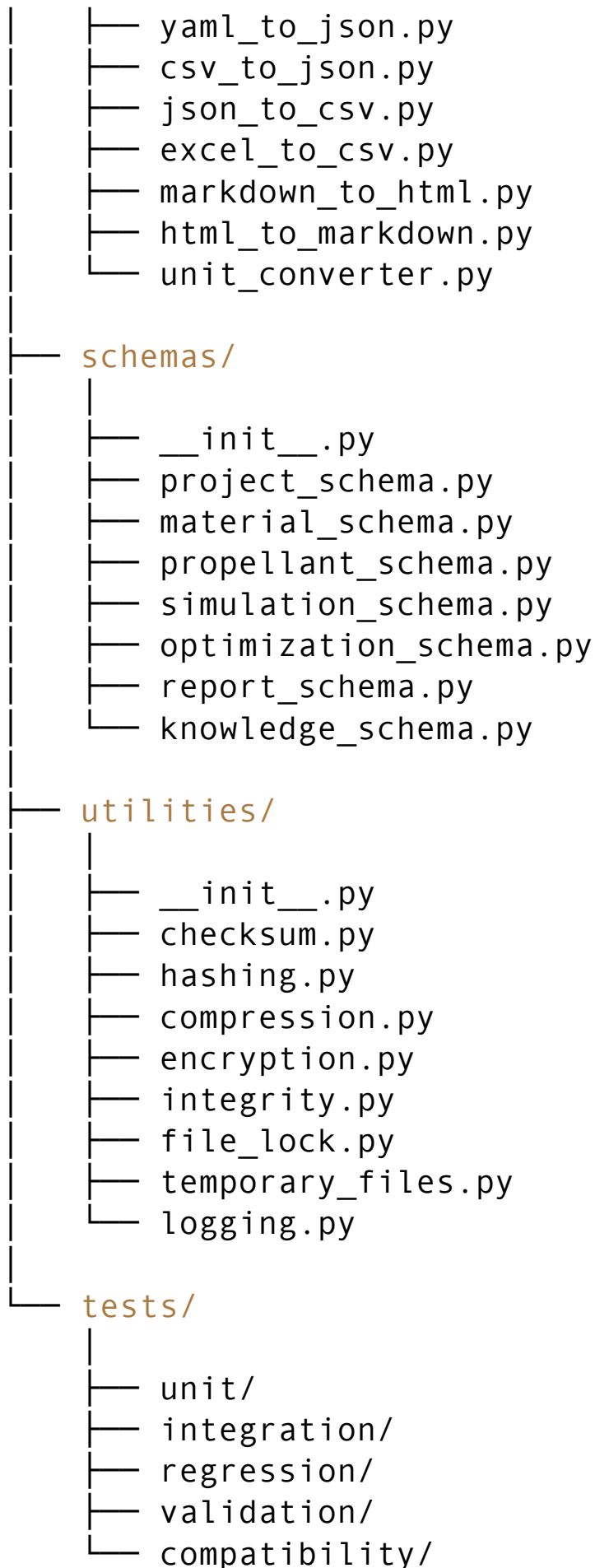
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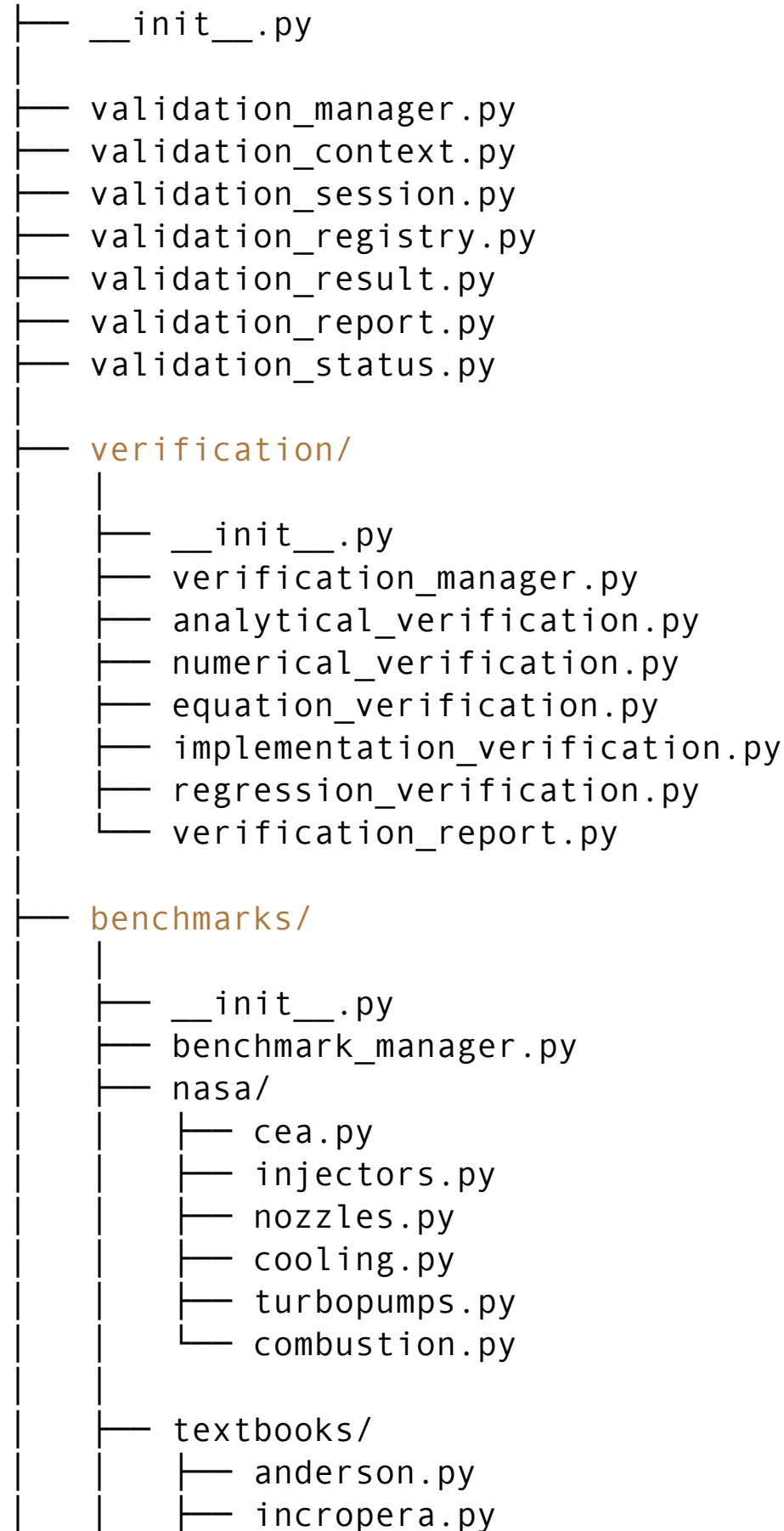
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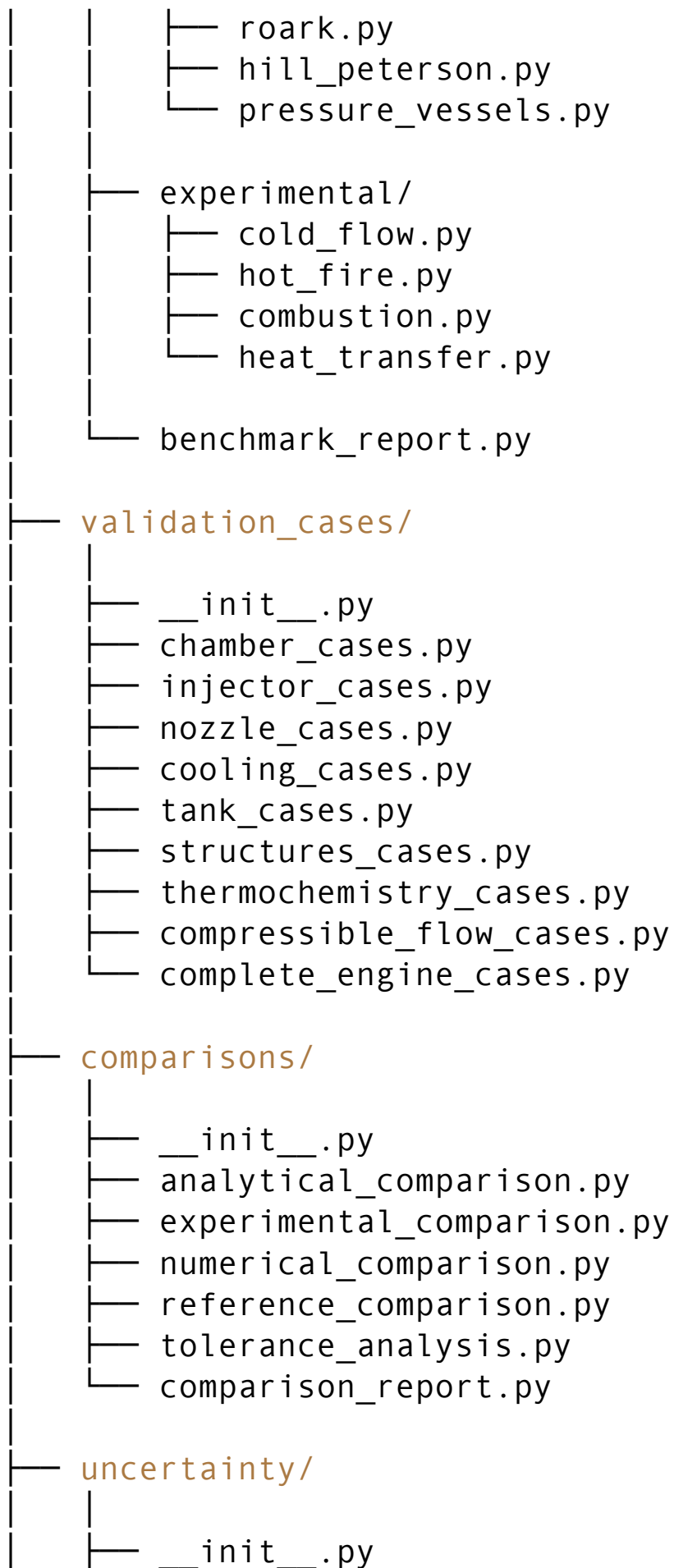
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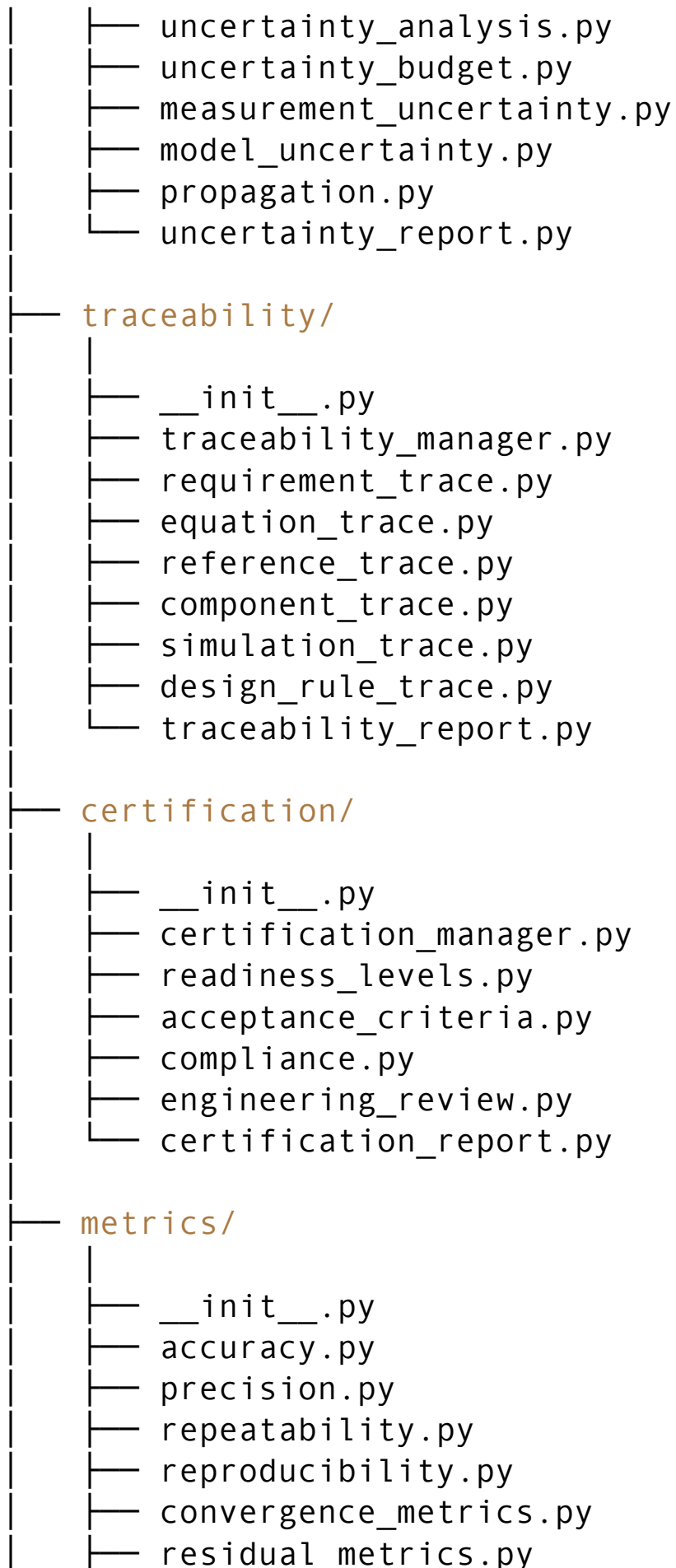
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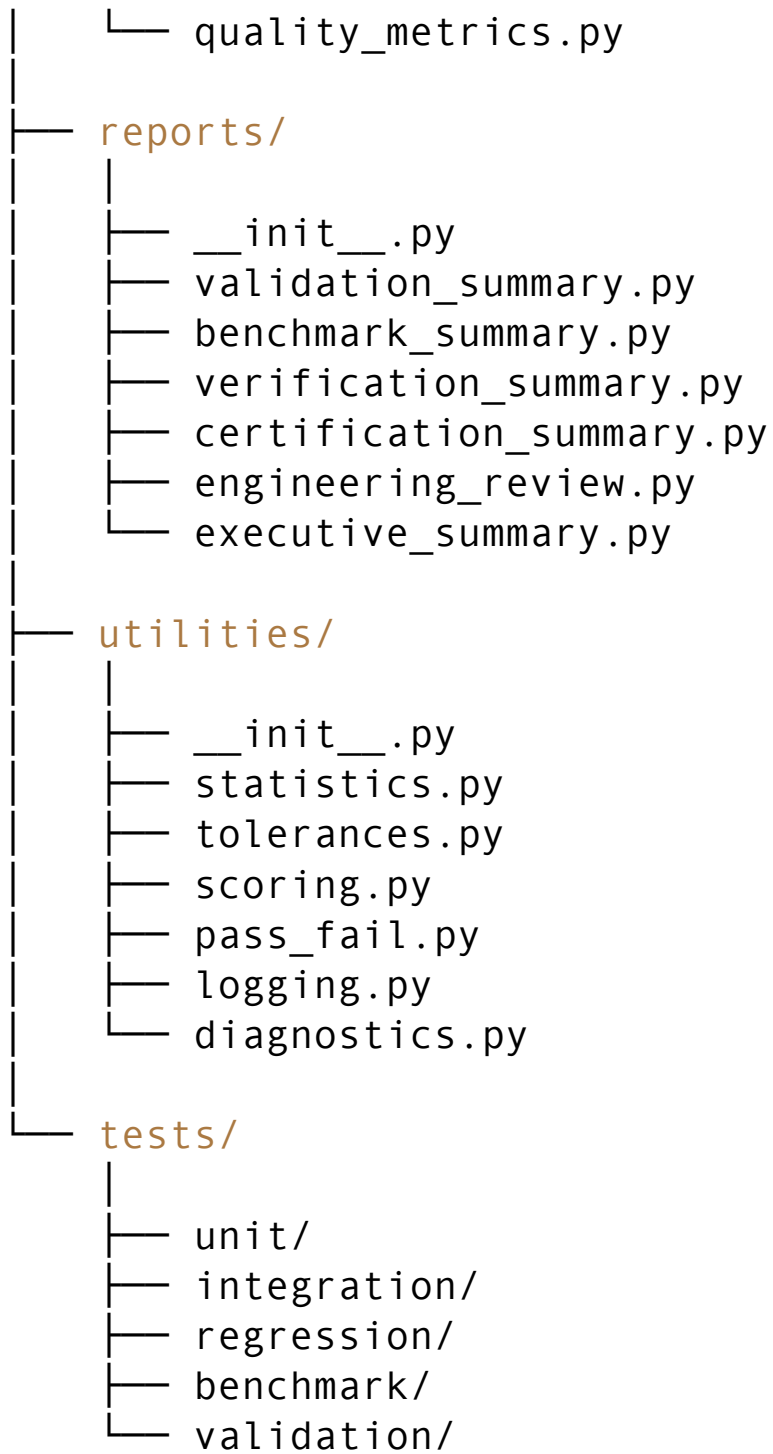
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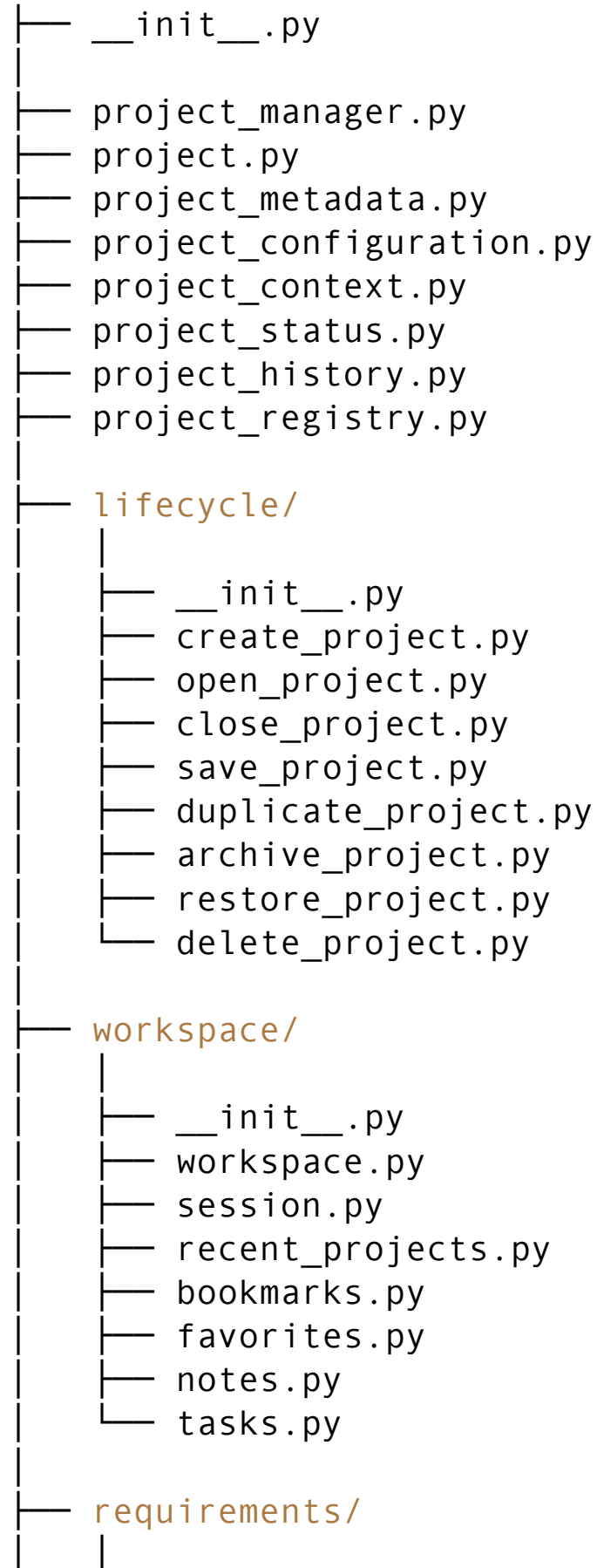


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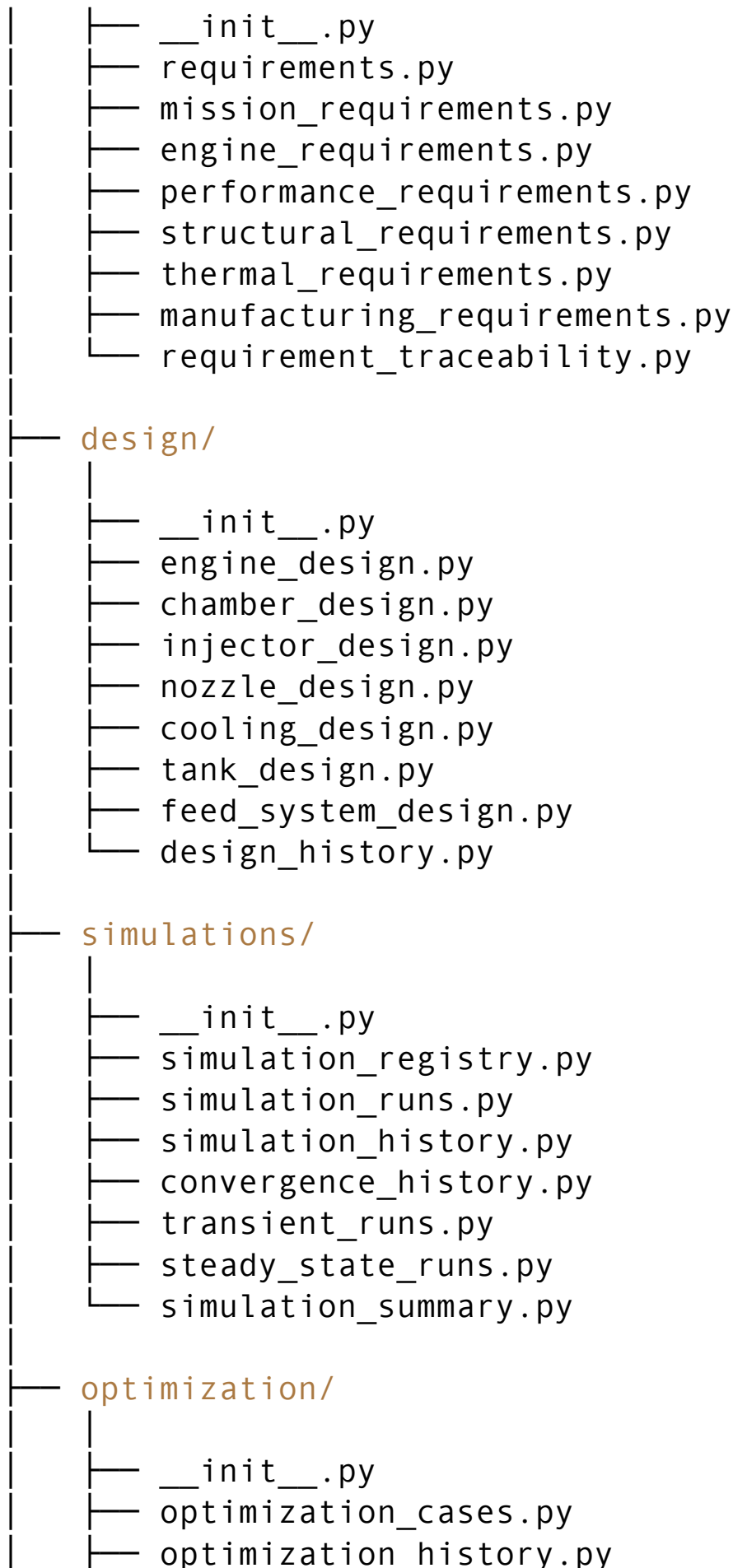


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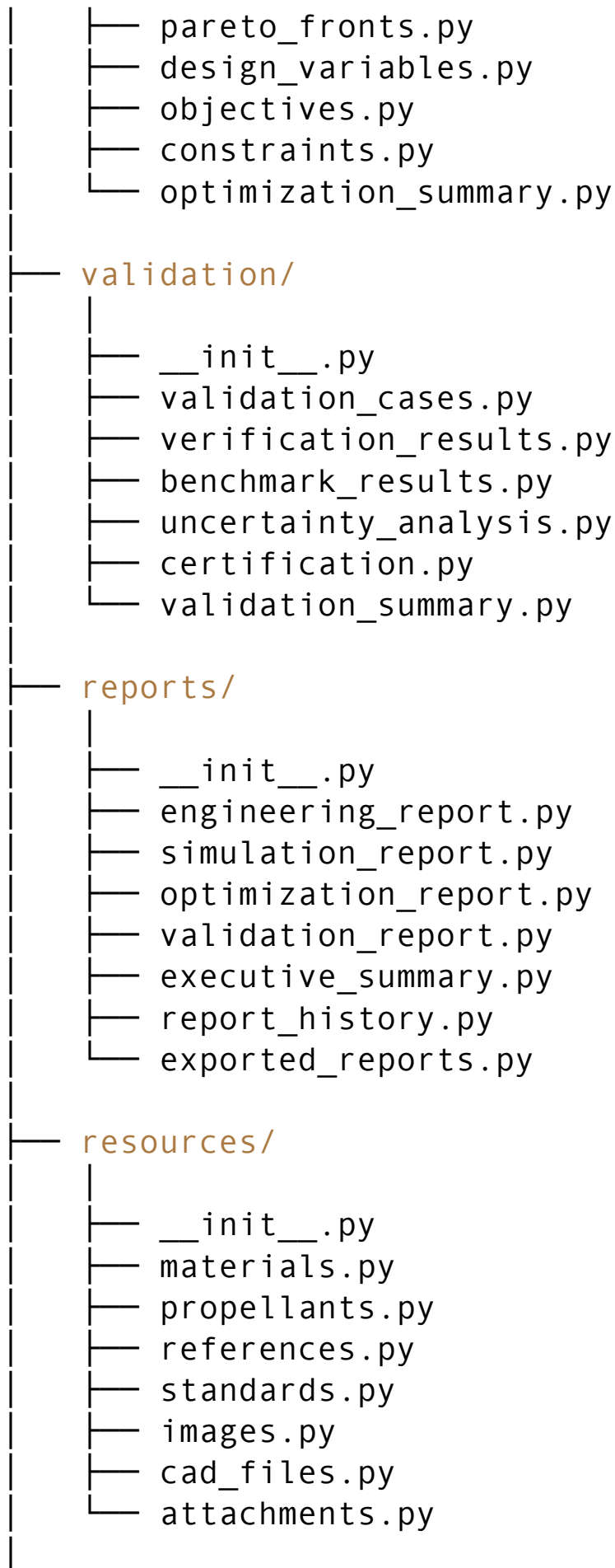
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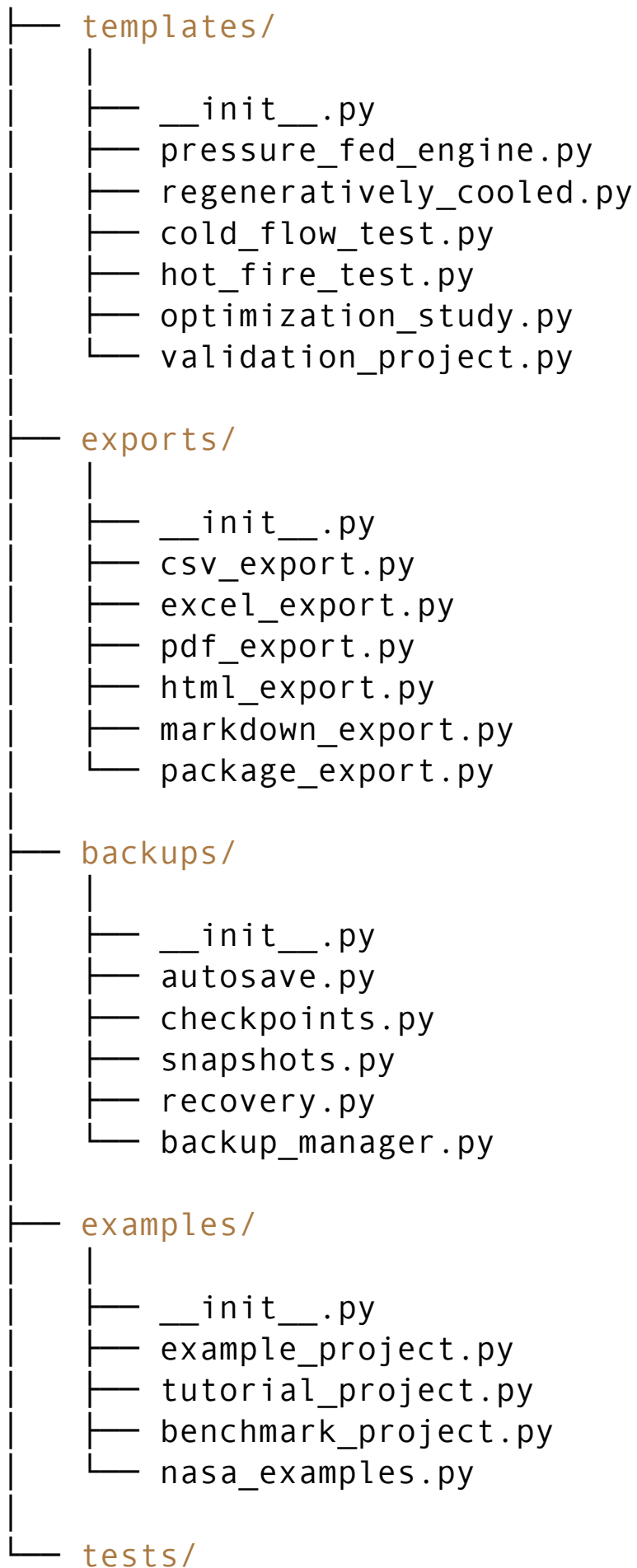
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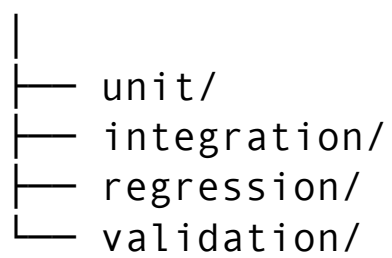
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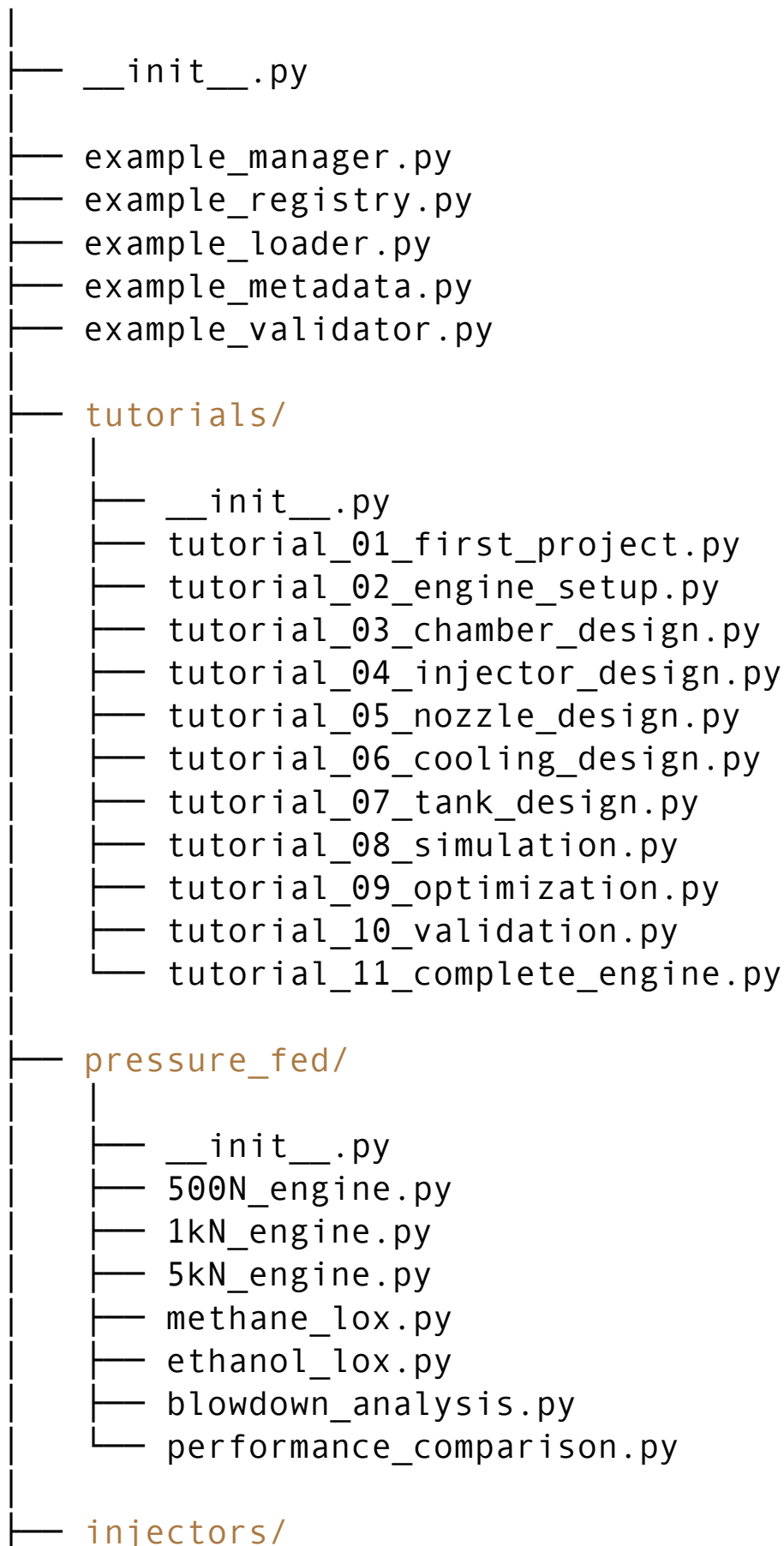
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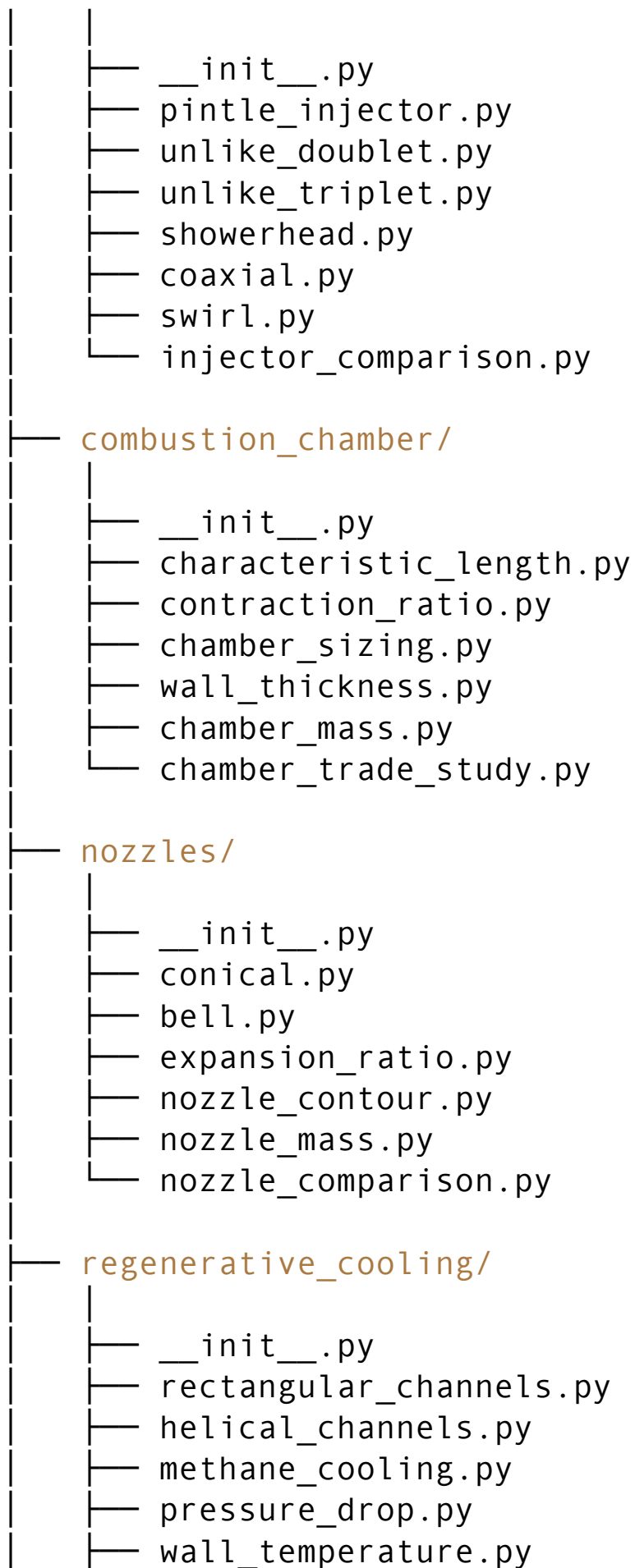
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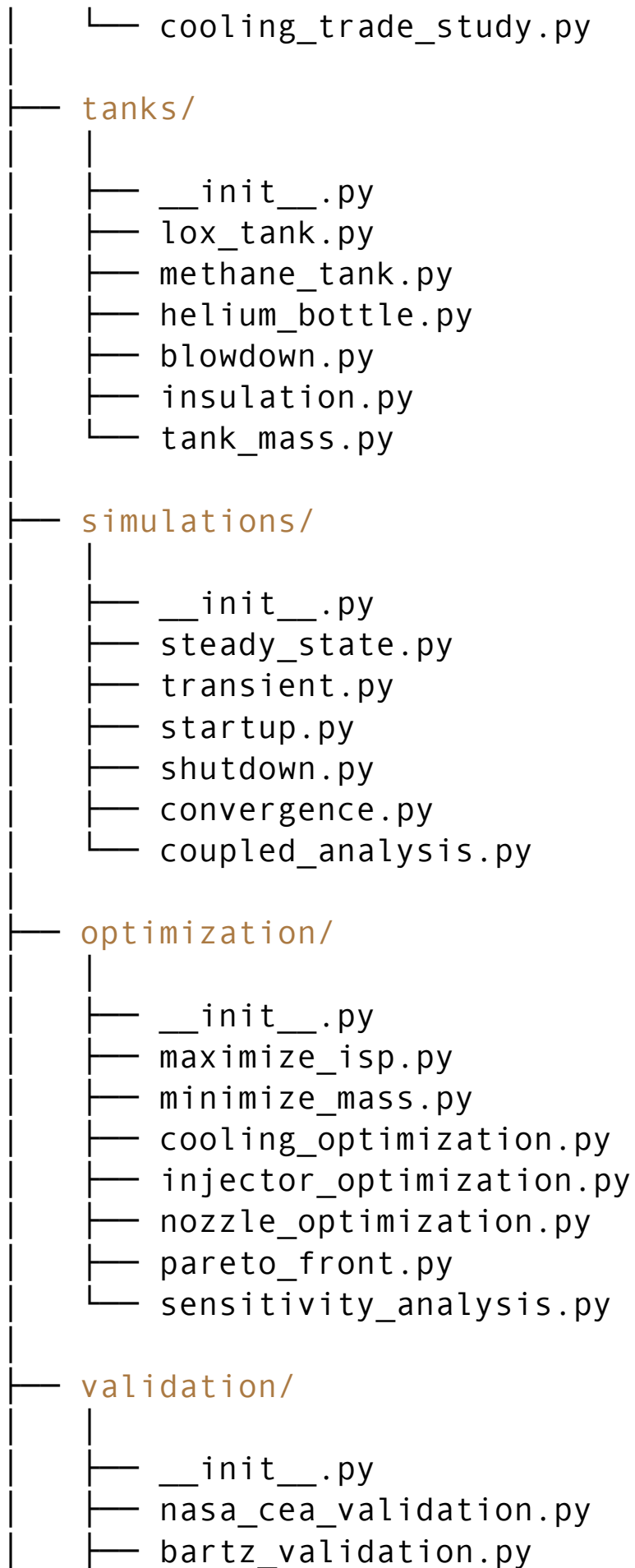
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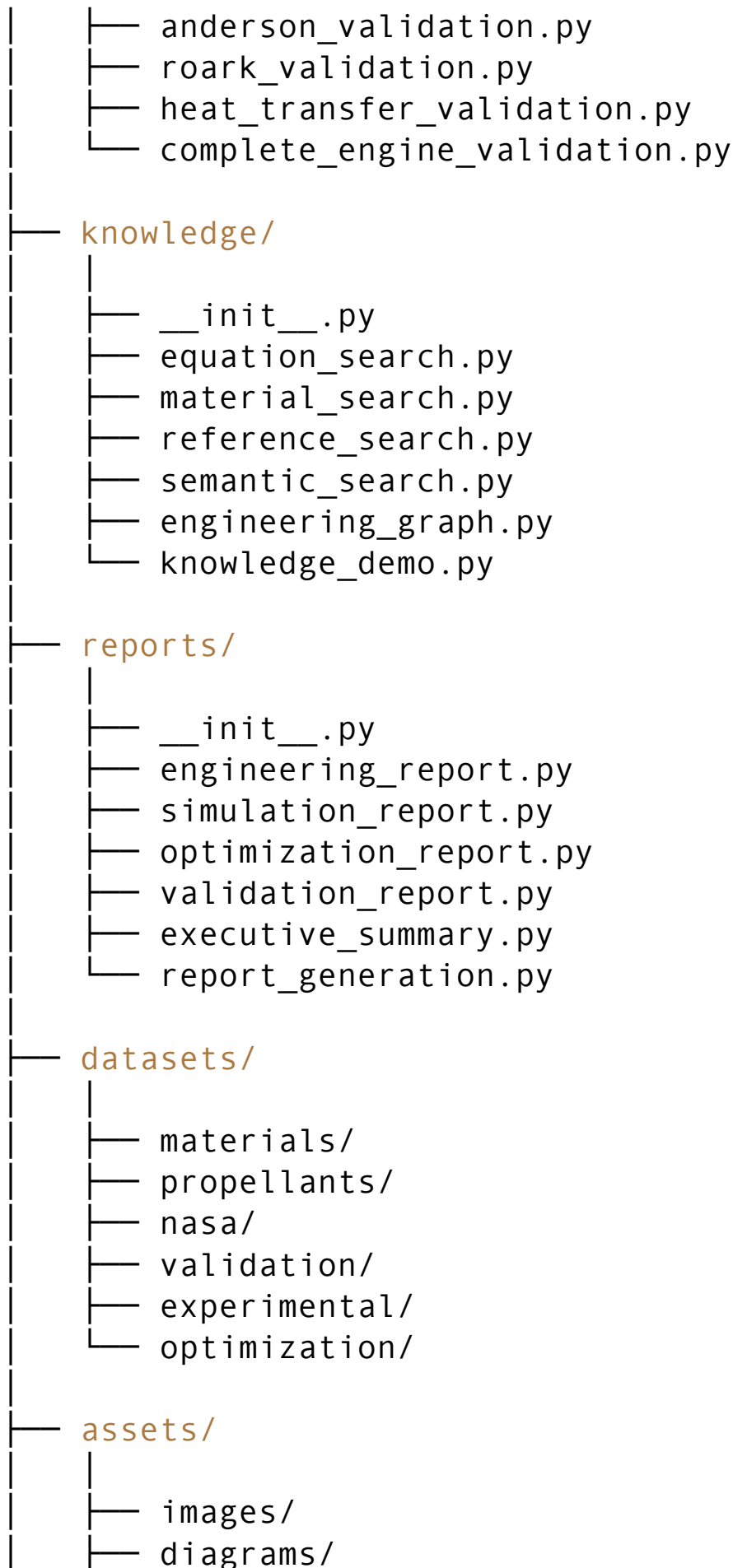
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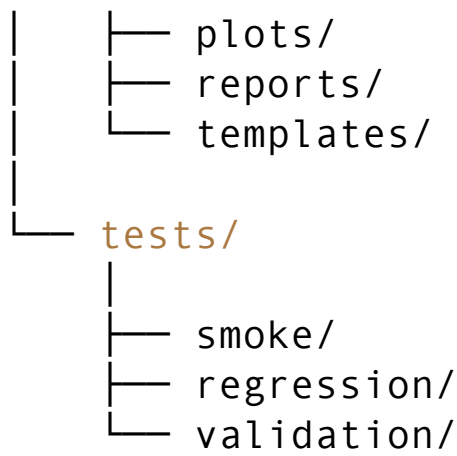
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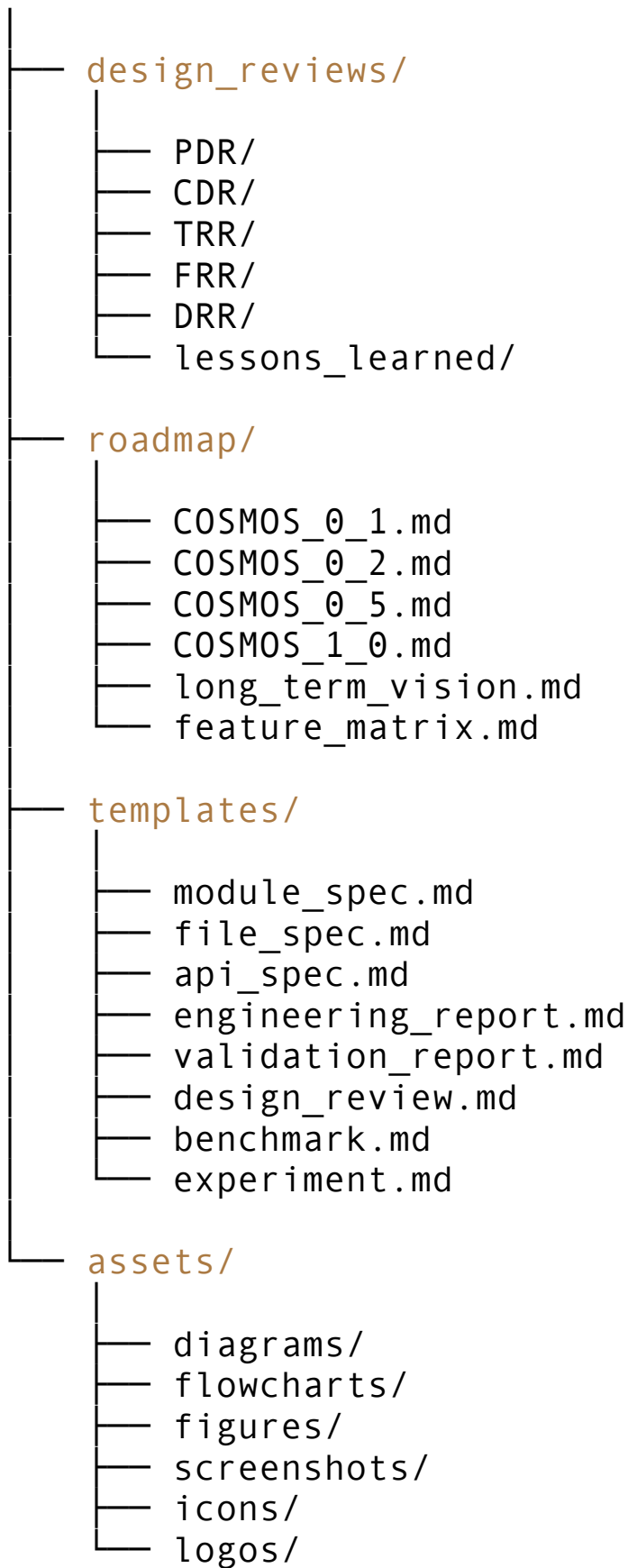
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- `infrastructure_manager.py`
- `startup.py`
- `shutdown.py`
- `bootstrap.py`
- `dependency_injection.py`
- `service_locator.py`
- `lifecycle.py`
- `health_check.py`

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- `__init__.py`
- `configuration_manager.py`
- `environment.py`
- `profiles.py`
- `defaults.py`
- `overrides.py`
- `settings_loader.py`
- `validation.py`

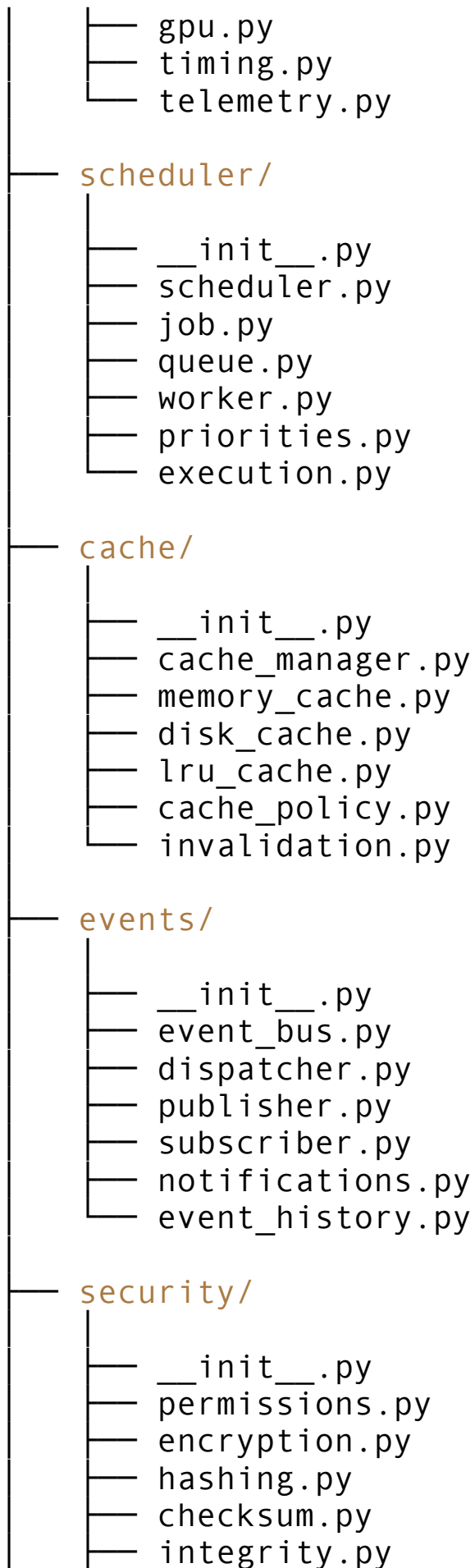
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- `__init__.py`
- `logger.py`
- `handlers.py`
- `formatters.py`
- `filters.py`
- `log_rotation.py`
- `performance_logger.py`
- `diagnostics.py`

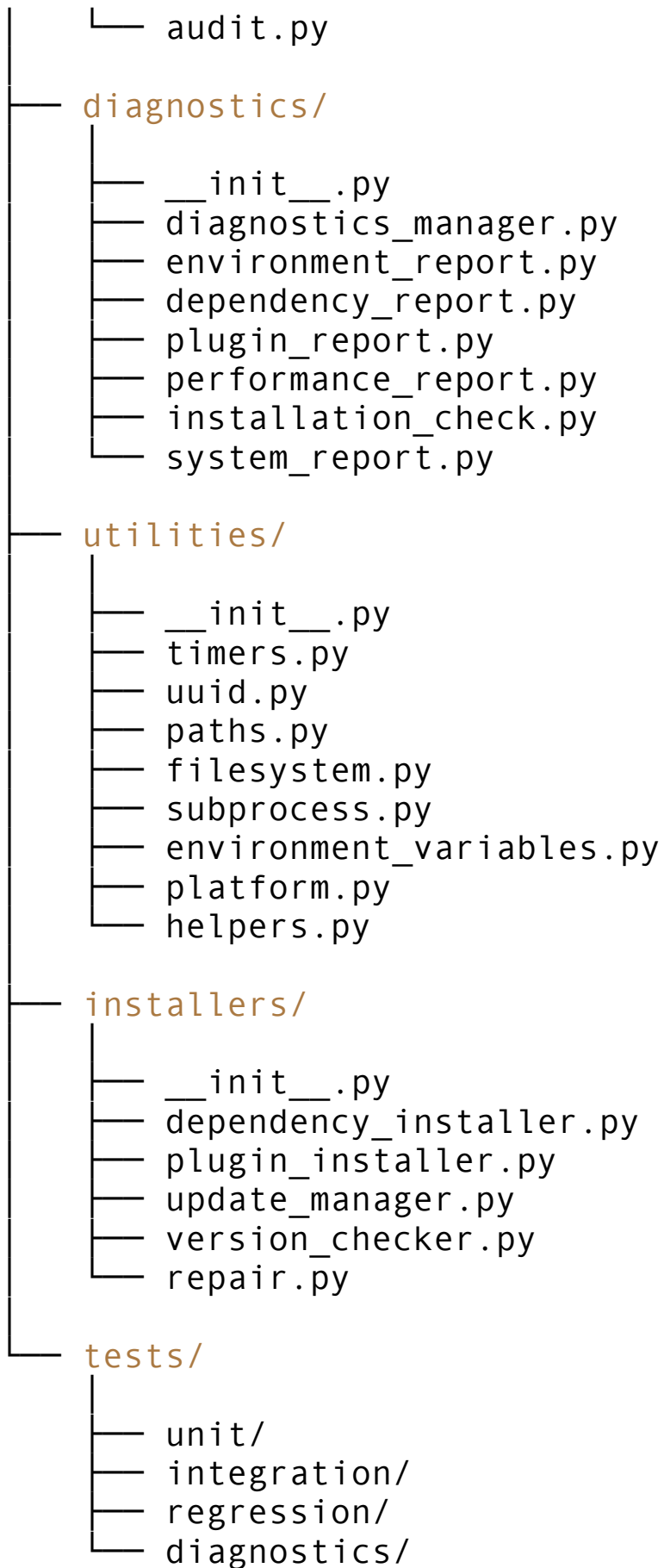
monitoring/

- `__init__.py`
- `resource_monitor.py`
- `cpu.py`
- `memory.py`
- `disk.py`

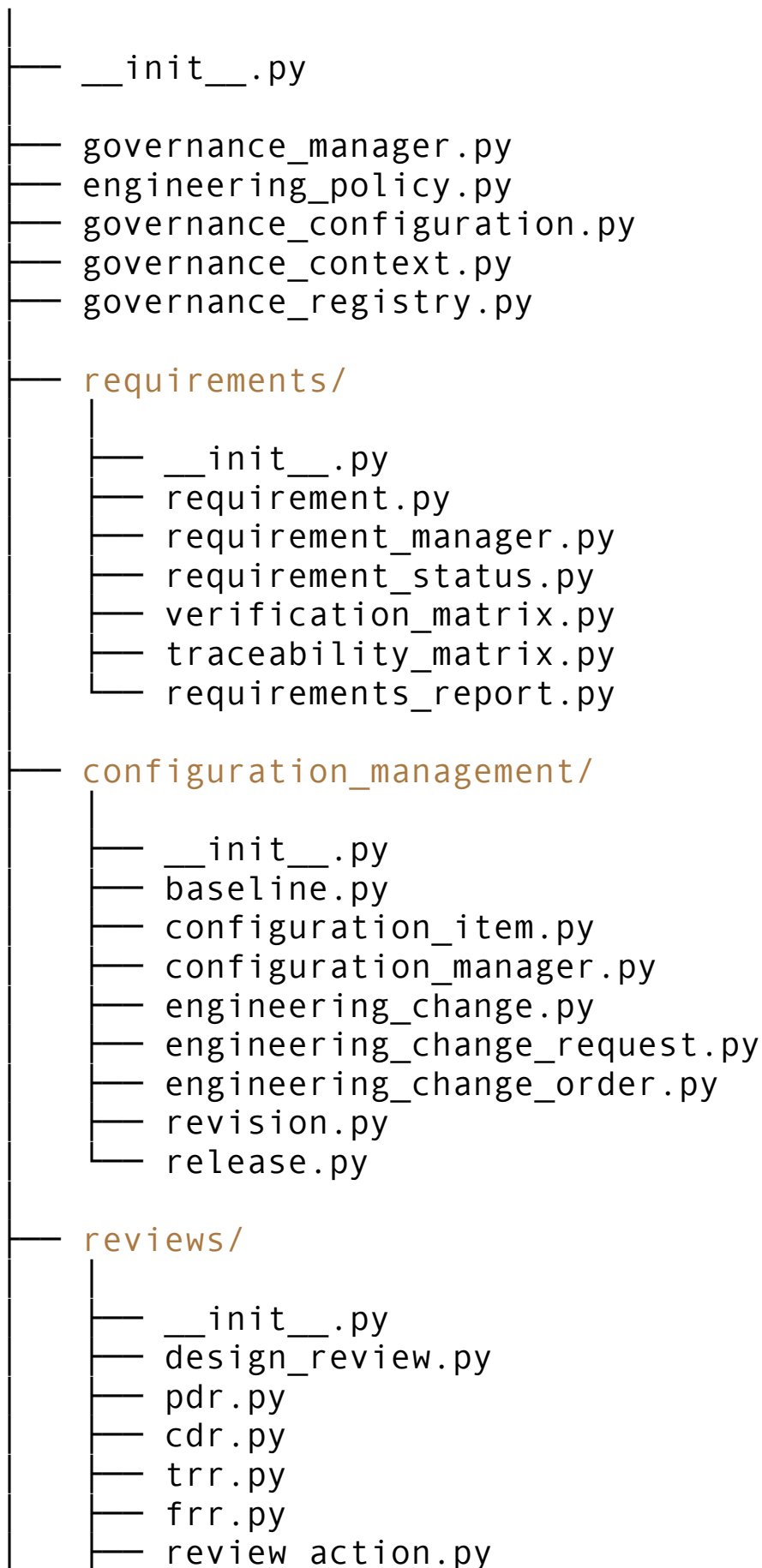
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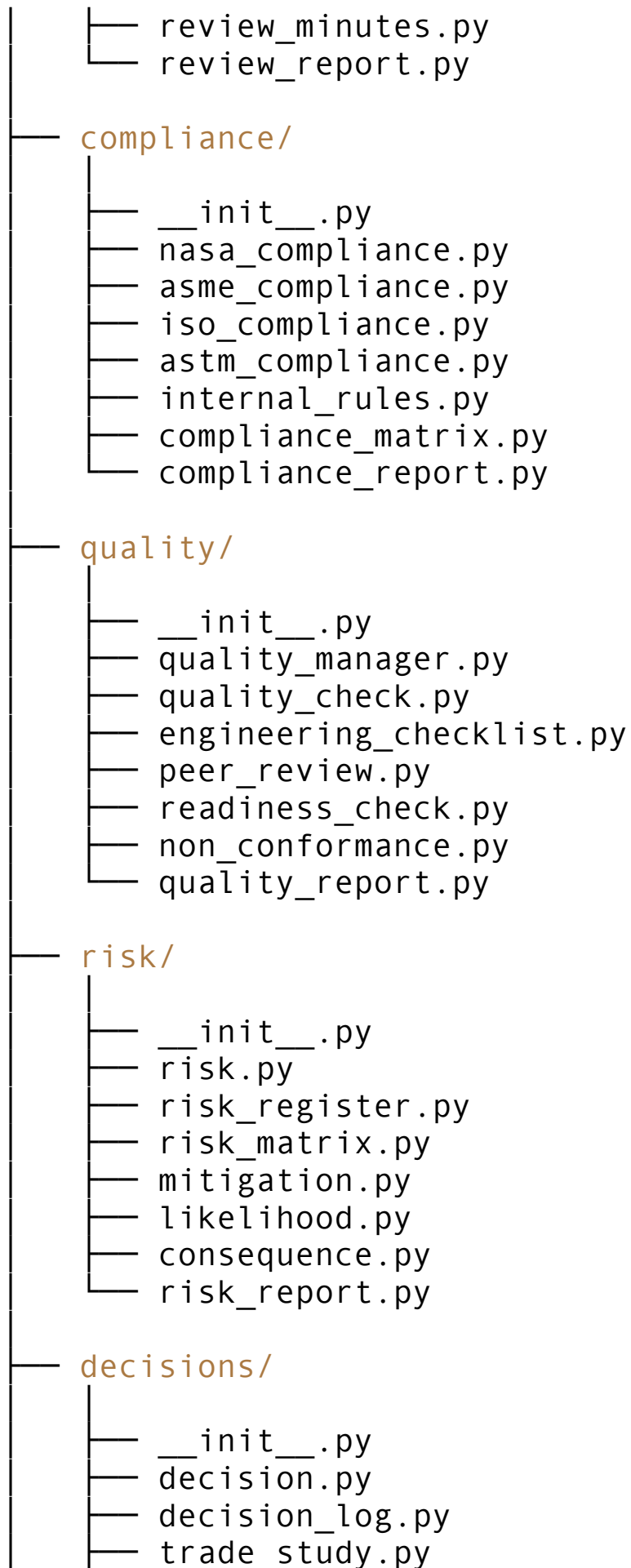
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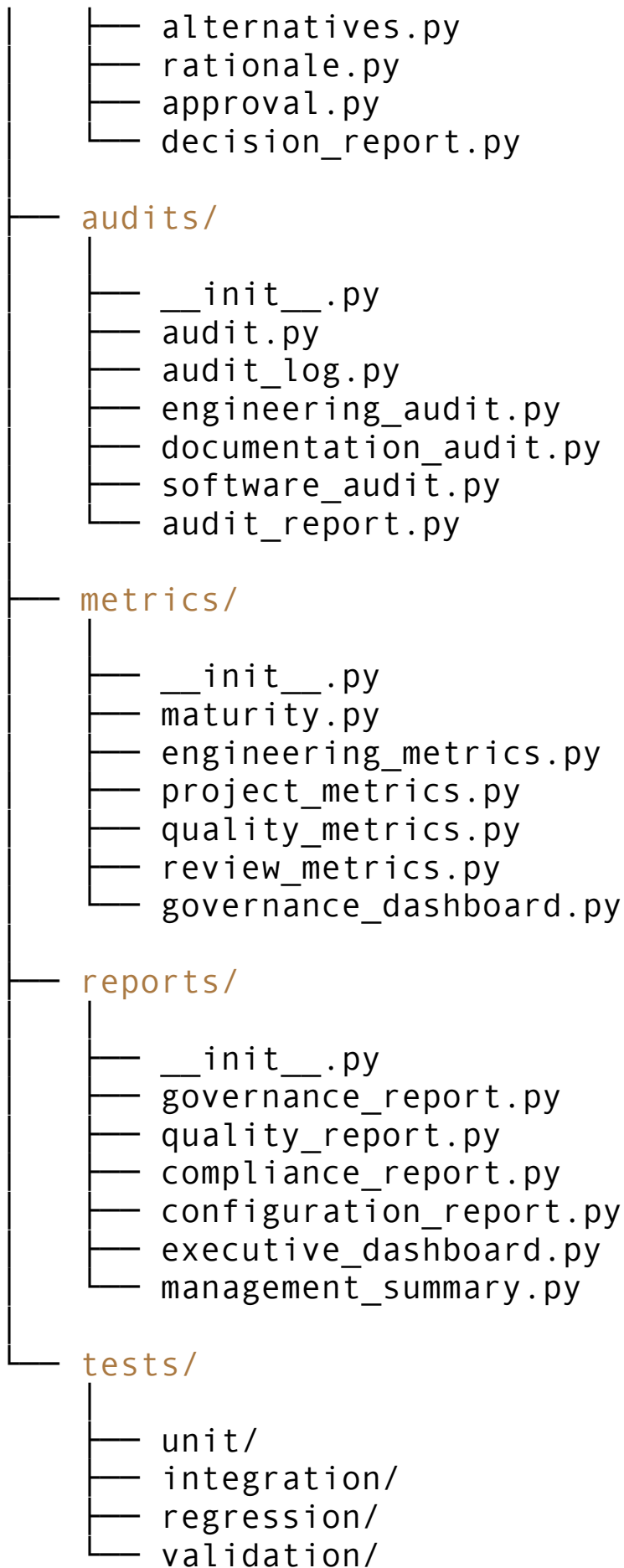
[governance/](#)



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- `__init__.py`
- `bootstrap.py`
- `setup_environment.py`
- `install_dependencies.py`
- `verify_installation.py`
- `update_dependencies.py`
- `clean_workspace.py`
- `generate_version.py`

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- `__init__.py`
- `build_material_database.py`
- `build_propellant_database.py`
- `build_equation_database.py`
- `import_nasa_data.py`
- `import_material_properties.py`
- `validate_database.py`
- `backup_database.py`
- `migrate_database.py`

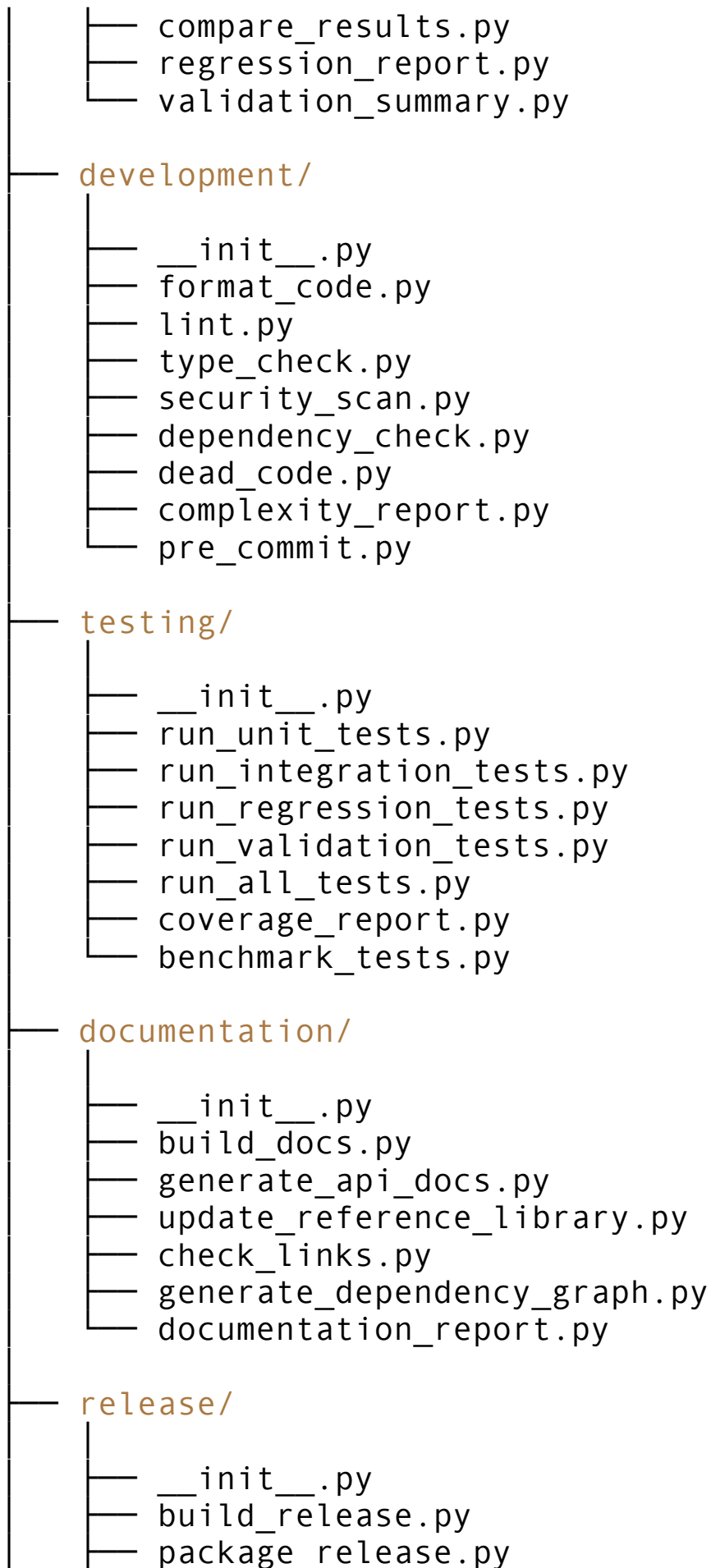
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- `__init__.py`
- `build_ontology.py`
- `build_knowledge_graph.py`
- `import_documents.py`
- `import_equations.py`
- `import_references.py`
- `rebuild_embeddings.py`
- `validate_knowledge.py`
- `export_knowledge.py`

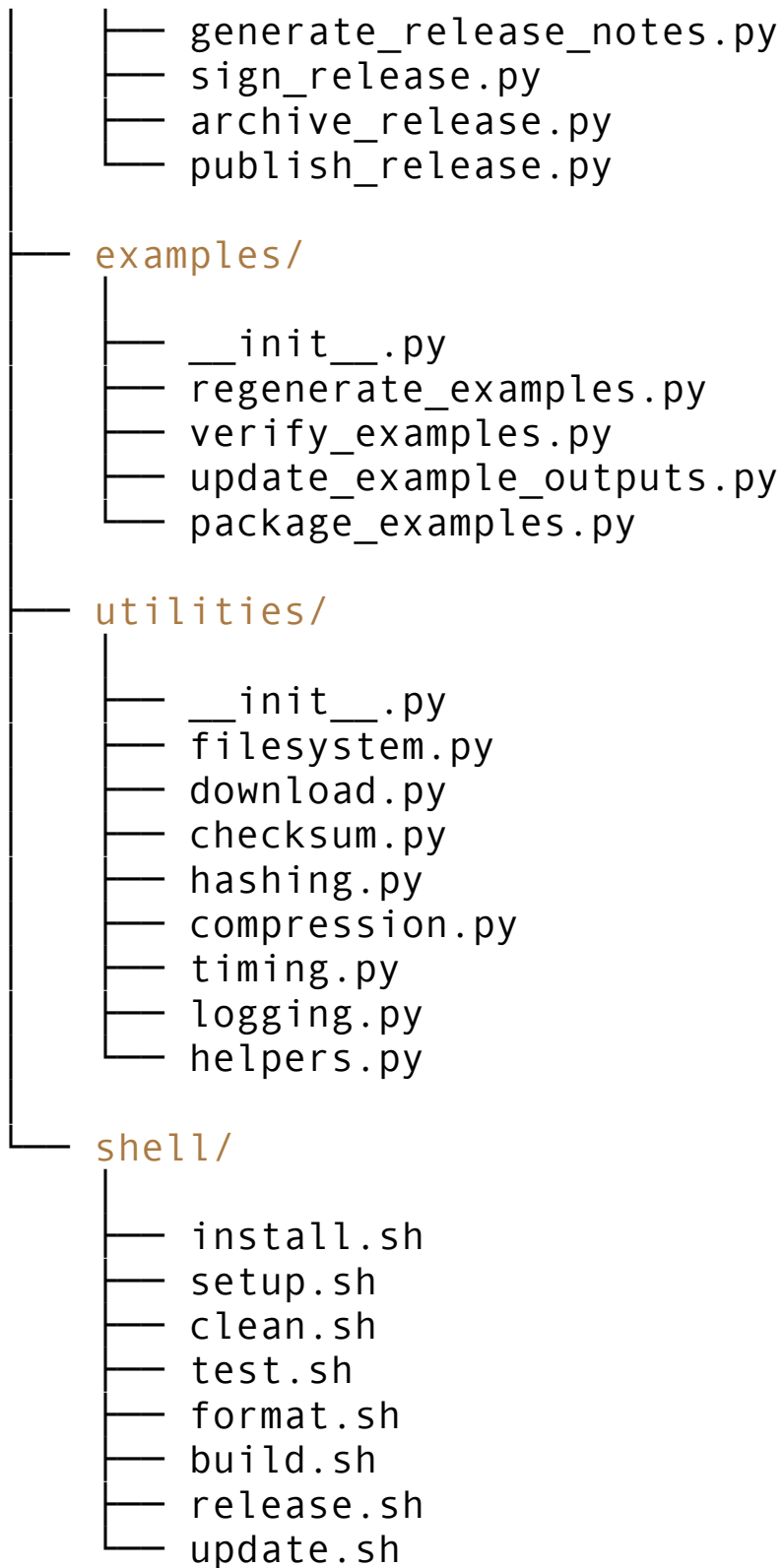
validation/

- `__init__.py`
- `generate_benchmarks.py`
- `import_nasa_benchmarks.py`
- `verify_reference_data.py`

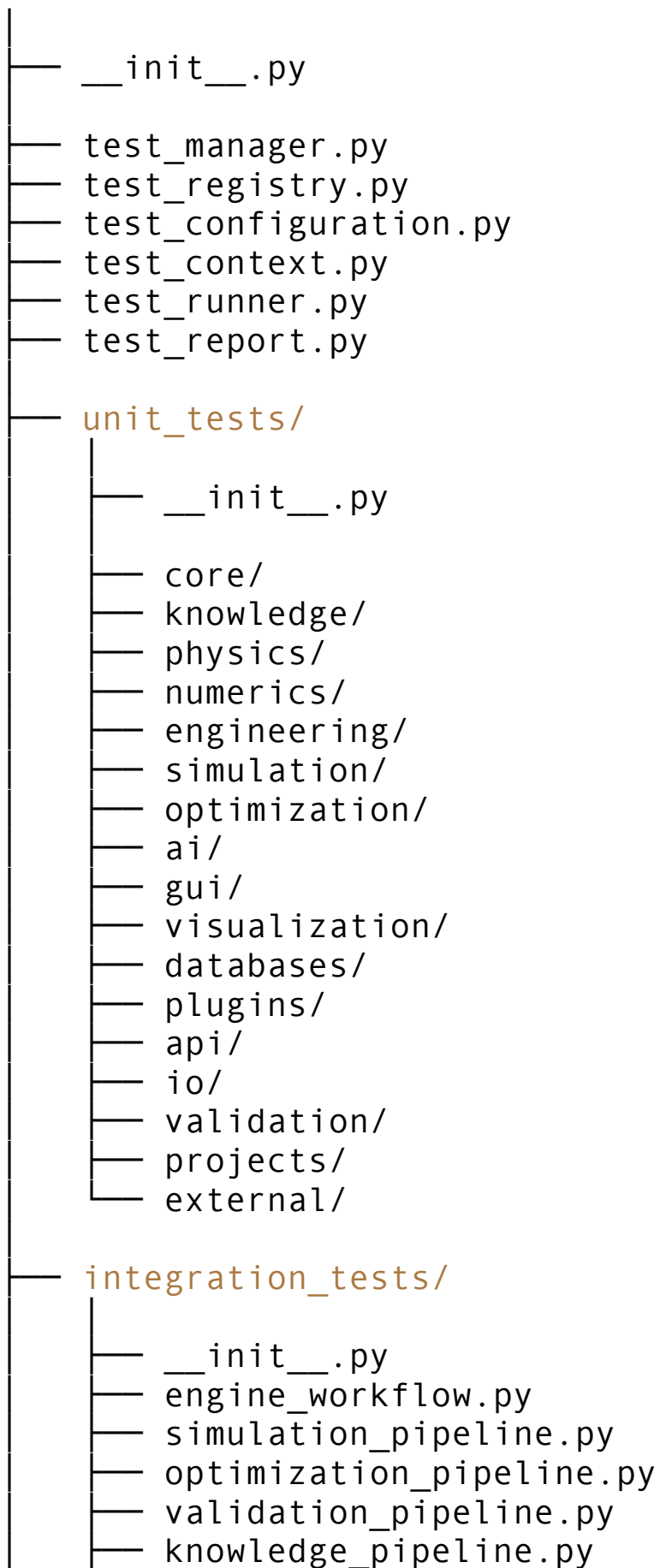
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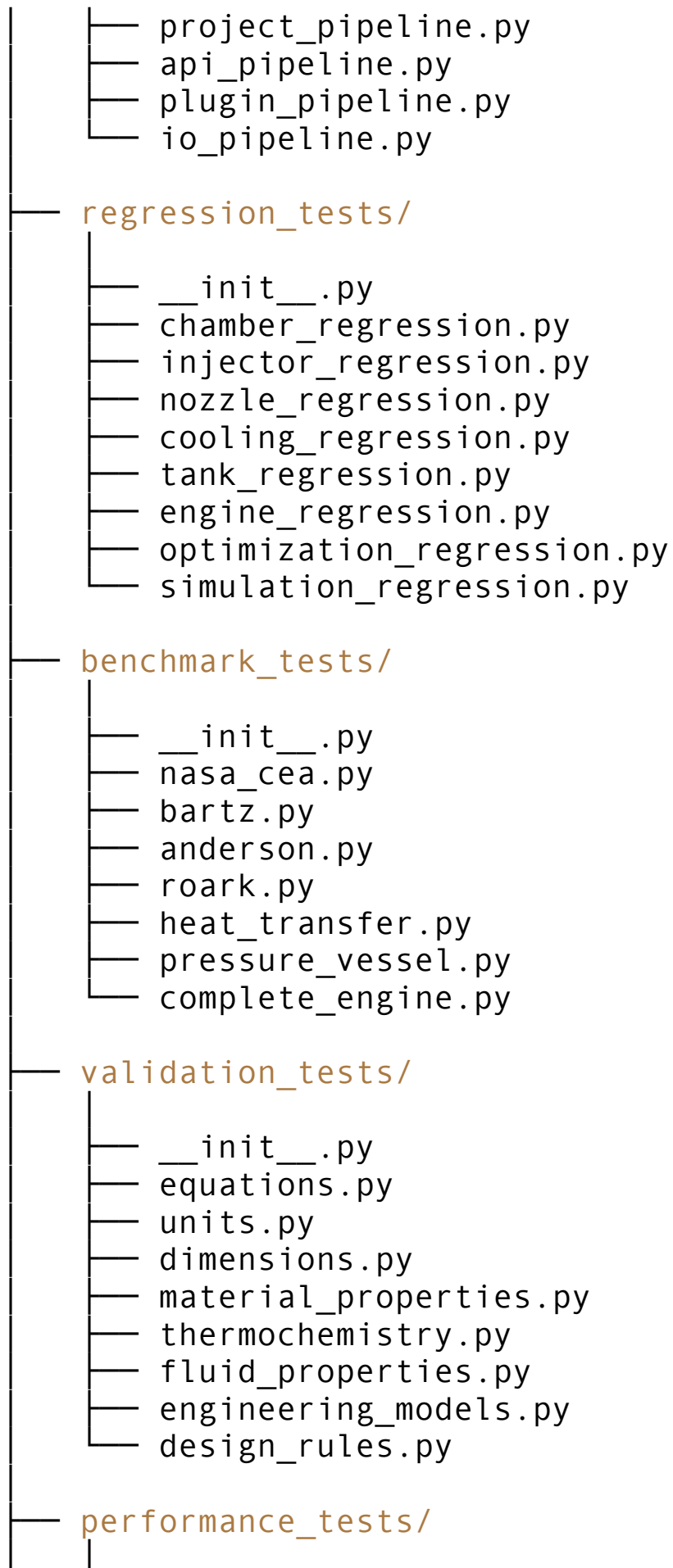
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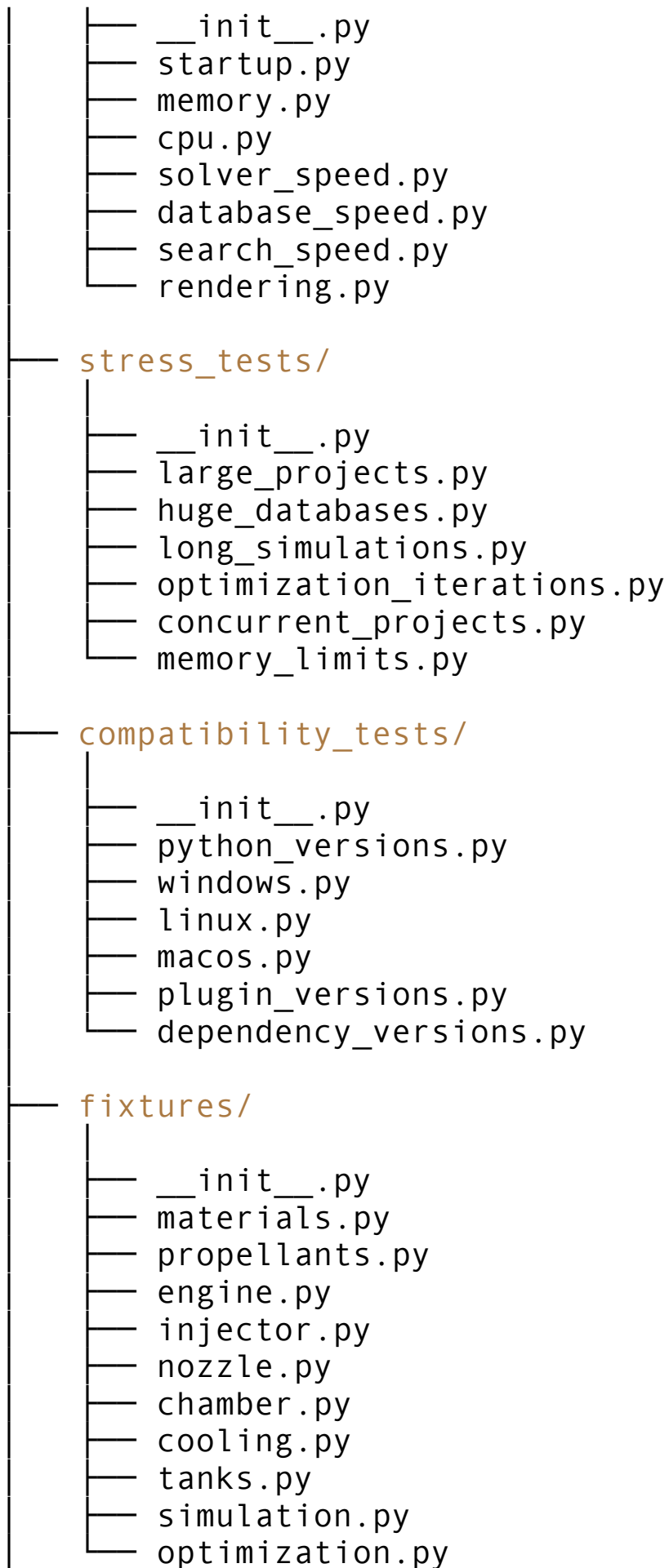
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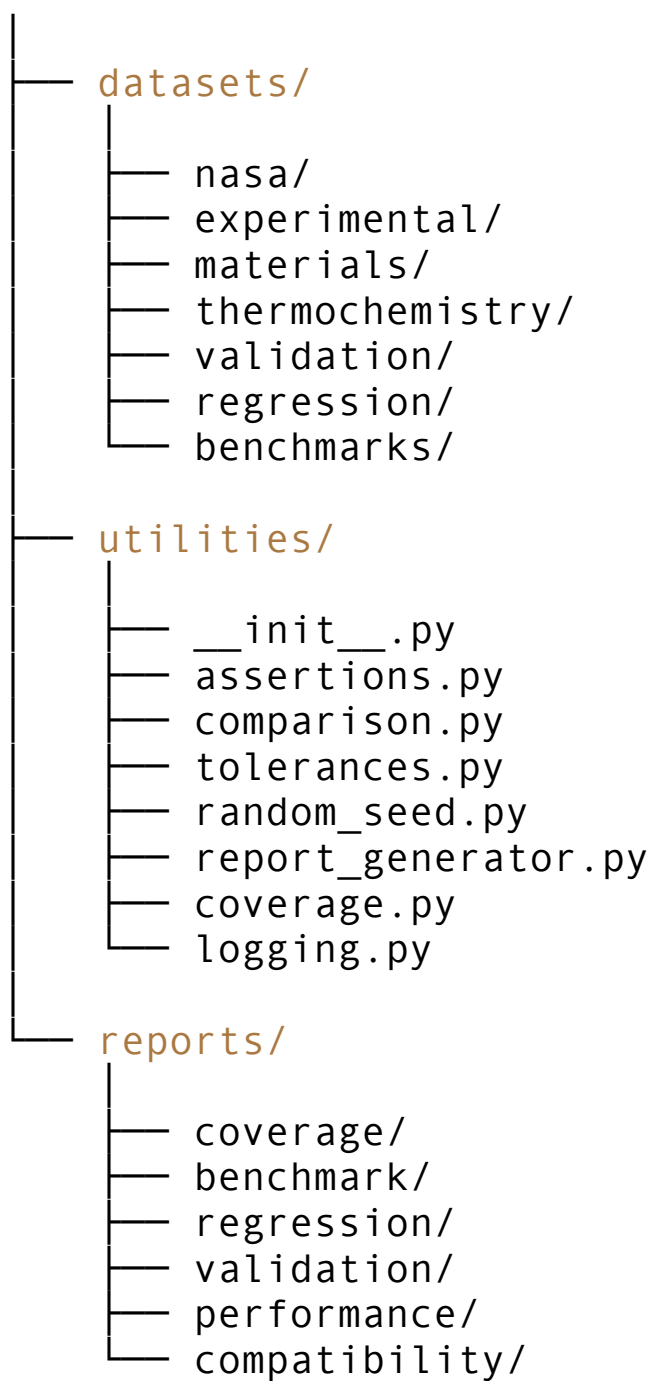
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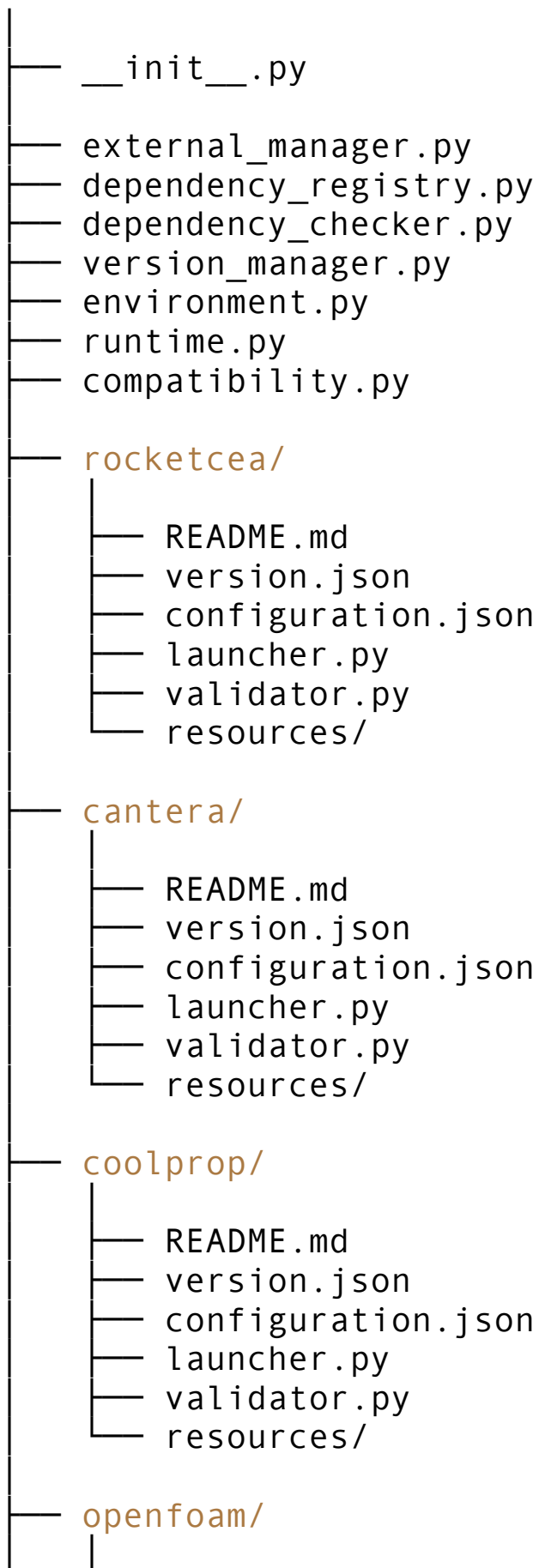
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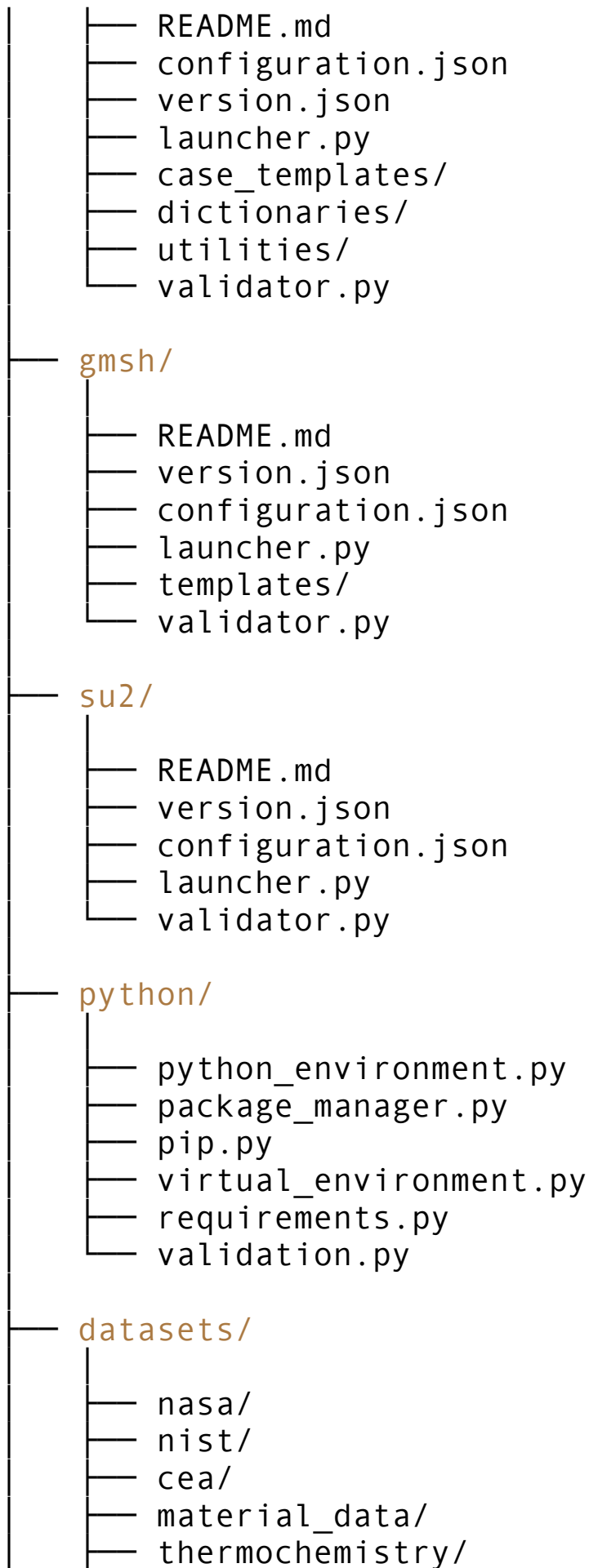
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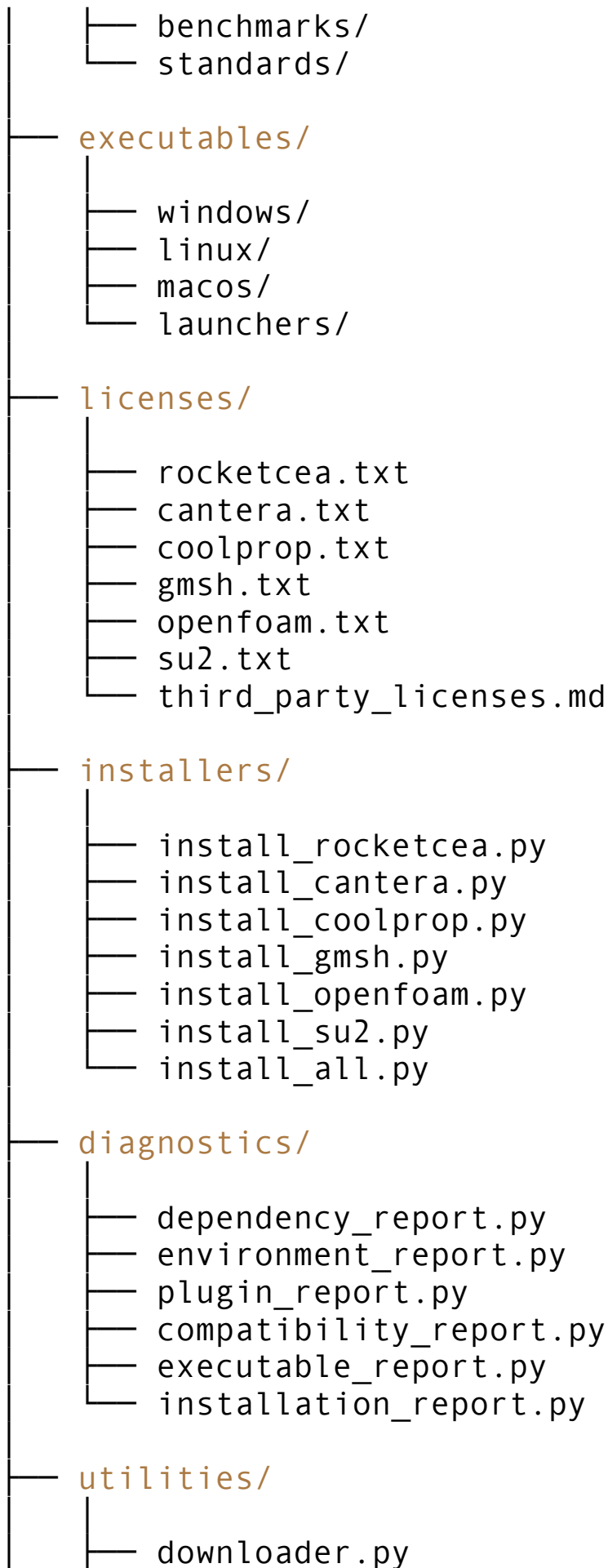
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